

UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.

Before the Honorable Doris Johnson Hines
Administrative Law Judge

In the Matter of

CERTAIN WEARABLE DEVICES

Inv. No. 337-TA-1478

COMPLAINANTS' PRE-HEARING BRIEF

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TABLE OF ABBREVIATIONS

Abbreviation	Phrase
Oura or Respondents	Respondents Ouraring Inc. and Oura Health Oy
Samsung or Complainants	Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc.
'359 Patent	U.S. Patent No. 10,642,359
'677 Patent	U.S. Patent No. 10,945,677
'675 Patent	U.S. Patent No. 10,231,675
'789 Patent	U.S. Patent No. 10,978,789
Sensor Patents	'359, '677, '675 Patents
Asserted Claims	'359 patent: claims 1 and 4 '677 patent: claims 1, 2, 3, and 9 '675 patent: claims 13 and 15 '789 patent: claims 6 (technical DI), and 7,9 (infringement)
DI	Domestic Industry
TDI	Technical prong of Domestic Industry
Accused Products	Oura Ring 4 Oura Ring 5 (a/k/a [REDACTED]) [REDACTED]
DI Product(s)	One or more of: - Galaxy Watch 4, 5, 6, 7, 8, Ultra - Galaxy Ring - Samsung Galaxy S22, S22+, S22 Ultra, S23, S23+, S23 Ultra, S24, S24+, S24 Ultra, S25, S25+, and S25 Edge
WDI Product(s)	One or more of the Wearable DI Products: - Galaxy Watch 4, 5, 6, 7, 8, Ultra - Galaxy Ring
PDI Product(s)	One or more of the Phone DI Products: Samsung Galaxy S22, S22+, S22 Ultra, S23, S23+, S23 Ultra, S24, S24+, S24 Ultra, S25, S25+, and S25 Edge
PCB	Printed Circuit Board
HR	Heart rate
SpO2	Blood oxygen (or blood oxygenation)
IR	Infrared

TABLE OF ASSERTED CLAIMS

'675 Patent	
[13.pre]	A method comprising:
[13.a]	determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device;
[13.b]	acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body;
[13.c]	activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;
[13.d]	detecting a biometric information of the user using the activated at least one second sensor;
[13.e]	during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor;
[13.f]	and in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.
[15.pre]	The method of claim 13, further comprising:
[15.a]	acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and
[15.b]	activating the at least one second sensor based on at least a part of the additional information.

'677 Patent	
[1.pre]	A wearable electronic device comprising:
[1.a]	a motion sensor;
[1.b]	one or more optical sensors includes a light emitting unit and a light receiving unit; and
[1.c]	at least one processor configured to:
[1.d]	obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,
[1.e]	emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition,
[1.f]	obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user,
[1.g]	obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and

[1.h]	refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.
[2.pre]	The wearable electronic device of claim 1, wherein the at least one processor is further configured to:
[2.a]	determine whether the wearable electronic device is attached to the user using the one or more optical sensors; and
[2.b]	perform the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user.
[3.pre]	The wearable electronic device of claim 2, wherein the at least one processor is further configured to:
[3.a]	perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.
[9.pre]	A wearable electronic device comprising:
[9.a]	a first sensor;
[9.b]	a second sensor; and
[9.c]	at least one processor configured to:
[9.d]	obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,
[9.e]	activate the second sensor based at least in part of the movement,
[9.f]	obtain, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and
[9.g]	refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range.

'359 Patent	
[1.pre]	A wearable device comprising:
[1.a]	a motion sensor;
[1.b]	a biological sensor; and
[1.c]	a controller configured to:
[1.d]	obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor,
[1.e]	determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, and
[1.f]	perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition.
[4.pre]	The wearable device of claim 1, wherein the controller is further configured to:
[4.a]	determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.

'789 Patent	
[1.pre]	A portable communication device comprising:
[1.a]	a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;
[1.b]	a wireless communication circuitry
[1.c]	a first printed circuit board (PCB) including
[1.c][i]-[iv]	a first PCB portion; a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion; a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion; a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion
[1.c][v][a]	a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line,
[1.c][v][b]	the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.
[6.pre]	The portable communication device of claim 1, further comprising:
[6.a]	a second PCB on which the wireless communication circuitry is located,
[6.b]	wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.
[7.pre]	A portable communication device comprising:
[7.a]	a first electronic component;
[7.b]	a second electronic component; and
[7.c]	a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:
[7.c][i]	a first rigid PCB portion;
[7.c][ii]	a first flexible PCB portion in contact with the first rigid PCB portion;
[7.c][iii]	a second rigid PCB portion in contact with the first flexible PCB portion;
[7.c][iv]	and a second flexible PCB portion in contact with the second rigid PCB portion,
[7.c][v]	wherein each of the first rigid PCB portion and the second rigid PCB portion includes:

[7.c][v][a]	a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion;
[7.c][v][b]	a first conduction layer located on or above the first insulating layer;
[7.c][v][c]	a second insulating layer located on or above the first conduction layer and having a first thickness;
[7.c][v][d]	a second conduction layer located on or above the second insulating layer; and
[7.c][v][e]	a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion,
[7.c][vi]	and wherein each of the first flexible PCB portion and the second flexible PCB portion includes:
[7.c][vi][a]	a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;
[7.c][vi][b]	a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the first rigid PCB portion and the first conduction layer of the second rigid PCB portion;
[7.c][vi][c]	a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;
[7.c][vi][d]	a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and
[7.c][vi][e]	a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.
[9.pre]	The portable communication device of claim 7, wherein:
[9.a]	each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer,
[9.b]	and a third conduction layer located between the third insulating layer and the fourth insulating layer.

I. INTRODUCTION

A violation of Section 337 is compelled by the great weight of evidence that Samsung has presented versus the complete absence of contrary evidence or even any rebuttal from Oura across many issues. Samsung is asserting infringement of four patents, falling into two groups: (1) three patents directed to the operation of sensors on wearable devices, U.S. Patent Nos. 10,231,675 (“’675 Patent”), 10,945,677 (“’677 Patent”), and 10,642,359 (“’359 Patent”) (hereinafter “Sensor Patents”), and (2) U.S. Patent No. 10,978,789 (“’789 Patent”), which is directed to the structure of a printed circuit board (“PCB”). For Technical DI (“TDI”), Oura has no rebuttal at all for some claims. For infringement, Oura’s rebuttals rely on an improper “practicing the prior art defense” or belated, incorrect claim constructions. Oura also cannot carry its burden on invalidity. Its invalidity grounds rely on references missing critical claim elements and dubious motivations to combine, and Oura disputes only a subset of the secondary considerations of non-obviousness.

For both infringement and TDI for all three Sensor Patents, Samsung’s theory is largely the same, focusing on how SpO2 (blood oxygen) detection is triggered based on motion signals in the Accused Products and the Wearable DI (“WDI”) Products. In both the Accused and WDI Products, an accelerometer tracks motion for wear detection and [REDACTED]. [REDACTED]. If little or no motion is detected, as is common during periods when the user is at rest, the products [REDACTED]. [REDACTED]. Conversely, if the accelerometer detects motion indicating that the user is active, the products [REDACTED]. [REDACTED]. Oura does not dispute the Accused and WDI Products perform SpO2 detection, nor does Oura dispute how that detection works, including that it is [REDACTED].

Indeed, for the two related '675 and '677 Patents, Oura has presented *no rebuttal* evidence or opinion from its expert, Dr. Mortazavi, on noninfringement. Instead, Oura responds with an improper “practicing the prior art defense,” contending that if the Accused Products practice the claims, then so too must Oura’s own Ring 1 product, which Oura contends is prior art. Leaving aside the stark material differences between Ring 1 and 4, the Federal Circuit and the ALJ have rejected “practicing the prior art” as a defense to infringement. Similarly, for TDI, Oura presents no rebuttal for all asserted claims of the '675 Patent and Claim 9 of the '677 Patent. For '677 Patent, Claim 1 (and Claims 2-3 that depend from it), Oura asserts failure of proof, claiming that Samsung’s expert did not fully explain what aspects of the WDI Products correspond to the “first motion signal” and “biological signal.” But Oura’s contention is belied by Samsung’s expert report, which fully details how these “signal” requirements are met.

Oura’s invalidity arguments for the '677 and '675 Patents also provide no reason to deny relief under Section 337. Oura is relying heavily on its own Ring 1 product as prior art. But, as a threshold matter, Oura has failed to provide evidence that the Ring 1 even qualifies as prior art, given that it only went on sale in late 2015, *after* the August 12, 2015 priority dates. The source code Oura attempts to rely on likely postdates the priority date of the '677 and '675 Patents by over a year. Beyond these threshold failings, it is undisputed that the Ring 1 could *not* measure SpO2—the very feature that Samsung accuses of infringement for Ring 4. Thus, Oura cannot possibly claim that Ring 1 satisfies the claims for the same reasons as Ring 4. Oura’s other prior art references all suffer from the same flaw: none inactivates the alleged “biological sensor” in response to detecting no motion as required by the claims. Rather, the “biological sensor” continues to operate at a reduced level, which does not satisfy the claim requirements to “*inactivate*” the “biological sensor” or “refrain from the emitting of the light.”

[REDACTED]

For the '359 Patent, asserted Claim 1 requires that a device determine a “select portion” of the “biological signal” that is obtained during a corresponding “select period” and then perform a “specified function” using that “select portion.” And its asserted dependent Claim 4 requires evaluating whether the “biological signal” is “invalid” based on its magnitude. Both the Accused and TDI Products’ SpO2 detection processes satisfy these requirements. For Claim 1, the products perform the “[REDACTED]

[REDACTED] (“select portion” of the “biological signal”) [REDACTED]

[REDACTED] (i.e., “select period”). For Claim 4, [REDACTED]

[REDACTED] In rebuttal, Oura’s *only* argument for both noninfringement and TDI is that the “select portion” of the “biological period” refers to a smaller portion of the signal *within* that “select period.” But Oura’s rebuttal argument ignores source code evidence Samsung has presented showing infringement even under Oura’s interpretation of “select portion.”

The prior art reference that Oura presents for the '359 Patent are also deficient. None of the references teaches the performing a “specified function” using the “select portion” requirement of both Claims 1 and 4. And for Claim 4’s additional requirement of checking for “invalid” values, Oura resorts to obviousness combinations with prior art references from disparate fields that lack motivation to combine and that, at any rate, do not disclose the claimed invalidity check.

For the '789 Patent, Samsung is asserting Claims 7 and its dependent Claim 9 for infringement and Claim 6 (which depends from Claim 1) for DI. Claims 7 and 9 recite a printed circuit board (“PCB”) structure composed of alternating rigid-flexible portions, with multiple conductive and insulating layers across those portions and each layer having particular thicknesses and materials. Based on extensive teardown analysis of the Oura Ring 4, Samsung has demonstrated that all of Claims 7 and 9’s structural requirements are met. In response, Oura’s sole

[REDACTED]

defense again relies on belated claim interpretation that different insulating materials cannot be grouped together and mapped to a single “insulating layer” as recited in the claims. But Oura’s interpretation is undermined by the ’789 Patent specification, which expressly describes individual “insulating layers,” each made of multiple insulating materials. For TDI, Oura similarly relies on belated claim interpretations that are either contradicted by the specification or irrelevant in view of the undisputed technical facts as described in Samsung’s documentation.

The prior art that Oura presents to the ’789 Patent does not come close to meeting the claims’ requirements. For Claims 7 and 9, none of Oura’s prior art grounds disclose both claims’ requirement that (1) the “second insulating layer” in the rigid portion of the PCB be *thicker* than the “second insulating layer” in the PCB’s flexible portion; and (2) that a *different* insulating material be used to form the exterior, outermost layer of the PCB’s rigid portion versus its flexible portion. And for Claim 6, Oura presents only one prior art ground, which relies a combination of three disparate references: the Galaxy S5 smartphone as a primary reference and two publication references having nothing to do with smartphones.

Oura also presents Section 101 defenses for the three Sensor Patents, but these defenses fail under Federal Circuit precedents. The Sensor Patents’ claims are directed to wearable devices with a specific sensor configuration (e.g., motion sensor and a biological sensor), and deactivating the biological sensor when motion is too high to (1) improve the reliability of measurements taken by the biological sensor; and (2) conserve power from not having the biological sensors in constant use. The patents’ claims are no different from those upheld by the Federal Circuit in multiple cases about improved sensor systems.

Regarding economic domestic industry, Samsung has made, and will continue to make, significant domestic investments in labor and capital, as well as plant and equipment, under 19

[REDACTED]

U.S.C. §1337(a)(3) with respect to the domestic industry products. Samsung's domestic investments in the domestic industry products are significant and substantial in the context of SEA's mobile division, in the context of Samsung's foreign expenditures, and in the context of the industry. Samsung's domestic investments are expected to continue.

II. BACKGROUND

A. Procedural History

Samsung filed a complaint under Section 337 against Oura on December 12, 2025, asserting infringement of the four Asserted Patents. JDX-0030C. Samsung filed an amended complaint on January 2, 2026. JDX-0032.

On January 14, 2026, the Commission instituted this Investigation based on the amended complaint's allegations of Section 337 violations arising from Oura's importation into the U.S. and the sale for importation of certain wearable devices, including Oura's Ring 4 product, that infringe claims of the Asserted Patents. JDX-0034.

On June 15, 2026, the ALJ issued an order construing certain disputed terms of the Asserted Patents. JX-0009.

On July 13, 2026, Samsung moved for summary determination regarding infringement and TDI for the '677 and '675 Patents ("Samsung's MSD"). EDIS ID 887878 (Mot. No. 1478-009). On July 13, 2026, Oura filed two motions: (1) a motion for summary determination of noninfringement by Oura's alternative products, Ring 5 and [REDACTED] ("Oura's MSD") and (2) a motion strike deposition testimony from Samsung's experts regarding Ring 5 and [REDACTED] ("Oura's MTS"). EDIS ID 887851 (Mot. No. 1478-007); EDIS ID 887862 (Mot. No. 1478-008). The parties have completed briefing on all three motions, which are pending before the ALJ.

On July 29, 2026, the ALJ granted Samsung's motion to partially terminate the

Investigation. EDIS ID 889677. The remaining asserted claims are:

U.S. Patent No.	Remaining Claims at Issue
10,642,359	1 and 4
10,945,677	1-3 and 9
10,231,675	13 and 15
10,978,789	6 (for DI) 7 and 9 (for infringement)

The evidentiary hearing is scheduled for September 16-18, 21, and 22, 2026, with a target date set for May 17, 2027. EDIS ID 881268.

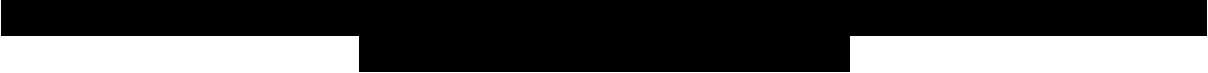
B. The Parties

1. Samsung Complainants

Complainant Samsung Electronics Co., Ltd. (“SEC”) is a corporation organized and existing under the laws of South Korea. JDX-0032, ¶8. SEC is the assignee of the Asserted Patents with the right to sue for all past, present, and future infringement thereof. *Id.* Complainant Samsung Electronics America, Inc. (“SEA”) is a corporation organized and existing under the laws of the state of New York, having its principal place of business in New Jersey. *Id.*, ¶9. SEA is a wholly-owned subsidiary of SEC. *Id.* SEA has over 40 facilities across the U.S. and employs thousands of personnel tasked with commercializing U.S. products, including its smartphones, wearables, and smart rings in the U.S. JDX-0002C, ¶28.

2. Oura Respondents

Ouraring Inc. is a Delaware corporation with its principal place of business in San Francisco, California. RDX-0034C, ¶40. Oura Health Oy is a Finnish company with its principal place of business in Oulu, Finland. *Id.*, ¶41. Ouraring, Inc. is a wholly owned subsidiary of Oura Health Oy. *Id.*, ¶40. Oura formed a new parent company, Oura Inc., a Delaware corporation. *Id.*, ¶41.



C. The Technology

The technology at issue in this Investigation falls into two categories. *First*, the three Sensor Patents relate to wearable devices that use biological sensors to measure a user's health characteristics (e.g., SpO2), and selectively activate the biological sensors during periods when a user is determined to be at rest based on motion signals detected by a separate motion sensor, while deactivating the biological sensors when the user is determined to be in motion also based on motion signals. JX-0001-JX-0003. This selective activation of the biological sensors during periods of rest (and deactivation during periods of activity) improves the accuracy of measurements and reduces power consumption. *See, e.g.*, JX-0002, 31:15-24.

Second, the '789 Patent relates to the internal structure of a rigid-flex PCB within a device, including its layers, thicknesses, and material composition, and the use of that PCB to transmit high frequency signals within the device. JX-0004, 1:19-24, 2:24-34. The patent explains that prior art used separate coaxial cables to transmit such high frequency signals within a device, and the patent's use of a PCB to transmit such signals instead presented many benefits, including saving space, making the claimed invention particularly advantageous for smaller-form factor devices, such as a ring. *Id.*, 1:36-48, 52-55.

D. The Asserted Patents

1. The '675 Patent

The '675 Patent, entitled "Method for Detecting Biometric Information and Electronic Device Using Same," issued on March 19, 2019, to inventors Gihung Chung, Jooman Han, So Hyun Chung, Dae-Yong Lee, Hongsig Kim, Hyunwoo Jang, Cheolho Cheong, Jae-Woong Chun, and Won Suk Choi, and claims priority to Korean Patent App. No. 10-2015-0113972, filed on August 12, 2015. JX-0003, Cover.

[REDACTED]

2. The '677 Patent

The '677 Patent, entitled “Method for Detecting Biometric Information and Electronic Device Using Same,” issued on March 16, 2021, to inventors Gihung Chung, Jooman Han, So Hyun Chung, Dae-Yong Lee, Hongsig Kim, Hyunwoo Jang, Cheolho Cheong, Jae-Woong Chun, and Won Suk Choi, and claims priority to Korean Patent App. No. 10-2015-0113972, filed on August 12, 2015. JX-0002, Cover. The '677 Patent is a continuation of the '675 Patent, and therefore shares a substantively identical specification.

3. The '359 Patent

The '359 Patent, entitled “Wearable Biosignal Interface and Method Thereof,” issued on May 5, 2020, to inventors Chang Mok Choi, Sang Joon Kim, and Seung Keun Yoon, and claims priority to Korean Patent App. No. 10-2013-0098476, filed on August 20, 2013. JX-0001, Cover.

4. The '789 Patent

The '789 Patent, entitled “Electronic Device Including High-Frequency Transmission Circuit,” issued on April 13, 2021, to inventors Han-Min Cho, Chan-Gi Park, Yeon-Sang Yun, Tae-Wook Ham, Hei-Seong Kwak, Byoung-Il Son, and Sung-Chul Park, and claims priority to Korean Patent App. No. 10-2016-0073355, filed on June 13, 2016. JX-0004, Cover.

E. The Products At Issue

1. The Accused Products

Samsung accuses the Ring 4 and accompanying software applications, including the Oura App, of infringing the Asserted Claims. JDX-0004C, ¶3. Samsung does not believe the Ring 5 or [REDACTED] are proper for adjudication (*see infra*), but addresses them herein where they would be

relevant should the ALJ disagree.¹

a. Ring 4

The Ring 4 is a finger-worn wearable device, released in October 2024, that operates together with the Oura App. JDX-0004C, ¶72. The Ring 4 is pictured below. CX-0710C.



b. Ring 5

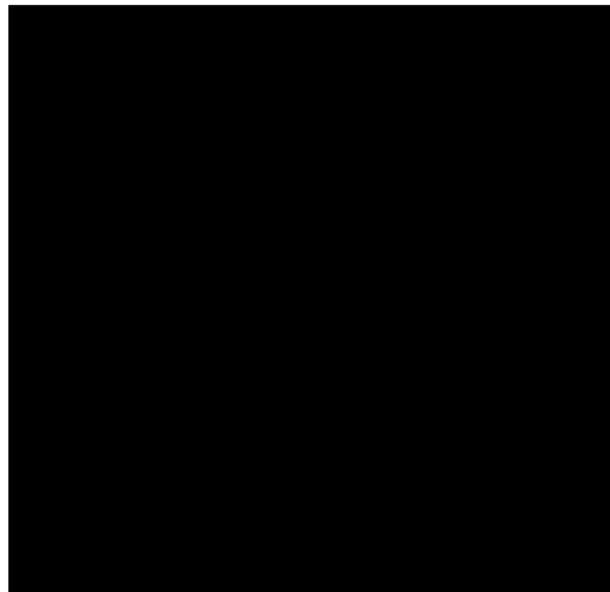
Ring 5 is another finger-worn wearable device that was announced publicly on May 28, 2026, with sales and shipments starting on June 4, 2026. JDX-0028C, 63:18-64:6. The Ring 5 is pictured below. CX-1148C.

¹ As detailed in the parties' MSD and motion to strike briefing, Samsung requests that the Oura Ring 5 and [REDACTED] be excluded from adjudication because they are redesigns or alternative products that Oura never disclosed as such during fact discovery, and Oura failed to comply with the *Oligosaccharides* factors. EDIS ID 889193; EDIS ID 889200. Samsung also intends to file a motion *in limine* excluding both products from the evidentiary hearing. But to avoid waiving its rights at this time, Samsung's pre-hearing brief addresses both products.



c. [REDACTED]

The ‘[REDACTED]’ ring is a future Oura ring product, projected for release [REDACTED] JDX-0028C, 63:18-64:13. [REDACTED] is pictured below. CX-1139C.



2. The DI Products

Samsung relies on two sets of Domestic Industry Products. For the three Sensor Patents, Samsung relies on the Samsung Galaxy Watch4, Galaxy Watch5, Galaxy Watch6, Galaxy Watch7, Galaxy Watch8, Galaxy Watch Ultra, and Galaxy Ring (“Wearable DI Products” or “WDI Products”). JDX-0004C, ¶4; CPX-0003 to CPX-0009. For the ’789 Patent, Samsung relies on the

Samsung Galaxy S22, S22+, S22 Ultra, S23, S23+, S23 Ultra, S24, S24+, S24 Ultra, S25, S25+, and S25 Edge smartphones (“Phone DI Products” or “PDI Products”). JDX-0003C, ¶313; CPX-0010 to CPX-0021.

a. Samsung Galaxy Watches

The Galaxy Watch Products are electronic smartwatches worn on a user’s wrist. JDX-0004C, ¶653. A representative Galaxy Watch, Galaxy Watch8, is pictured below. CX-0227.



b. Samsung Galaxy Ring

The Galaxy Ring is an electronic smart ring worn on a user’s finger. JDX-0004C, ¶¶126, 653. The Galaxy Ring is pictured below. CX-1513.



c. Samsung Galaxy Smartphones

The Galaxy Phone Products are smartphones with cellular and other wireless communication capabilities. JDX-0003C, ¶¶317-18. A representative Galaxy smartphone, Samsung S25+, is pictured below. CX-1304C.



F. Claim Construction

The table below establishes the adopted constructions for the Asserted Claims. *See* JX-0009 (Order Construing Claim Terms, Order No. 10). The parties identified three disputed terms and no agreed constructions. *Id.* at 1.

Claim Terms	ALJ Construction
“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body” (’675 Patent, Claim 13)	Within this phrase, “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body” means: second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn. Claim 13 is not indefinite; Samsung’s proposed

	construction was adopted.
“third period of time” (’675 Patent, Claim 15)	Not indefinite. No further construction is needed at this point.
“the biometric signal” (’359 Patent, Claim 14)	The claim term “the biometric signal” should correctly read “the biological signal,” which is a signal associated with biological information of the user.

III. JURISDICTION

The Commission has subject matter jurisdiction over certain acts of unfair importation. 19 U.S.C. § 1337(a)(1)(B)(i); *see also Amgen, Inc. v. Int’l Trade Comm’n*, 902 F.2d 1532, 1536 (Fed. Cir. 1990). The Commission has *in rem* jurisdiction over products that have been imported into the US. *Sealed Air Corp. v. Int’l Trade Comm’n*, 645 F.2d 976, 985 (C.C.P.A. 1981).

The Respondents have stipulated that the Accused Products at issue in this Investigation were manufactured abroad and imported into the US. *See* EDIS ID 883875. For example, the Respondents stipulate that since December 12, 2025, Respondents “or one or more of its U.S. subsidiaries have imported, sold for importation, and/or sold after importation into the United States at least one unit of [the Accused Products]” and that Respondents “do not and will not dispute that the importation requirement within the meaning of 19 U.S.C. § 1337(a)(1)(B) is satisfied.” *Id.* Accordingly, the importation requirement has been satisfied in this Investigation as to the Accused Products.

But as explained in Samsung’s Opposition to Oura’s MSD (EDIS ID 889193) (“Samsung’s MSD Opposition”), Ring 5 and [REDACTED] should be excluded from adjudication because Oura fails to meet its burden of satisfying three of the four *Oligosaccharides* factors. Samsung’s MSD Opposition, 13; *Certain Human Milk Oligosaccharides & Methods of Producing the Same* (“*Oligosaccharides*”), Inv. No. 337-TA-1120, Comm’n Op., 2020 WL 3073788, at *11 (June 8,

2020).

First, as to importation, Oura's discovery responses stated that Ring 5 would be [REDACTED]. *Id.* Oura's reliance on the parties' import stipulation, filed on June 1, 2026, after the close of fact discovery, is misplaced, as it fails to timely correct Oura's misleading discovery responses. *Id.*

Second, as to Ring 5 and [REDACTED] being sufficiently fixed, Oura's corporate witness testimony suggesting otherwise, that Ring 5's design was always evolving and [REDACTED] was [REDACTED]. *Id.* at 14-15. This failure to provide "fair notice that these products were at issue, or show that these products were sufficiently fixed during fact discovery" warrants declining adjudication. *See Certain Portable Battery Jump Starters & Components Thereof (II)*, Inv. No. 337-TA-1359, Comm'n Op., 2024 WL 4145130, at *26 (July 5, 2024).

Third, as to sufficient discovery, the crucial question is whether "the timing of the production deprived [complainant] of the opportunity to develop infringement contentions, to obtain expert opinions, and to prepare for trial on that product." *Certain Unmanned Aerial Vehicles & Components Thereof*, Inv. No. 337-TA-1133, Comm'n Op., 2022 WL 20402976, at *35 (Mar. 1, 2022). This factor weighs decisively against adjudication because Oura failed to timely produce discovery sufficient to permit Samsung to analyze infringement of Ring 5 and [REDACTED]. Samsung's MSD Opposition, 15. While Oura cited the technical materials it produced for Ring 5 and [REDACTED] before the last day of fact discovery, none of those materials provided the information needed to evaluate the relevant internal PCB structures of Ring 5 and [REDACTED] for the '789 Patent. *Id.*, 18. Oura waited until the last day of fact discovery to finally produce teardown images of Ring 5 and [REDACTED] PCBs, but this was too little too late, as it failed to give Samsung any opportunity to analyze those images before its final infringement contentions were due. *Id.*,

[REDACTED]

17. Critically, Oura failed to produce the most important technical discovery that Samsung needed to actually assess infringement—physical samples of Ring 5 or [REDACTED] for Samsung to teardown, despite such samples being specifically requested by Samsung’s Request for Production. *Id.*, 16.

IV. U.S. PATENT NO. 10,231,675

A. The Accused Products Infringe the ’675 Patent²

Samsung’s expert Dr. Welch and the evidence will demonstrate that the Accused Products infringe Claims 13 and 15 of the ’675 Patent and Oura has offered no rebuttal expert opinion or evidence demonstrating otherwise.

Oura’s attorneys argue, for the first time in Opposition to Samsung’s MSD Regarding Infringement, that method Claims 13 and 15 are not infringed because Samsung fails to show proof of actual use or performance of the method steps. EDIS ID 889195, 20-21. Oura’s position lacks credibility. It is beyond question that the Accused Products perform each step of method Claims 13 and 15 when worn in the ordinary course by a user. Oura provided extensive evidence in its affirmative -1398 and -1468 Investigations of its own use and testing in the United States in alleging it satisfied the DI requirement in those cases. Oura’s own testing and use constitutes direct infringement.³

Before its attorney argument, all that Oura and its expert, Dr. Moratzavi, mustered in opposition was a faulty “practicing the prior art” defense based on a legally improper comparison

² The analysis below focuses on Ring 4. But both parties’ experts agree that Ring 4 is representative of newer Oura products such as Ring 5 and [REDACTED] with no differences that are material to infringement. JDX-0007C, ¶¶552-54.

³ In fact, Oura’s activities regarding the Accused Products were the entire basis behind its economic domestic industry arguments in the -1468 Investigation. *See, e.g.*, EDIS ID 886733. Moreover, Oura advertises that the Ring 4 uses red and IR LEDs to measure SpO2 levels during sleep. JDX-0007C, ¶¶88. Oura’s advertising of the patented method, inducing users to use the patented method to measure SpO2 during sleep, constitutes induced infringement.

[REDACTED]

of Ring 4 to an alleged prior art product, Ring 1. JDX-0007C, ¶¶138, 162, 384, 440. The Federal Circuit has made “unequivocally clear” that “there is no ‘practicing the prior art’ defense to literal infringement.” *Tate Access Floors, Inc. v. Interface Architectural Res., Inc.*, 279 F.3d 1357, 1365 (Fed. Cir. 2002). At any rate, Ring 1 and the accused Ring 4 are far from the same—they are multiple generations and nearly 10 years apart—and Ring 1 lacks the primary feature forming the basis of infringement—the ability to measure SpO2. JDX-0026C, 237:24-238:8 (Oura’s engineer Michael Larson confirming that “[REDACTED]”).

1. The Accused Products Infringe Claim 13

a. Limitation 13.pre (“A method comprising:”)

The Accused Products perform the method below when worn on a user’s finger. JDX-0004C, ¶¶524-526.

b. Limitation 13.a (“determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device;”)

The Accused Products determine whether the electronic device is worn on a part of a user’s body using an accelerometer (“at least one first sensor”), such as the Ring 4’s [REDACTED] accelerometer. JDX-0004C, ¶¶72-77, 527-539; CX-0997C, OURA1468_00096111.

[REDACTED]

CX-0997C, OURA1468_00096111.

[REDACTED]

JDX-0004C, ¶¶72-111, 527-539; CX-1500C, OURA_1478_SC_0000185-187, 191, 197; CX-1686C, OURA_1468_SC_0000178, 363; JDX-0026C, 173:22-174:18; 178:17-24, 180:11-18, 185:5-18; 186:14-17; 188:6-189:9.

- c. **Limitation 13.b (“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body;”)**

The Accused Products acquire first information [REDACTED]

[REDACTED]

[REDACTED]. JDX-0004C, ¶¶72-111, 540-547; CX-1500C, OURA_1478_SC_0000185-187, 191, 197; CX-1686C,

[REDACTED]

OURA_1468_SC_0000178, 363; JDX-0026C, 173:22-174:18; 178:17-24, 180:11-18, 185:5-18; 186:14-17; 188:6-189:9; 213:6-214:6; 215:16-20.

[REDACTED]

d. Limitation 13.c (“activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;”)

The Accused Products activate at least one second sensor [REDACTED] based on at least a part of the first information on [REDACTED] of the electronic device. JDX-0004C, ¶¶72-111, 548-564; CX-0994C, OURA1468_00029608; CX-0996C, OURA1468_00096106; CX-1653; CX-1649; CX-1651; CX-0241, SAMS382-0027803 (“Red and infrared LEDs measure blood oxygen while you sleep”); CX-0221 (“Please note that you may see the red LEDs light up whenever your Oura detects that you may be asleep”). The SpO2 sensor in the Accused Products is the “second sensor” and includes light emitting units and light receiving units, and it is thus an “optical sensor.” *Id.* For example, the Ring 4’s SpO2 sensor includes red and IR LEDs from [REDACTED] [REDACTED] [REDACTED] (the “light emitting units”), and three [REDACTED] [REDACTED] (the “light receiving units”), as depicted below. *Id.*



CX-0994C, OURA1468_00029608.

The Accused Products determine whether a user is active or at rest by [REDACTED]

[REDACTED]

[REDACTED]. JDX-0004C, ¶¶72-111, 548-564; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 191-92, 197, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178, 363. [REDACTED]

[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. *Id.* [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.* [REDACTED]

[REDACTED]

[REDACTED] *Id.*

e. Limitation 13.d (“detecting a biometric information of the user using the activated at least one second sensor;”)

The Accused Products detect biometric information of the user [REDACTED] using

[REDACTED]

the activated second sensor. *See* JDX-0004C, ¶¶72-111, 565-572; CX-0241, SAMS382-0027803; CX-0221, SAMS382-0027593; JDX-0025C, 66:15-68:22; JDX-0026C, 78:24-79:5, 233:23-234:15, 234:18-237:10; JX-0003, 1:29-35; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 331, 351-53, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. Once the SpO2 sensor is activated, the Accused Products use it to detect the user’s biological information, specifically SpO2. JDX-0004C, ¶¶72-111, 565-572.

f. Limitation 13.e (“during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor;”)

While detecting the biometric information [REDACTED] the Accused Products acquire second information on [REDACTED] using at least a part of the first sensor. *See* JDX-0004C, ¶¶72-111, 573-575. In the same manner as described in limitation 13.c, the Accused Products [REDACTED]
[REDACTED], which is “biometric information.” *See* Section IV.A.1.d.

g. Limitation 13.f (“and in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.”)

In response to identifying that [REDACTED]
[REDACTED], the Accused Products [REDACTED] the second sensor (i.e., SpO2 sensor). *See* JDX-0004C, ¶¶72-111, 576-579; CX-0241, SAMS382-0027803; CX-0221; JDX-0026C, 233:23-234:15, 234:18-237:10; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. As Samsung’s expert Dr. Welch confirmed based on his review of the firmware code, the Accused Products [REDACTED]
[REDACTED]

[REDACTED]
[REDACTED]
[REDACTED] JDX-0004C, ¶¶72-111, 576-579; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

[REDACTED] *Id.*

For the first time in opposing Samsung’s MSD, Oura argues this claim element is not met because “the ring’s PPG sensor” [REDACTED].
EDIS ID 889195, 4-8. Oura’s eleventh-hour attorney argument is wrong. [REDACTED]
[REDACTED]

[REDACTED] See Section IV.A.1.d. Oura’s insistence that an unaccused sensor (whether it defines as the HR sensor or PPG sensor) may be active does not change this fact. [REDACTED]

[REDACTED] JDX-0004C, ¶¶434-438; JDX-0026, 220:25-221:6 (Oura firmware engineer Michael Larson confirming based on Oura documentation that [REDACTED]
[REDACTED]); CX-1219C, 93:23-95:10 (Oura engineer Ketan Patel confirming that the [REDACTED]
[REDACTED]); CX-0241, SAMS382-0027803 (disclosing [REDACTED]
[REDACTED]).

[REDACTED]

2. The Accused Products Infringe Claim 15

- a. Limitation 15.pre (“The method of claim 13, further comprising:”)
- b. Limitation 15.a (“acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and”)
- c. Limitation 15.b (“activating the at least one second sensor based on at least a part of the additional information.”)

The Accused Products “acquir[e] additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes.” *See* JDX-0004C, ¶¶72-111, 583-590. As explained in limitation 13.b, [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. *See* Section IV.A.1.c.

[REDACTED] *See* Section IV.A.1.g. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *See* Section IV.A.1.d-e. [REDACTED]

[REDACTED]

[REDACTED] See JDX-0004C, ¶¶72-111, 583-590.

B. TDI Is Met for the '675 Patent

Samsung's expert Dr. Welch and the evidence will demonstrate that the WDI Products practice Claims 13 and 15 of the '675 Patent as detailed below. Oura has not argued to the contrary or otherwise attempted to rebut this showing.

1. The WDI Products Practice Claim 13

a. Limitation 13.pre ("A method comprising:")

The WDI Products practice the method below when worn on a user's wrist (Galaxy Watch products) or finger (Galaxy Ring). JDX-0004C, ¶¶1023-1026.

b. Limitation 13.a ("determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device;")

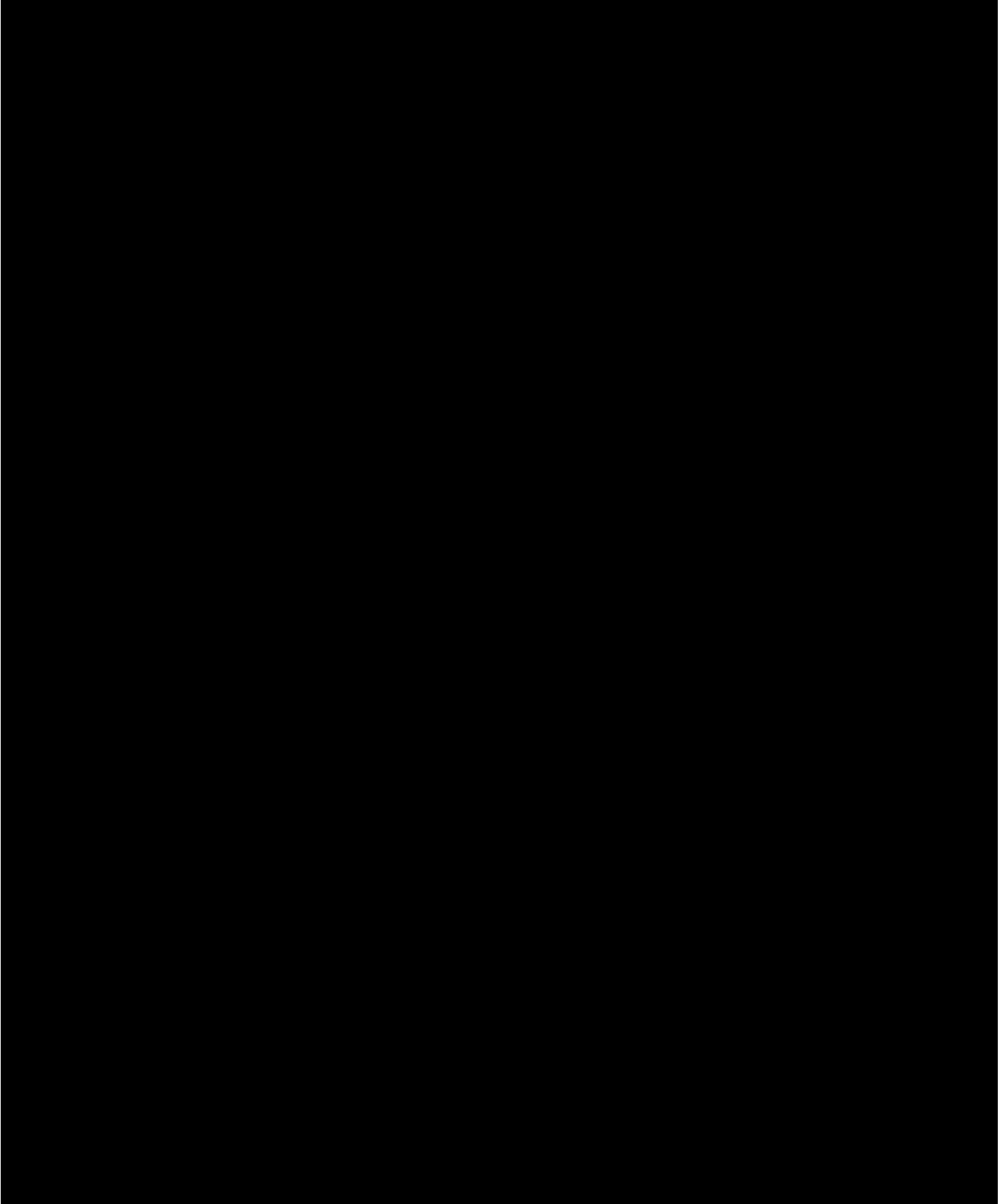
The WDI Products determine whether the electronic device is worn on a part of a user's body using "at least one first sensor," which is an accelerometer, such as:

- [REDACTED] in Galaxy Watch4, Watch5, Watch6, and Watch7
- [REDACTED] in Watch Ultra;
- [REDACTED] in Watch8;
- [REDACTED] in the Galaxy Ring.

JDX-0004C, ¶¶112-132, ¶¶1027-1043; CX-0148C, SAMS382-0012471; CX-0160C, SAMS382-0013172; CX-0179C, SAMS382-0014881; CX-0150C, SAMS382-0012801; CX-0151C, SAMS382-0012809; CX-0152C, SAMS382-0012815; CX-0156C, SAMS382-0013147; CX-0157C, SAMS382-0013153; CX-0158C, SAMS382-0013161; CX-0164C, SAMS382-0013288; CX-0165C, SAMS382-0013296; CX-0159C, SAMS382-0013169; CX-0176C, SAMS382-0013620; CX-0177C, SAMS382-0013626; CX-1640C, SAMS382-0011270; CX-0992C, SAMS406-0074870; CX-1643. For example, the accelerometers for the Galaxy Ring and Galaxy

[REDACTED]

Watch6 are annotated with a red box in each of the block diagram schematics below.



[REDACTED]

[REDACTED]

The WDI Products use accelerometer data collected over a period of time in combination with an IR wear detection sensor to determine whether the product is being worn. JDX-0004C, ¶¶112-196, ¶¶1027-1043; CX-1499C, SAMS382-SC-0000306, 0318-0319, 0329, 0373-0374, 0384, 0416, 0428-0429, 0439, 0471, 0484-485, 0495, 0527, 0540-541, 0551-0552, 0701-0702, 0714-0715, 0725, 0736. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*

- c. **Limitation 13.b (“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body;”)**

The WDI Products acquire first information on a motion of the electronic device using at least a part of the first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the device is worn. JDX-0004C, ¶¶112-196, 1044-1057; CX-0829C, SAMS406-0071843-845; CX-0142, SAMS382-0011184; CX-1639, SAMS406-0001878; CX-1685C, SAMS406-SC-0000007, 0323-0325, 0331-0332; CX-1499C, SAMS382-SC-0000017-18, 0027-0029, 0048, 0051, 0054, 0057-0058, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300, 0955-0958, 1011-1012; JDX-0020C, 46:16-48:25, 76:14-78:4, 78:18-79:5; JDX-0012C, 49:16-24, 50:15-52:4, 53:25-54:13, 55:7-14, 56:2-57:3, 62:5-64:17, 71:15-72:16; JDX-0013C, 50:3-19, 145:11-148:10. [REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶112-196, 1044-1057; CX-1639,

[REDACTED]

SAMS406-0001878; CX-1685C, SAMS406-SC-0000007, 0323-0325, 0331-0332; CX-1499C, SAMS382-SC-0000017-18, 0027-0029, 0048, 0051, 0054, 0057-0058, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300, 0955-0958, 1011-1012; *see* Section IV.B.1.b.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*, *see* Section IV.B.1.d.

d. Limitation 13.c (“activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;”)

The WDI Products activate at least one second sensor based on at least a part of the first information on the motion of the electronic device. JDX-0004C, ¶¶112-196, 1058-1071; CX-0829C, SAMS406-0071843-845; CX-0142, SAMS382-0011184; CX-1639, SAMS406-0001878; CX-1685C, SAMS406-SC-0000007, 0323-0325, 0331-0332; CX-1499C, SAMS382-SC-0000017-18, 0027-0029, 0057-0058, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300, 0955-0958; JDX-0020C, 46:16-48:25, 76:14-78:4, 78:18-79:5; JDX-0012C, 49:16-24, 50:15-52:4, 53:25-54:13, 55:7-14, 56:2-57:3, 62:5-64:17, 71:15-72:16; JDX-0013C, 50:3-19; 145:11-148:10. [REDACTED]

[REDACTED] JDX-0004C, ¶¶112-196, 1058-1071; CX-1685C, SAMS406-SC-0000007, 0323-0325, 0331-0332; CX-1499C, SAMS382-SC-0000017-18, 0027-0029, 0057-0058, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300, 0955-0958. [REDACTED]

[REDACTED]

[REDACTED]

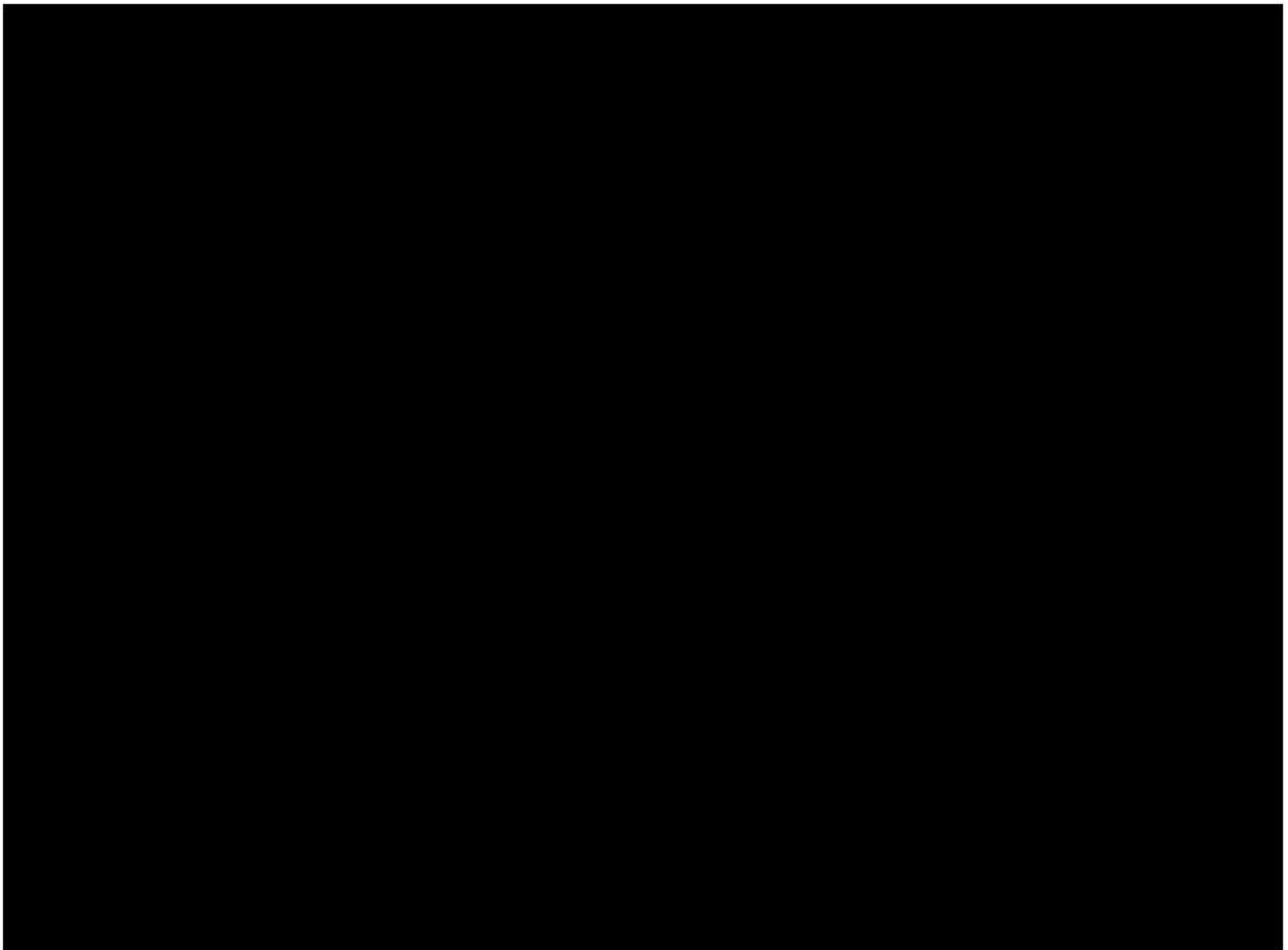
[REDACTED]

[REDACTED]. *Id.*

[REDACTED]

[REDACTED]

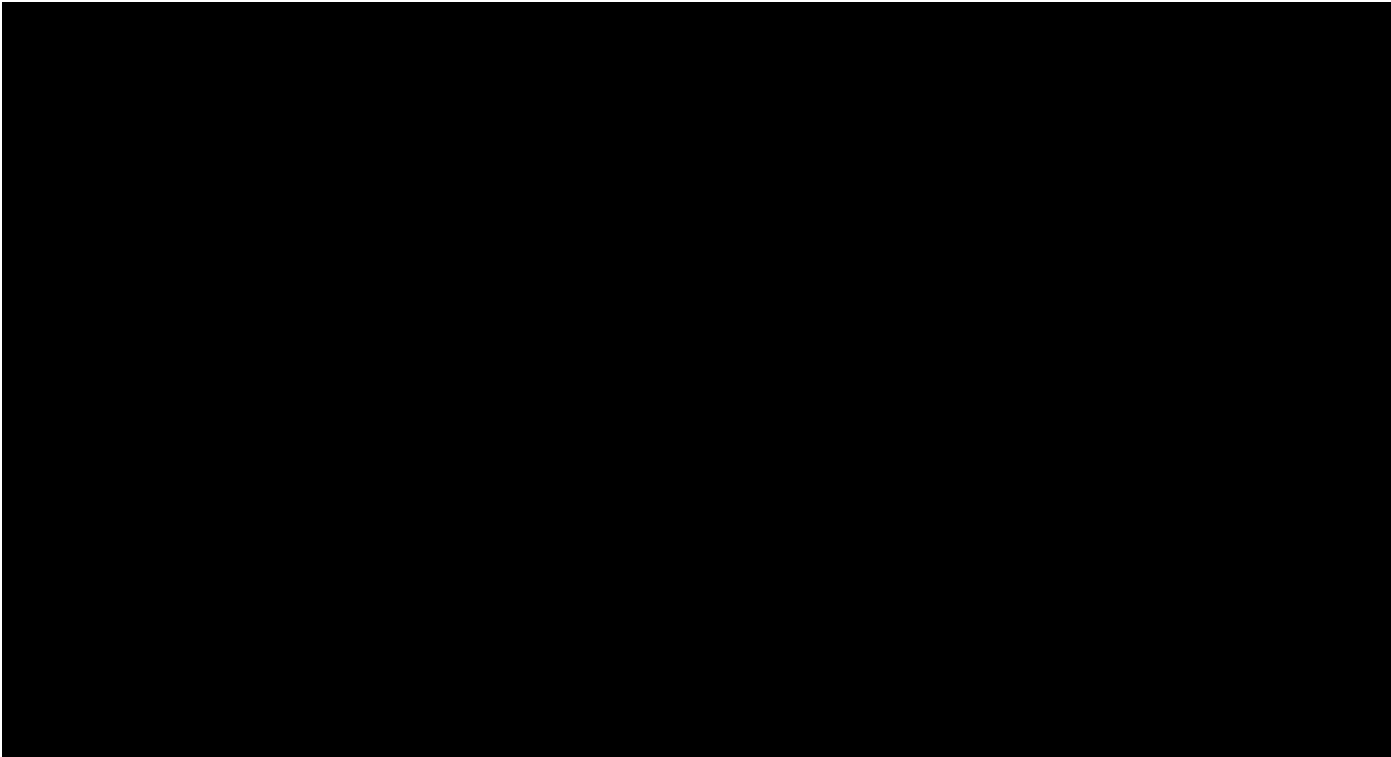
[REDACTED] *Id.*, see CX-0261C, SAMS382-0029425-426; CX-1640C, SAMS382-0011274. These SpO2 sensors are identified with a green box in each of the block diagram figures for the WDI Products below.

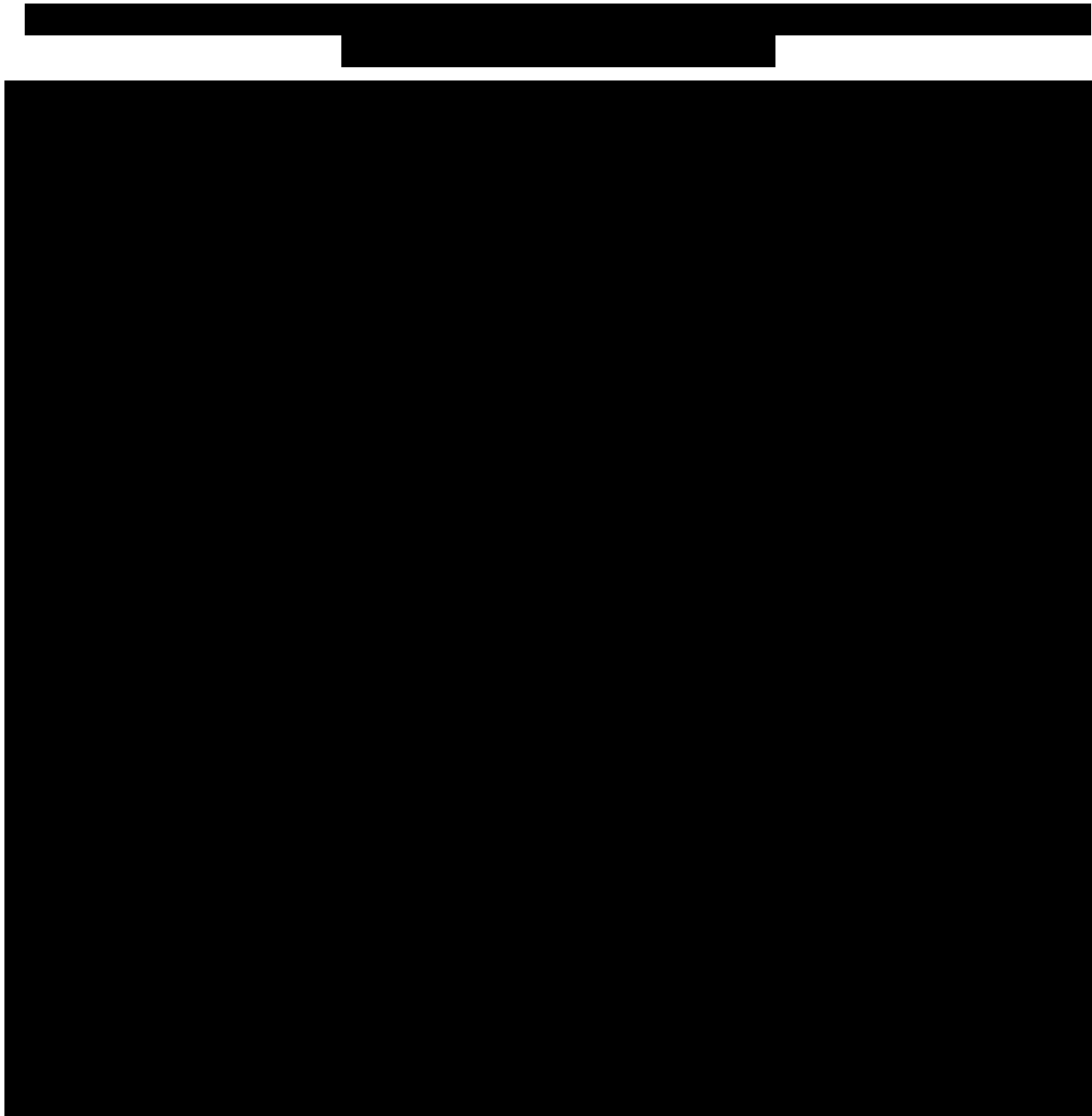




CX-0135C, SAMS382-0010444.

The light emitting and receiving units comprising the SpO2 sensor are further depicted below:





e. Limitation 13.d (“detecting a biometric information of the user using the activated at least one second sensor;”)

The WDI Products detect biometric information of the user using the activated second sensor. JDX-0004C, ¶¶112-196, 1072-1078; CX-0142, SAMS382-0011184; CX-1639, SAMS406-0001878; CX-1685C, SAMS406-SC-0000327-0328, 0331-0332; CX-1499C, SAMS382-SC-0000011-0012, 0017-0018, 0027-0029, 0117, 0124, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300. [REDACTED]

[REDACTED], which is “biometric information.” JDX-0004C, ¶¶112-196,

[REDACTED]

1072-1078.

- f. Limitation 13.e (“during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor;”)**

While detecting the biometric information, the WDI Products acquire second information on a motion of the electronic device using at least a part of the first sensor. JDX-0004C, ¶¶112-196, 1079-1081. As discussed for 13.c, [REDACTED]

[REDACTED]. See Section IV.B.1.d. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶112-196, 1079-1081; see Section IV.B.1.d.

- g. Limitation 13.f (“and in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.”)**

If the identified motion of the electronic device included in the second information exceeds a threshold motion, then the WDI Products inactivate the second sensor. JDX-0004C, ¶¶112-196, 1082-1091; JDX-0022C, 76:14-78:4 78:18-29:5; CX-0142, SAMS382-0011184; CX-1499C, SAMS382-SC-0000679-0680, 0955-0958; CX-1685C, SAMS406-SC-0000007, 0327-0328, 0331-0332. [REDACTED]

[REDACTED] See Section IV.B.1.d-f. [REDACTED]

[REDACTED]

[REDACTED]

2. **TDI Is Met For Claim 15**

- a. **Limitation 15.pre (“The method of claim 13, further comprising:”)**
- b. **Limitation 15.a (“acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and”)**
- c. **Limitation 15.b (“activating the at least one second sensor based on at least a part of the additional information.”)**

The WDI products practice the method of claim 15 when worn on a user’s wrist (Galaxy Watch products) or finger (Galaxy Ring). JDX-0004C, ¶¶112-196, 1096-1103.

In the context of Claim 15, the “first period of time” [REDACTED]

[REDACTED]

[REDACTED] the “second period of time” [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

The WDI Products then acquire additional motion information using the accelerometer for a “third period of time” immediately after the second period of time. JDX-0004C, ¶¶112-196, 1096-1103. [REDACTED]

[REDACTED]

See Section IV.B.1.c-d. [REDACTED]

[REDACTED]

[REDACTED] See Section IV.B.1.d. [REDACTED]

[REDACTED]

[REDACTED]

C. The Asserted Claims of the '675 Patent Are Valid

Samsung's expert Dr. Welch and the evidence will demonstrate that the asserted claims are valid. Oura incorrectly asserts that the asserted claims are invalid based on the following prior art grounds:

- Ground 1 – Claims 13 and 15 as anticipated by Oura Ring Gen 1 (“Ring 1”)
- Ground 2 – Claims 13 and 15 as anticipated by U.S. Patent No. 9,848,832 (“Hong”)
- Ground 3 – Claim 13 as obvious in view of Hong and Utter II (“U.S. Patent App. Pub. No. 2015/0186609”)
- Ground 4 – Claim 13 as anticipated by Utter II
- Ground 5 – Claims 13 and 15 as obvious in view of Utter II and Hong

As for Ring 1, Oura will not be able to show that it is prior art or that the code or documents to which it points actually describe the features of the Ring 1 prototype Oura alleges was demonstrated before the priority date. Nor did the Ring 1 activate and deactivate a HR sensor (i.e., the “biological sensor”) based on motion, as required by Claim 13 (and dependent Claim 15).

Hong also fails to disclose the elements of Claim 13 (and dependent Claim 15). Hong may teach immediately activating a HR sensor when a device is worn, and immediately deactivating that HR sensor when it is no longer worn. But Hong does not teach the claimed activating a biological sensor based on motion detection triggered by the device being worn.

Utter II fails to disclose Claim 13 (or dependent Claim 15) because it does not teach: (1) wear detection, much less wear detection using a motion sensor; (2) wear detection that acts as a trigger for subsequent motion-based sensor activation; and (3) the claimed sensor activation/deactivation.

[REDACTED]

1. Ground 1: Ring 1 Does Not Anticipate the Asserted Claims of the '675 Patent

a. Ring 1 Is Not Prior Art

Oura cannot show by clear and convincing evidence that Ring 1 is prior art. “[A] party claiming his own prior inventorship must proffer evidence corroborating his testimony.” *Sandt Tech., Ltd. v. Resco Metal & Plastics Corp.*, 264 F.3d 1344, 1350 (Fed. Cir. 2001). Contemporaneous documentary evidence “provides the most reliable proof” of corroboration. *Id.* at 1351. “In contrast to contemporaneous documentary evidence, however, post-invention oral testimony is more suspect, as there is more of a risk that the witness may have a litigation-inspired motive to corroborate the inventor’s testimony.” *Id.*

Oura has not produced any contemporaneous evidence corroborating the Ring 1 code. Instead, code to which Oura points is from at least late 2016, *more than a year after* the August 12, 2015 priority date. See JDX-0009C, ¶¶280-302; CX-1500C, OURA 1478 SC 0000361 [REDACTED]; [REDACTED]; OURA_1478_SC_0000543 [REDACTED]; see also CX-1500C, OURA_1478_SC_0000317 [REDACTED]; OURA_1478_SC_0000365 [REDACTED]. Although Oura points to portions of code with comments stating [REDACTED]

[REDACTED] JDX-0009C, ¶¶208-283; JDX-0026C, Larson Tr. 134:8-15 [REDACTED]

[REDACTED] Critically, Oura has not produced a version of the code predating August 12, 2015 such as one that would have been stored by version control software.

Oura continued to modify Ring 1 code after the product was released, as reflected by Oura’s practice of releasing ongoing firmware updates. JDX-0009C, ¶285. The presence of dates from

[REDACTED]

at least 2016 suggest that the Ring 1 code represents a snapshot from on or after that year. As such, the code relied on by Oura reflects these post-release firmware updates. *Id.* [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Ring 1 was not sold to the public until November 2015 or later, after the August 12, 2015 priority date and Oura does not contend otherwise. *See* JDX-0004C, ¶300. Oura did not even launch its efforts to raise funds to develop the product, as part of a “Kickstarter campaign” until August 18, 2015, after the priority date. JDX-0004C, ¶300; JDX-0001C, ¶1306. Oura itself targeted delivery of its product for November 2015. RX-0023, OURA1478_00001519 (“Estimated delivery: Nov. 2015”). Oura did not receive FCC RF certification until December 12, 2015, which is a prerequisite for sale in the United States. JDX-0004C, ¶300, CX-1416. And Oura’s corporate witnesses could not identify the month or year Gen 1 was actually sold. JDX-0026C, 63:2-13.

In an attempt to backfill that clear hole in proof, Oura and its expert rely on a conversation from June 2026 that its expert had with two Oura employees, neither of who was deposed in this case and neither of whom will testify at the hearing. *See* . JDX-0001C, ¶¶1321, 1329. Nor does the information either provided have any indicia of reliability since neither is named as authors of the Ring 1 code and one of the employees, Mr. Vallius, did not start working at Oura until May 2015, a mere three months before the priority date and the other, Ms. Uustialo, did not start until

[REDACTED]

the month later as a marketing communications manager.

Trying to buttress its failed code evidence, Oura points to demonstrations of a Ring 1 prototype it alleges to have performed in June and July 2015. JDX-0001C, ¶¶1322-1334. Like the code evidence, Oura’s effort falls well short as it offers no corroborating evidence identifying the sensors or code on those prototypes. In particular, Oura offers no evidence showing that the Ring 1 product as demonstrated in mid-2015 used the same source code dated from late 2016 that Oura cites to evidence the Ring 1’s operation. The Federal Circuit has repeatedly held that even disinterested sworn witness testimony can be insufficient to corroborate assertions of public use of prior art products, particularly with respect to which exact features existed in the prior art product by particular dates. *See, e.g., Woodland Tr. v. Flowertree Nursery, Inc.*, 148 F.3d 1368, 1373 (Fed. Cir. 1998) (oral testimony insufficient to prove public use, regardless of evidence from “undated photographs” where “their lack of detail and clarity cannot have provided documentary support”); *Finnigan Corp. v. Int’l Trade Comm’n*, 180 F.3d 1354, 1366 (Fed. Cir. 1999) (despite published article by disinterested third party describing all but one element of asserted claim, finding oral testimony of prior public use uncorroborated). Here, Oura points only to unsworn discussions between its expert and two employees—far from the type of “disinterested” testimony the Federal Circuit has directed to be viewed with skepticism.

[REDACTED]

b. Ring 1 does not disclose motion-based heart rate sensor activation/deactivation :

[13.b]: “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body”

[13.c] “activating at least one second sensor based on at least a part of the first information on the motion of the electronic device”

Ring 1 does not disclose elements 13.b and 13.c. JDX-0009C, ¶¶400-408.

Dr. Mortazavi points to late 2016 Ring 1 code as allegedly showing motion-based activation/inactivation of an HR sensor. Not only does that code not evidence functionality by mid-2015, but the available *contemporaneous* evidence from mid-2015 shows Dr. Mortazavi is wrong and that contemporaneous evidence is the “most reliable proof” of how Ring 1 code may have worked. *Sandt Tech.*, 264 F.3d at 1351. Oura produced a May 12, 2015 technical specification that appears to be an early-stage and evolving design document, as evidenced by the dates and version numbers. [REDACTED]

[REDACTED] CX-1376C, OURA1478_00001491.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]: JDX-0001C, ¶2406 [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0009C, ¶401. Oura’s corporate witness, Michael Larson, similarly confirmed that the relied-upon Ring 1 code does not [REDACTED]. JDX-0026C, 138:5-9, 138:22-141:7, 145:12-148:5.

Oura further fails to show that the ALJ’s construction is met. Specifically, the construction requires a “second period of time,” triggered by wear detection, in which motion is measured. JX-0009, 7. This motion must trigger activation of the second sensor (limitation 13.c). As discussed above, Oura has failed to show any motion that triggers activation of the HR sensor much less that that motion is measured at the appropriate time. As illustrated in the diagram above, [REDACTED]

[REDACTED]

[REDACTED]

c. Ring 1 does not disclose claim 15

Limitation 15.pre (“The method of claim 13, further comprising:”)

Limitation 15.a (“acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and”)

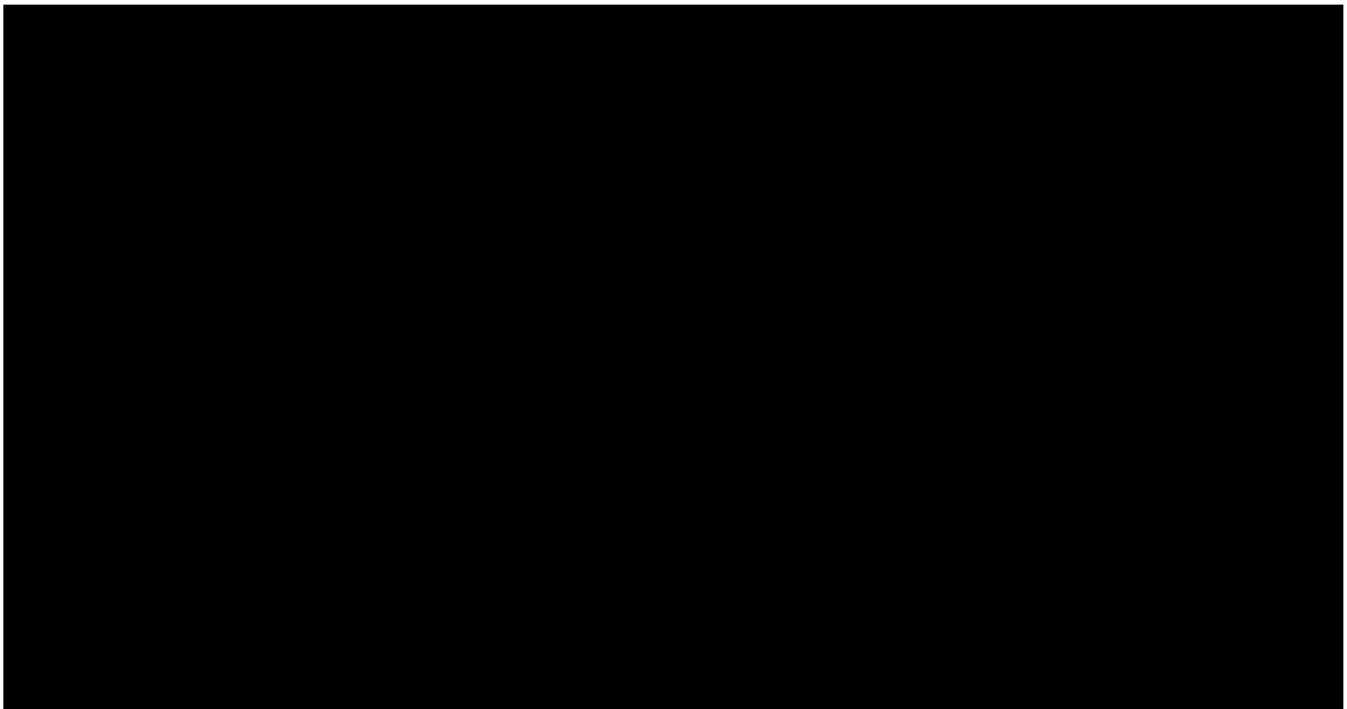
Limitation 15.b (“activating the at least one second sensor based on at least a part of the additional information.”)

Ring 1 fails to disclose claim 15. JDX-0009C, ¶¶400-411; *see* Section IV.C.1.b.

Claim 15 requires measuring motion in a “third period of time” for determining whether to “activat[e] the at least one second sensor.” Claim 15 requires a *re-activation* of the biological sensor that was both activated and deactivated in Claim 13.

But Dr. Mortazavi fails to show how Ring 1 maps onto a re-activation of a biological signal. Dr. Mortazavi relies on the timing diagram that he draws up, excerpted below, to assert that the

[REDACTED]:



¶2433. However, this mapping does not correspond to Claim 15. As discussed above, Oura fails to identify any activation and deactivation based on motion because none of the states of operation in which HR is measured are entered or exited based on motion. *see* Section IV.C.1.b. Oura similarly fails to show re-activation, after activation and deactivation, based on motion. Oura's expert further fails to specify what states of operation in which "subsequent motion measurement[s]" are obtained and used to re-activate the sensor. JDX-0009C, ¶¶400-411; JDX-0001C, ¶2433.

2. Ground 2: Hong Does Not Anticipate Claims 13 and 15 of the '675 Patent

a. Hong Does Not Disclose Wear Detection Triggering Motion Detection To Activate A Biometric Sensor

[13.b]: "acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body"

[13.c] "activating at least one second sensor based on at least a part of the first information on the motion of the electronic device"

Hong does not disclose limitations 13.b and 13.c. JDX-0009C, ¶¶448-464. These limitations require: (1) a "first period of time" allocated to wear detection; (2) a "second period of time," triggered by wear detection, in which motion is measured; and (3) "activating" a biological sensor based on motion information measured during the "second period of time."

Dr. Mortazavi's theory is that Hong teaches a "first period of time" when motion is used for wear detection, followed by a "second period of time" when motion is used to detect that the user is asleep or sedentary, after which the HR sensor will turn on. JDX-0001C, ¶¶2764-2770, citing RX-0072, Fig. 18A. However, this misrepresents Hong.

[REDACTED]

Hong instead teaches that, upon determining that the device is worn, the device *immediately* begins measuring HR information. In doing so, Hong bypasses the required “second period” of time in which motion is measured in response to wear determination. Figure 18A, the figure Dr. Mortazavi cites and annotated below, shows that upon detecting a worn state in Step 1806 (in yellow), the device *immediately* begins operating a HR monitor in Step 1808 (in green). As such, motion during a time period *after* wear detection (as required by the claims) plays no role in determining whether or not the HR sensor will be activated.

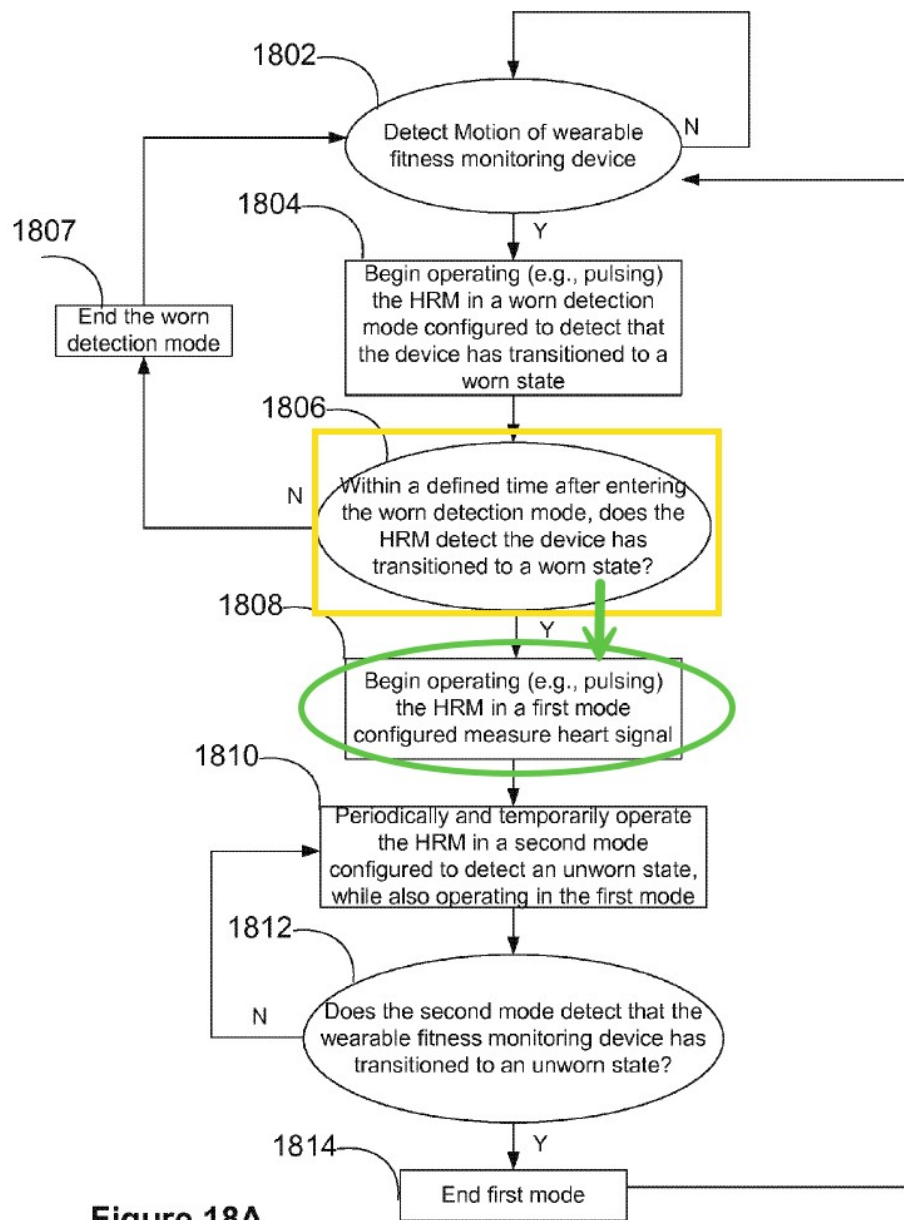


Figure 18A

Dr. Mortazavi argues that Hong discloses using motion to activate a biometric sensor because it teaches a device that “continues to monitor motion via the accelerometer and determines that the user has become sedentary or asleep, culminating in activation of a higher-rate sampling mode.” JDX-0001C, ¶2766; RX-0072, 16:36-40. But Dr. Mortazavi is wrong—the only timing evidence Dr. Mortazavi identifies, block 1808 of Fig. 18A is directed to activating a HR monitor, not activating a motion sensor. Because the claims require wear detection triggering a period of

[REDACTED]

motion detection for the purpose of determining whether to activate or use a biometric sensor, and because Hong is silent on whether and how wear detection would trigger any sort of motion-based sleep detection as required by the ALJ’s claim construction, Hong cannot satisfy limitation [13.b].

Further, Dr. Mortazavi’s statement that Hong teaches “activation of a higher sampling mode” (JDX-0001C, ¶2766, citing RX-0072, 16:36-40) does not mean that Hong teaches activation of the HR sensor based on sleep. Changing a sampling rate of a sensor means that the sensor is already active. Hong at 16:36-40 actually teaches that where motion is below a threshold, the sensor may “*reduce* power consumption by *reducing* the sensor sampling rate,” which is certainly not activating a sensor, even under Oura’s erroneous interpretation of Hong. *See* RX-0072, 16:36-49.

That Hong does not teach activation/deactivation based on sleep is confirmed by the fact that sleep detection is based on data from a variety of sensors besides the motion sensor; for example, “a reduction in user heart rate and/or a change in heart rate variability” may indicate sleep; “changes in heart rate variability and galvanic sleep response” may indicate sleep stage transitions, and “an elevated heart rate and/or a change in heart rate variability” may indicate wakeup. RX-0072, 19:2-13. Given Hong’s disclosure of biological sensors including the HR sensor for sleep monitoring, sleep is not the trigger for activating the HR sensor.

b. Hong Does Not Disclose Activating Or Inactivating A Biometric Sensor Based On Motion

[13.c] “activating at least one second sensor based on at least a part of the first information on the motion of the electronic device”

[13.f]: “in response to identifying that motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.”

Limitation 13.c requires activating a sensor based on motion, and 13.f requires deactivating

[REDACTED]

that sensor when motion is more than a threshold motion. Hong does not disclose these limitations. JDX-0009C, ¶¶455-462.

Nothing in Hong discloses or suggests activating/deactivating a biological “second sensor” based on motion. Instead, Hong teaches continuous HR monitoring regardless of motion. As discussed above, Hong teaches that the HR monitor activates immediately upon wear detection. Hong further teaches that a HR monitor will continuously operate in a first mode that “involves pulsing the light source in the heart rate monitor at a first mode frequency and detecting light from the light source at the first mode frequency.” RX-0072, 2:30-38; see *id.* 2:24-29 (“ . . . *continuously* operating the heart rate monitor in the first mode.”), 3:7-11 (“In some embodiments, operating the heart rate monitor in the skin characterization mode involves periodically operating the heart rate monitor in the skin characterization mode while *continuously* operating the heart rate monitor in the first mode.”), 4:64-5:3, 6:9-14.

Dr. Mortazavi’s arguments that Hong teaches these limitations are unavailing. For 13.c, activating a sensor based on motion, Dr. Mortazavi relies on two potential interpretations that fail. *First*, Dr. Mortazavi argues that, because wear detection occurs based on motion and Hong immediately activates the HR sensor, the sensor is also activated based on motion. JDX-0001C, ¶2773. As discussed above, Dr. Mortazavi’s argument fails because turning on a HR monitor immediately after motion-based wear detection does not satisfy the claim limitations—the same motion cannot be used to both detect wear and turn on the biological sensor. See JDX-0009C, ¶¶459-460.

Second, Dr. Mortazavi argues that 13.c is met, asserting that “when the device determines the user is sedentary or asleep based on motion data, it activates the higher-rate sampling mode.” JDX-0001C, ¶2773. As an initial matter, Dr. Mortazavi’s opinions are internally inconsistent—

[REDACTED]

for 13.b, Dr. Mortazavi relies upon Hong’s teaching of *lowering* power consumption during sleep (JDX-0001C, ¶2766, citing RX-0072, 16:36-40), whereas for 13.f, Dr. Mortazavi relies on Hong’s teaching of operating biometric sensors in a *higher sampling rate mode* to measure HR variability. Changing power consumption and sampling rate presumes that the HR monitor is still active and operating continuously. It is not the same as “activating” the sensor as the claim requires and the only examples of activating the sensor taught by Hong are in response to wear detection. RX-0072, Fig. 18A. Given that HR monitoring happens in conjunction with motion to detect sleep and wakeup (RX-0072, 19:2-13), it is clear that the HR sensor is active regardless of whether motion levels indicate the user is asleep or awake.

Finally, 13.f requires that “more than a threshold motion” inactivates a biological sensor. Dr. Mortazavi argues that, when a user wakes up, the device will enter into a lower power state, which Dr. Mortazavi equates to inactivating a sensor. JDX-0001C, ¶2791-2791, citing RX-0072, 15:33-45 and 16:33-37. While limitation 13.f requires inactivating a sensor when motion is “more than a threshold motion,” Hong discloses reducing sampling rate when motion is *below* a threshold. RX-0072, 16:33-37. Oura’s argument also improperly relies on disclosures that have nothing to do with the sleep detection functionality Oura maps to other claim limitations. RX-0072, 15:33-45.

c. Hong Does Not Disclose “a third period of time” For Activating A Biometric Sensor”

[15.pre]: “The method of claim 13, further comprising:”

[15.a] “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information.”

Hong does not teach dependent claim 15 at least because Hong does not teach independent claim 13. JDX-0009C, ¶¶463.

Hong also does not teach a third period of time, immediately after a second period of time, for activating a biological sensor. JDX-0009C, ¶¶464. Dr. Mortazavi’s theory is that the “third period of time” is a period of time when the user is asleep informing the sampling rate of a biological sensor. *Id.*, citing JDX-0001C ¶¶2803-04. Contrary to Dr. Mortazavi’s theory, Hong teaches *decreasing* sampling rate, or sensor activity, of the HR monitor when the user is asleep. Thus, if anything, Hong teaches making the sensor less active rather than “activating the at least one sensor” as claimed. RX-0072, 16:33-37. Additionally, even if Hong taught increasing sampling rate when the user was asleep (it does not), changing power consumption or sampling rate is not activating/deactivating a biological sensor. *see* Section IV.C.2.a-b.

3. Ground 3: Hong In View Of Utter II Does Not Render Obvious Claim 13 of the ’675 Patent

Hong in view of Utter II does not render obvious Claims 13 of the ’675 Patent. Oura and Dr. Mortazavi do not offer a plausible motivation to combine these references and, even if they had, the combination would not disclose all elements of Claim 13..

a. No Motivation to Combine

Dr. Mortazavi fails to articulate a plausible motivation to combine Utter II and Hong. JDX-

0009C, ¶466-469.

First, Dr. Mortazavi does not identify any specific deficiency in Hong that would lead a POSITA to combine it with Utter II. JDX-0009C, ¶467. Dr. Mortazavi argues a POSITA would be motivated to combine Utter II into Hong because Utter II discloses deactivating a biological sensor when motion is high. JDX-0001C, ¶¶2887-94. But Dr. Mortazavi claims *both* references are already directed to devices that manage the operation of biological sensors to optimize power consumption. And Dr. Mortazavi also argues that Hong teaches a PPG sensor that is not continuously active because it discloses modifying power consumption of the sensor based on motion and turning the sensor on/off based on wear. JDX-0001C, ¶¶2864-65. It is entirely inconsistent to argue that Hong teaches activating/deactivating a sensor based on motion while also arguing that a POSITA would be motivated to combine Hong with other prior art specifically because it does not teach activating/deactivating a sensor based on motion.

Second, Dr. Mortazavi does not sufficiently explain how to add Utter II's functionality into Hong's system or why doing so would yield predictable results. JDX-0009C, ¶468; JDX-0001C, ¶¶2891-93. Dr. Mortazavi acknowledges that "Hong already contemplates reducing power to biometric sensors when undesired motion is detected." JDX-0001C, ¶2892. Hong acknowledges that motion impacts biometric sensor measurements and that increasing sensor use increases power consumption. *See generally* RX-0072 15:33-45, 19:20-13, 72:33-73:21, Fig. 18A. Despite doing so, Hong intentionally adjusts sensor operation rather than entirely shutting sensors off. A POSITA would understand that Hong does so because the system of Hong requires and uses continuous HR measurements in all motion conditions. *See, e.g.,* RX-0072 2:24-38, 3:7-11, 4:64-5:3, 6:9-14 (disclosing constant HR monitoring when device is worn). Dr. Mortazavi assumes, without explanation, that Hong's system would continue to operate successfully without these

measurements. Additionally, as discussed below, Utter II, under Dr. Mortazavi's invalidity read, teaches away from such a combination because it teaches using separate sensors for wear and motion detection. RX-0071, [0035]-[0037], Fig. 1A-B; *see* Section IV.C.5.a.

b. Hong In View of Utter II Does Not Render Obvious Claim 13

[13.f]: “in response to identifying that motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.”

Dr. Mortazavi uses Utter II to supplement limitation 13.f that is absent from Hong. JDX-0001C, ¶¶2887-2899. However, this combination fails to disclose Claim 13. JDX-0009C, ¶466-470. Utter II does not disclose claim limitations [13.c] and [13.f] at least because Utter II does not disclose activating/inactivating a biological sensor based on motion as discussed below. *See* Section IV.C.4. Hong similarly fails to teach claim limitations [13.c] and [13.f] as discussed above. *see* Section IV.C.2. Thus, Hong does not disclose Claim 13, including claim limitation [13.f], whether alone or in combination with Utter II. In addition, the combination fails to disclose the limitations not disclosed by either reference individually (*see* Sections Section IV.C.4 and Section IV.C.2) and not addressed by Oura's proposed combination.

4. Ground 4: Utter II Does Not Anticipate Claim 13 of the '675 Patent

a. Utter II Does Not Teach Wear Detection

[13.a]: “determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device”

Limitation 13.a recites wear detection, specifically “determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device.” But Utter II does not teach wear detection. JDX-0009C, ¶¶472-475.

Oura and Dr. Mortazavi rely on Utter II's disclosure of “[s]ensors 110 [that] generate one

[REDACTED]

or more signals S indicative of force acting on them via a coupling or contact with a body portion 101 of a user.” JDX-0001C, ¶2948; RX-0071, [0035]. But “sensors 110” are pressure sensors “for detecting/sensing force, pressure, or other metric associated with tissues of a body indicative of inflammation and/or contraction. RX-0071, [0035]. Utter II nowhere describes using this pressure sensor to determine wear. *Id.*

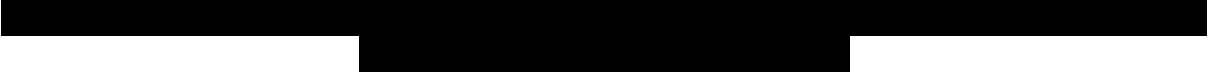
b. Utter II Does Not Disclose A Single “First Sensor” Performing Wear Detection And Motion Detection

[13.a]: “determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device”

[13.b]: “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body”

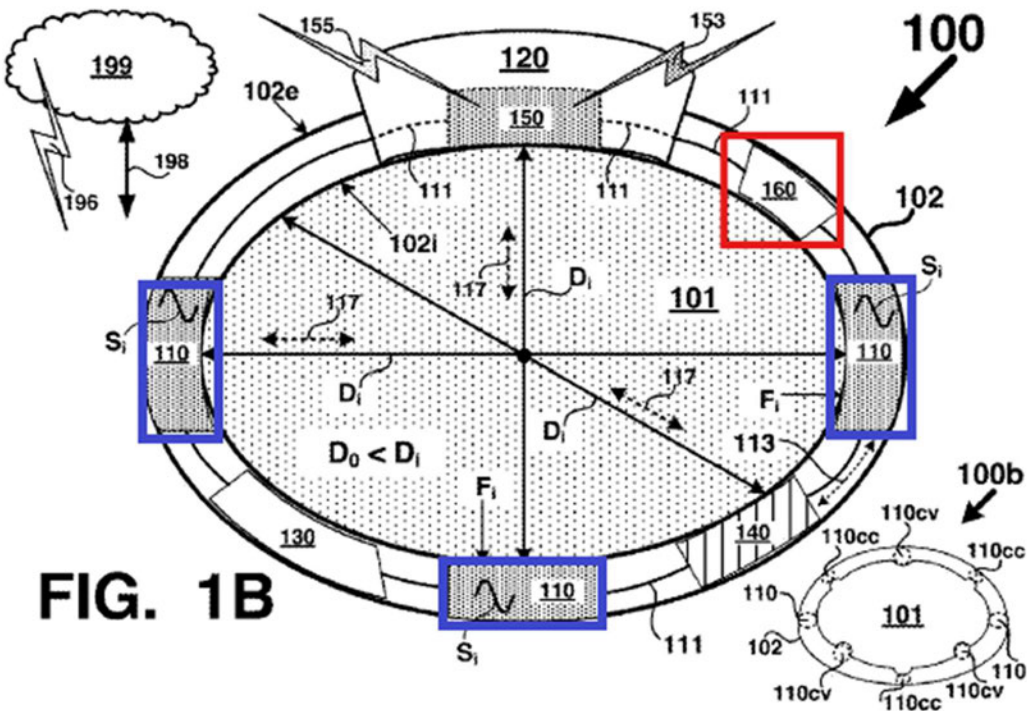
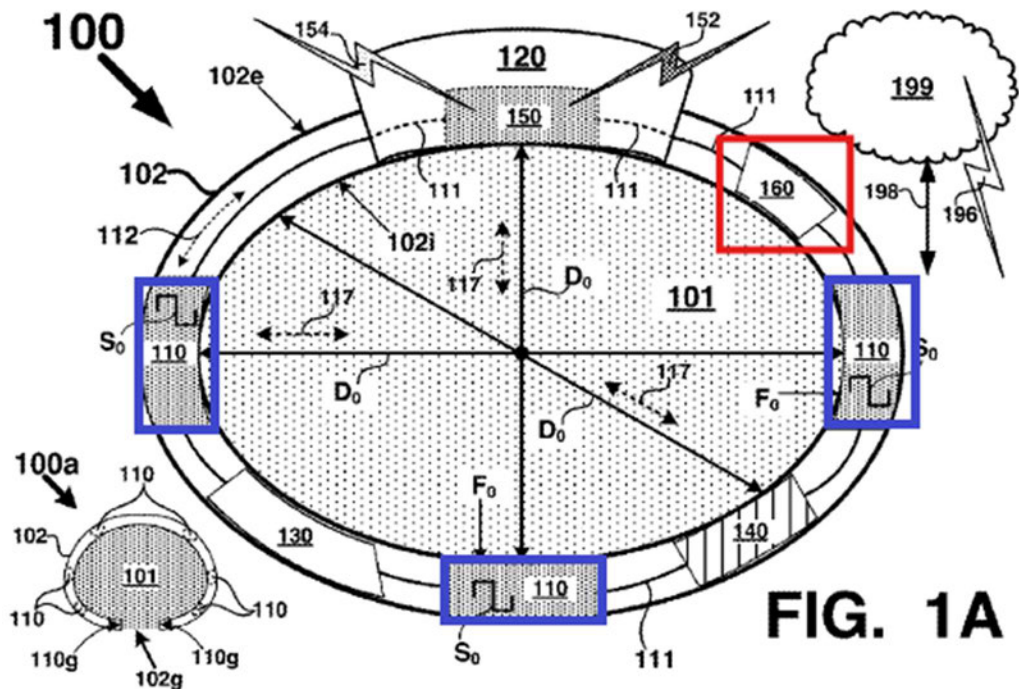
Limitations 13.a and 13.b together require that the same “first sensor” used for motion detection is used to measure both wear and motion. JDX-0009C, ¶475-482. Specifically, Claim 13 requires “determining whether an electronic device is worn on a part of a user’s body *using at least one first sensor* of the electronic device” and “acquiring first information on a motion of the electronic device *using at least a part of the at least one first sensor.*” The, antecedent reference requires that the same first sensor measure both motion and wear.

Utter II does not disclose a single sensor that both detects motion and wear. The pressure sensor that Dr. Mortazavi cites for alleged wear detection in 13.a, does not detect motion. JDX-0009C, ¶¶478. In Utter II, however, the motion sensor is separate from the pressure sensor. Utter II makes clear that the pressure detection Dr. Mortazavi points to as the claimed wear detection is conducted using a sensor “for detecting/sensing force pressure, or other metric associated with tissues of a body indicative of inflammation and/or contraction” rather than an accelerometer. RX-



0071, [0035].

Utter II further describes an “accelerometer” as *another* sensor (160) distinguishable from the pressure sensor (110) described in [0035]. RX-0071 [0037] (discussing “*sensors 110, other sensors,*” and separate “*motion sensors 160 (e.g., single or multi-axis accelerometer, gyroscope, piezoelectric device).*”) Figure 1 below of Utter II also depicts the force sensors as sensors 110 (blue boxes) and the motion sensors as sensors 160 (red box) as separate.



Under Oura's theory, the "wear detection" pressure sensor is not the same as the motion sensor that detects motion and thus Utter II does not disclose a "first sensor" that allegedly does wear detection and motion detection.

c. Utter Does Not Disclose Wear Detection Triggering Motion Detection To Activate A Biometric Sensor

[13.b]: “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body”

[13.c] “activating at least one second sensor based on at least a part of the first information on the motion of the electronic device”

Limitation 13.b requires sensing wear detection in a “first period of time” that triggers motion detection in a “second period of time” resulting in the “activating” of limitation 13.c. Utter II does not disclose these requirements. JDX-0009C, ¶¶476-486.

Utter II does not teach wear detection whatsoever. JDX-0009C, ¶¶477; *see* Section IV.C.4.a. Since Utter II does not disclose wear detection, it cannot disclose using such wear detection as a trigger for a second period of time for determining whether activate a sensor.

Even applying Oura’s theory, Utter II does not teach that any “wear detection” by the pressure sensor triggers motion detection for activating a sensor. According to Dr. Mortazavi, “[t]he second period of time corresponds to the motion evaluation at stage 1403 of flow 1400. Utter II teaches that at stage 1403, ‘signals from one or more sensors and/or sensor types for sensing motion may be analyzed to determine whether or not the user 800 is in motion.’” JDX-0001C, ¶2954 (quoting RX-0071 [0108]). But Utter II does not teach that any “wear detection” or any activity of Utter II’s pressure sensor triggers the motion detection at stage 1403. Utter II is silent as to the temporal relationship between any wear detection and subsequent motion detection.

d. Utter II Does Not Disclose Activating Or Inactivating A Biometric Sensor Based On Motion

[13.c] “activating at least one second sensor based on at least a part of the first information on the motion of the electronic device”

[13.f]: “in response to identifying that motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.”

Limitations 13.c and 13.f require activating and inactivating a biological sensor based on motion information. Utter II does not teach activating or deactivating a biological sensor whatsoever—instead, it teaches selecting which data to use for measuring true resting heart rate (TRHR). JDX-0009C, ¶¶483-486. Thus, in Utter II, data has *already* been obtained by a continuously active biological sensor.

Oura and Dr. Mortazavi point to Utter II at [0107]-[0109] as disclosing these elements. JDX-0001C ¶¶2958, 2968, citing RX-0071 [0107]-[0109]. These disclosures do not teach activating or deactivating biometric sensors. The descriptions at [0107]-[0109] are focused on what sensor data may be reliable for measuring TRHR and, at most, teach that certain motion conditions may mean that biometric sensor information is not reliable for measuring TRHR and may not be used. For example, Utter teaches that “motion being sensed (e.g., via motion sensors in 340) may indicate that the user 800 is not at rest and one or more biometric signals such as HR, heart rate variability (HRV), or arousal activity in the sympathetic nervous system (SNS) may not be reliably *used* in a determination of TRHR.” RX-0071 [0108]. But using/refraining from using measured sensor data for TRHR is distinct from activating/deactivating the biological sensor as the claims require.

5. Ground 5: Utter II In View of Hong Does Not Render Obvious Claims 13 and 15 of the '675 Patent

Utter II in view of Hong does not render obvious Claims 13 and 15 of the '675 Patent.

[REDACTED]

Oura and Dr. Mortazavi fail to offer a plausible motivation to combine these references, and the combination would not disclose all elements of Claims 13 and 15.

a. No Motivation to Combine

Dr. Mortazavi fails to articulate a plausible motivation to combine Utter II and Hong. JDX-0009C, ¶¶489-492.

Dr. Mortazavi does not identify any specific deficiency in Utter II that would lead a POSITA to combine it with Hong. JDX-0009C, ¶490. Dr. Mortazavi argues that both are directed to “determining device wear status and controlling biometric sensor” using motion and that a POSITA would be motivated to implement Hong’s accelerometer-based wear detection into Utter II’s monitoring framework “to provide explicit wear determination as a precondition for biometric sensing.” JDX-0001C, ¶¶3020-3025. His explanation is internally inconsistent. As discussed with respect to Utter II above, Utter II does not teach wear detection and, even if it did, incorporating Hong’s wear detection into Utter II would be unnecessary because Utter II would already perform wear detection.

Dr. Mortazavi also does not sufficiently explain how to add Hong’s functionality into Utter II’s system or why doing so would yield predictable results. JDX-0009C, ¶491; JDX-0001C, ¶¶3023-24. Dr. Mortazavi argues that combining the functionality would yield a predictable result because accelerometer-based wear detection was a known technique. *Id.*, ¶3024. But under Dr. Mortazavi’s invalidity mapping, Utter II teaches away from such a combination because it teaches using separate sensors for wear and motion detection. RX-0071, [0035]-[0037], Fig. 1A-B. Specifically, the functionality Dr. Mortazavi maps to wear detection is performed by pressure sensors separate from an accelerometer. *See* Section IV.C.4. Dr. Mortazavi does not show that the separate accelerometers of Utter II would be capable of wear detection, especially in

[REDACTED]

combination with the purported wear detection already provided by Utter II's pressure sensors. As such, combining Utter II with Hong would require using accelerometers and pressure sensors in a totally new way.

b. Utter II In View of Hong Does Not Render Obvious Claim 13

[13.a]: “determining whether an electronic device is worn on a part of a user’s body using at least one first sensor of the electronic device;”

[13.b]: “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body;”

Dr. Mortazavi uses Hong to supplement limitations 13.a and 13.b that are absent from Utter II. JDX-0001C, ¶¶3026-3038. However, this combination fails to disclose Claim 13. JDX-0009C, ¶493-494. Utter II does not disclose claim limitations [13.a], [13.b], and [13.c] at least because Utter II does not disclose wear detection, does not disclose wear detection and motion detection using the same “first sensor,” and does not disclose the specific sequence of steps in which wear detection in a “first period of time” triggers a “second period of time” for detecting motion to activate a sensor. *See* Section IV.C.4. Hong also fails to teach claim limitations [13.b] and [13.c] at least because Hong does not disclose the specific sequence of steps in which wear detection in a “first period of time” triggers a “second period of time” for detecting motion to activate a sensor. *see* Section IV.C.2. Thus, Utter II does not disclose Claim 13, including claim limitations [13.a] and [13.b], whether alone or in combination with Hong. In addition, the combination fails to disclose the limitations not disclosed by either reference individually (*see* Sections Section IV.C.4 and Section IV.C.2) and not addressed by Oura’s proposed combination.

c. Utter II In View of Hong Does Not Render Obvious Claim 15

[15.pre]: “The method of claim 13, further comprising:”

[15.a] “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information.”

Dr. Mortazavi does not contend that Utter II discloses Claim 15, but instead relies on Hong for that disclosure. JDX-0001C, ¶¶3039-3045. But, as discussed above, Hong fails to teach Claim 15 at least because (1) Hong teaches modifying the power consumption of a sensor rather than “activating” it and (2) even under Oura’s mapping, Hong teaches decreasing sensor activity during Oura’s purported “third period of time.” *See* Section IV.C.2. JDX-0009C, ¶¶493-494. Moreover, neither Utter II nor Hong teach Claim 13. *see* Sections IV.C.2, IV.C.4.

D. The Asserted Claims Are Directed to Patent Eligible Subject Matter Under 35 U.S.C. § 101

Oura incorrectly contends that the asserted claims are directed to the abstract idea of “using motion to activate and deactivate sensing of a biological signal.

Whether or not claims are abstract depends on whether the claims “focus on a specific means or method that improves the relevant technology or are instead directed to a result or effect that itself is the abstract idea and merely invoke generic processes and machinery.” *CardioNet, LLC v. InfoBionic, Inc.*, 955 F.3d 1358, 1368 (Fed. Cir. 2020). The asserted claims of the ’675 Patent are directed to a concrete improvement in the functionality of a multi-hardware sensor system—improving the battery life of multi-sensor systems by allowing one sensor to selectively activate and deactivate a second sensor, ensuring it is not continuously consuming power. JDX-0009C, ¶¶584-93; JX-0003, 1:44-63, 19:65-20:29, 31:24-58, 32:28-33, Claims 13, 15. For example, Claim 13 improves upon past multi-sensor systems by “activating at least one second

[REDACTED]

sensor” only “based on at least a part of the first information” obtained using a first sensor. JX-0003, Claim 13; JDX-0009C, ¶¶584-93.

The '675 Patent further explains its selective activation approach resolves power consumption problems in prior art multi-sensor systems. The specification notes that existing multi-sensor systems had “difficulty in measuring biometric information for a long period of time” because the biometric sensors “consume the battery power of the electronic devices. JX-0003, 1:44-64; JDX-0009C, ¶¶584-93. The '675 Patent solves this problem and provided a novel approach for doing so by “decreas[ing] the light-emitting time” of biological sensors by operating them only “when the previously measured motion (e.g., the average of the motion) is changed to [a] threshold value or less.” JX-0003, 19:65-20:29, 32:28-33.

The Federal Circuit and other courts have previously determined that sensor systems are not abstract. *See, e.g., CardioNet*, 955 F.3d 1358; *Thales Visionix Inc. v. U.S.*, 850 F.3d 1343, 1348–49 (Fed. Cir. 2017); *Uniloc USA, Inc. v. Samsung Elecs. Am., Inc.*, No. 2:17-CV-00651-JRG, 2018 WL 4927279, at *3–4 (E.D. Tex. Sept. 18, 2018). In *CardioNet*, the Federal Circuit determined that claims directed to improving HR monitors by tracking variability in beat-to-beat HR timing were not abstract because they were targeted at specific improvements to sensor capabilities such as detecting atrial fibrillation and atrial flutter. 955 F.3d at 1368-70. In *Thales*, the Federal Circuit similarly determined that claims directed to a multi-sensor system used to calculate the orientation of an object were not directed to an abstract idea because they addressed problems inherent in prior sensor systems to improve the accuracy with which inertial sensors tracked objects. 850 F.3d at 1348-49. Finally, the *Uniloc* court determined that a claim directed at improving accuracy of step counting by using an accelerometer to track the precise incline the user is stepping from was not abstract. *Uniloc*, No. 2:17-CV-00651-JRG at *3–4. The asserted

[REDACTED]

claims of the '675 Patent are analogous and are similarly directed to a specific and tangible improvement in sensor systems, improving battery life, by allowing one sensor to selectively activate/deactivate another. JDX-0009C, ¶¶584-93; JX-0003, 1:44-63, 19:65-20:29, 31:24-58, 32:28-33, Claims 13, 15.

Oura further argues that the asserted claims of the '675 Patent are directed to well-understood, routine and conventional technology based on the same prior art discussed above. JDX-0001C, ¶¶3081-086. Oura's argument conflates patent eligibility with anticipation and obviousness. Regardless, as discussed above, the asserted claims are neither anticipated nor obvious. *see* Section IV.C. The '675 Patent discloses a novel application of its multi-sensor system—selectively activating or deactivating a biological sensor based on motion information from a motion sensor—for a tangible improvement in sensor technologies—improved battery life. JDX-0009C, ¶¶584-93; JX-0003, 1:44-63, 19:65-20:29, 31:24-58, 32:28-33, Claims 13, 15.

V. U.S. PATENT NO. 10,945,677

A. The Accused Products Infringe the '677 Patent⁴

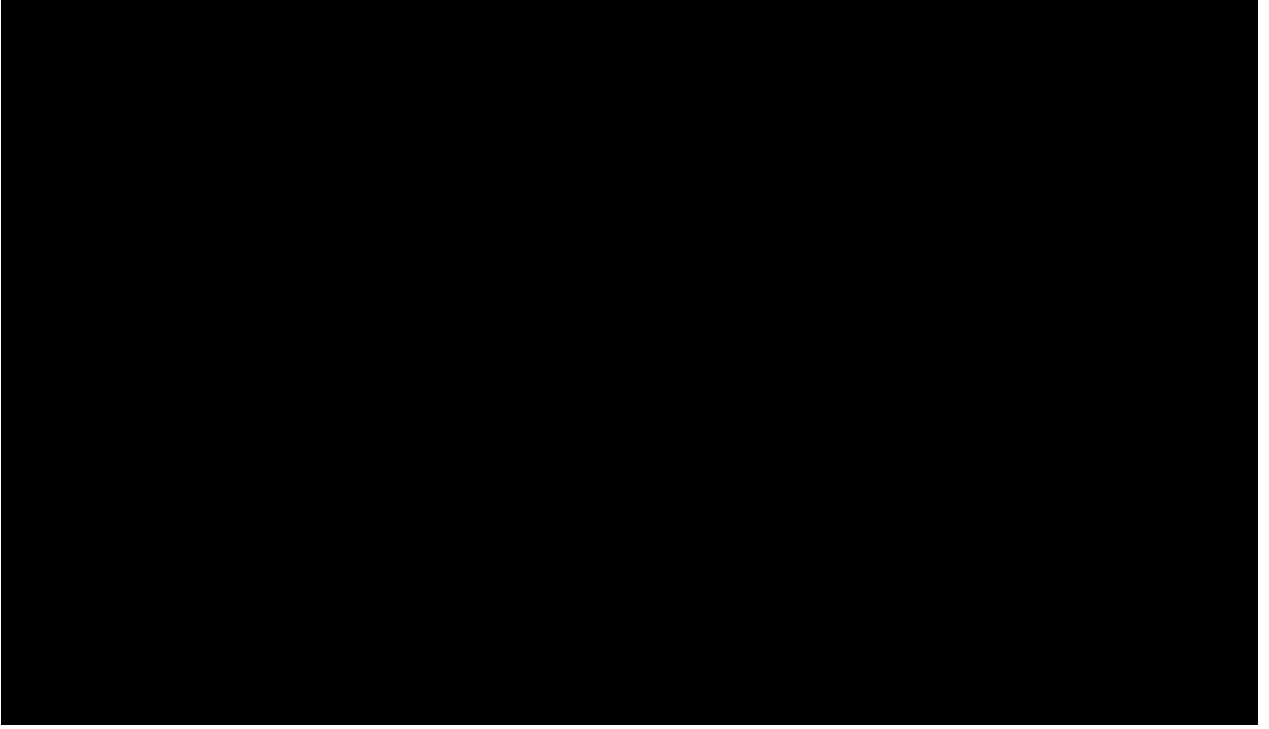
As set forth in more detail below, the Accused Products infringe Claims 1-3 and 9 of the '677 Patent. As with the '675 Patent, Oura has offered no rebuttal expert opinion or evidence to the contrary. Thus, Samsung's infringement positions, the opinions of Samsung's expert Dr. Welch, and supporting evidence all stand entirely unrebutted. All that Oura musters is the same legally improper practicing the prior art defense, even though Ring 1 lacked the very feature Samsung is accusing of infringement—SpO2 detection.

⁴ The analysis below focuses on Ring 4. But both parties' experts agree that Ring 4 is representative of newer Oura products such as Ring 5 and [REDACTED] with no differences that are material to infringement. JDX-0007C, ¶¶381-84, 552-54.

1. The Accused Products Infringe Claim 1

a. Limitation 1.pre (“A wearable electronic device comprising:”)

The Accused Products satisfy the preamble because they are “wearable electronic device[s],” i.e., smart rings. JDX-0004C, ¶¶72-77, 352-54, 441-43; CX-0996C, OURA1468_00096106.



b. Limitation 1.a (“a motion sensor;”)

The Accused Products include the claimed “motion sensor”—an accelerometer such as the  illustrated below. JDX-0004C, ¶¶72-77, 355-57, 444-46; CX-0997C, OURA1468 00096111.

[REDACTED]

[REDACTED]

[REDACTED]

c. Limitation 1.b (“one or more optical sensors includes a light emitting unit and a light receiving unit; and”)

The Accused Products meet this limitation as they include the claimed “one or more optical sensors” that include “a light emitting unit and a light receiving unit.” JDX-0004C, ¶¶72-77, 88-111, 358-62, 447-49; CX-0994C, OURA1468_00029608; CX-0996C, OURA1468_00096106; CX-1653; CX-1649; CX-1651. The Accused Products include a SpO2 sensor that is an “optical sensor.” The SpO2 sensor is comprised of the red and IR LEDs from [REDACTED] units (the “light emitting units”), and [REDACTED] (the “light receiving units”). *Id.*

[REDACTED]

[REDACTED]

[REDACTED]

CX-0994C, OURA1468_00029608.

d. Limitation 1.c (“at least one processor configured to:”)

The Accused Products meet this limitation because they include “at least one processor” such as the [REDACTED] JDX-0004C, ¶¶72-77, 363-68, 450-52; JDX-0025C, 68:24-69:1; JDX-0024C, 81:5-14, 82:4-85:2; CX-0994C, OURA1468_00029596. This processor is coupled to the accelerometer (i.e., “motion sensor”) and the SpO2 sensor (i.e., “one or more optical sensors”) such that the processor is configured to perform each subsequent claimed step through the sensors as described below. *Id.*; see Section V.A.1.e-i. [REDACTED]

[REDACTED] CX-1501, SAMS382-0028059; *see also id.* The diagram below depicts the [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0994C, OURA1468_00029596 (annotated).

- e. **Limitation 1.d (“obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,”)**

The motion sensor, or accelerometer, of the Accused Products is used to [REDACTED]

[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED]
[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED]
[REDACTED]

The accelerometer may [REDACTED]

[REDACTED]
[REDACTED] [REDACTED] [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

f. Limitation 1.e (“emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition,”)

The Accused Products “emit light using” red and IR LEDs (the “light emitting unit”) comprising the SpO2 sensor to measure SpO2 by [REDACTED]. CX-0221, SAMS382-0027593; *see also* JDX-0004C, ¶¶72-77, 88-111, 369-86, 450-73; CX-0241, SAMS382-0027803 (“Red and infrared LEDs measure blood oxygen while you sleep”); CX-0221 (“Please note that you may see the red LEDs light up whenever your Oura detects that you may be asleep (based on your heart rate, movement, and respiratory rate), during the day”); JDX-0026C, 233:23-234:15, 234:18-237:10; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178.

Activation of the SpO2 sensor is [REDACTED]. [REDACTED] JDX-0004C, ¶¶72-77, 88-111, 369-86, 450-73; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.* [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

g. Limitation 1.f (“obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user,”)

The Accused Products obtain a biological signal using the SpO2 sensor. The products measure reflected Red and IR LED light at photodiodes (the “light receiving unit”) in the SpO2 sensor to “measure[] peripheral oxygen saturation (SpO2) levels” (the “biological signal”) in the blood based on a determination that the user is asleep. CX-0221, SAMS382-0027593; *see also* JDX-0004C, ¶¶72-77, 88-111, 387-92, 450-73; CX-0241, SAMS382-0027803 (“Red and infrared LEDs measure blood oxygen while you sleep”); CX-0221 (“Please note that you may see the red LEDs light up whenever your Oura detects that you may be asleep (based on your heart rate, movement, and respiratory rate), during the day”); JDX-0026C, 233:23-234:15, 234:18-237:10; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*

h. Limitation 1.g (“obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and”)

The Accused Products use the [REDACTED] to “obtain” “a second motion signal.” JDX-0004C, ¶¶72-77, 88-111, 393-95, 450-73; CX-0994C, OURA1468_00029610; CX-0241, SAMS382-0027803, JDX-0026C, 178:17-24. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] CX-0994C, OURA1468_00029610; CX-0241, SAMS382-0027803; *see also id.* Because the [REDACTED] is [REDACTED], it obtains a “second motion signal” “during the obtaining of the biological signal.” *Id.*

i. Limitation 1.h (“refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.”)

The Accused Products “refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.” JDX-0004C, ¶¶72-77, 88-111, 396-402, 474-77; CX-0241, SAMS382-0027803; CX-0221; JDX-0026C, 233:23-234:15, 234:18-237:10; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *See* Section V.A.1.i. [REDACTED]

[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

[REDACTED] JDX-0004C, ¶¶72-77, 88-111, 396-402, 474-77; CX-1500C, OURA_1478_SC_0000080-81, 185-89, 192, 448-50, 457-58, 460-61, 464; CX-1686C, OURA_1468_SC_0000178. [REDACTED]

[REDACTED]

[REDACTED] *Id.*

For the first time in opposing Samsung’s MSD, Oura argues that this claim element is not met because [REDACTED]

[REDACTED] EDIS ID 889195, 4-8. Oura’s eleventh-hour argument is wrong for at least three

[REDACTED]

reasons. *First*, it is the SpO2 sensor, not the PPG sensor, that Samsung and Dr. Welch identifies as the claimed “one or more optical sensors.” And, as s shown above, the SpO2 sensor satisfies the claim limitation because the [REDACTED]. *See* Section V.A.1.c. Oura’s insistence that an unaccused sensor (whether it defines as the HR sensor or PPG sensor) may be active does not change this fact. *Second*, [REDACTED] JDX-0004C, ¶¶434-438; JDX-0026, 220:25-221:6 (Oura firmware engineer Michael Larson confirming based on Oura documentation that [REDACTED] [REDACTED]); CX-1219C, 93:23-95:10 [REDACTED]; CX-0241, SAMS382-0027803 (disclosing “Red and infrared LEDs measure blood oxygen;” not the green LED). Thus, Oura’s assertion that the Accused Products continue to measure HR fails to rebut Samsung’s showing that the [REDACTED]. *Third*, 1.h requires “refrain[ing] from emitting of *the* light” where “the light” refers back to the specific light emitted “based at least in part on a determination that the first motion signal satisfies a specified condition” in 1.e. Even if the PPG sensor was accused and emits other light under other conditions (e.g., light used to measure daytime HR), the specific light emitted and accused for 1.e is the specific red and IR light for measuring SpO2, which is [REDACTED] *See* Section V.A.1.f. There is no dispute that this specific light is no longer emitted when “the second motion signal does not satisfy the specified condition.”

2. The Accused Products Infringe Claims 2-3

a. Claim 2

Limitation 2.pre (“The wearable electronic device of claim 1, wherein the at least one processor is further configured to:”)

Limitation 2.a (“determine whether the wearable electronic device is attached to the user using the one or more optical sensors, and”)

Limitation 2.b (“perform the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user.”)

The Accused Products infringe claim 2. *First*, the Accused Products “determine whether the wearable electronic device is attached to the user using the one or more sensors” as claim 2 requires by [REDACTED].

JDX-0004C, ¶¶72-111, 403-410; CX-1500C, OURA_1478_SC_0000185-187, 191, 197; CX-1686C, OURA_1468_SC_0000178, 363; JDX-0026C, 173:22-174:18; 178:17-24, 180:11-18.

The Accused Products also “perform the [claimed] emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user” [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*; see also JDX-0026C, 213:6-214:6, 215:16-20.

a. Claim 3

Limitation 3.pre (“The wearable electronic device of claim 2, wherein the at least one processor is further configured to:”)

Limitation 3.a (“perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.”)

The Accused Products satisfy the added limitation of claim 3 because they “perform the

[REDACTED]

determining based at least in part on a determination that the first motion signal satisfies a specified condition” [REDACTED]

[REDACTED]

JDX-0004C, ¶¶72-111, 411-423; CX-1500C, OURA_1478_SC_0000185-187, 191, 197; CX-1686C, OURA_1468_SC_0000178, 363; JDX-0026C, 173:22-174:18; 178:17-24, 180:11-18, 185:5-18; 186:14-17; 188:6-189. Alternatively, a [REDACTED]

[REDACTED] *Id.*

3. The Accused Products Infringe Claim 9

a. Limitation 9.pre (“A wearable electronic device comprising:”)

9.pre is satisfied for the same reason as 1.pre. *See* Section V.A.1.a.

b. Limitation 9.a (“a first sensor;”)

The “motion sensor” described for 1.a is “a first sensor.” *See* Section V.A.1.b.

c. Limitation 9.b (“a second sensor; and”)

The “optical sensor” described for 1.b is “a second sensor.” *See* Section V.A.1.c.

d. Limitation 9.c (“at least one processor configured to:”)

9.c is satisfied for the same reason as 1.c. *See* Section V.A.1.d.

e. Limitation 9.d (“obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,”)

9.d is satisfied for the same reason as 1.d and 1.g. *See* Section V.A.1.e, V.A.1.h. The “first information corresponding to movement” of 9.d is motion information detected using the accelerometer, which includes motion signals as described for 1.d and 1.g. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

f. Limitation 9.e (“activate the second sensor based at least in part of the movement,”)

The Accused Products “activate the second sensor based at least in part of the movement” resulting in emission of light as described for 1.e. *See* Section V.A.1.f.

g. Limitation 9.f (“obtain, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and”)

9.f is satisfied for the same reasons as 1.f. *See* Section V.A.1.g. The “biological signal” corresponding to the “second information” is obtained by emitting light using the “activated” SpO2 sensor (“second sensor”).

h. Limitation 9.g (“refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range.”)

9.g is satisfied for the reasons described for 1.h. *See* Section V.A.1.i. For 9.g the “specified range” triggering the “refrain[ing]” is [REDACTED].

B. TDI Is Met for the '677 Patent

Samsung’s expert Dr. Welch and the evidence will demonstrate that WDI Products practice Claims 1-3 and 9 of the '677 Patent as detailed below. In fact, Oura has offered no rebuttal expert opinion or evidence regarding Claim 9.

1. TDI Is Met for Claim 1

a. Limitation 1.pre (“A wearable electronic device comprising;”)

The WDI Products are the claimed “wearable electronic device[s]” in the form of wearable electronic rings or watches. JDX-0004C, ¶¶112-32, 177, 823-26, 930-32.

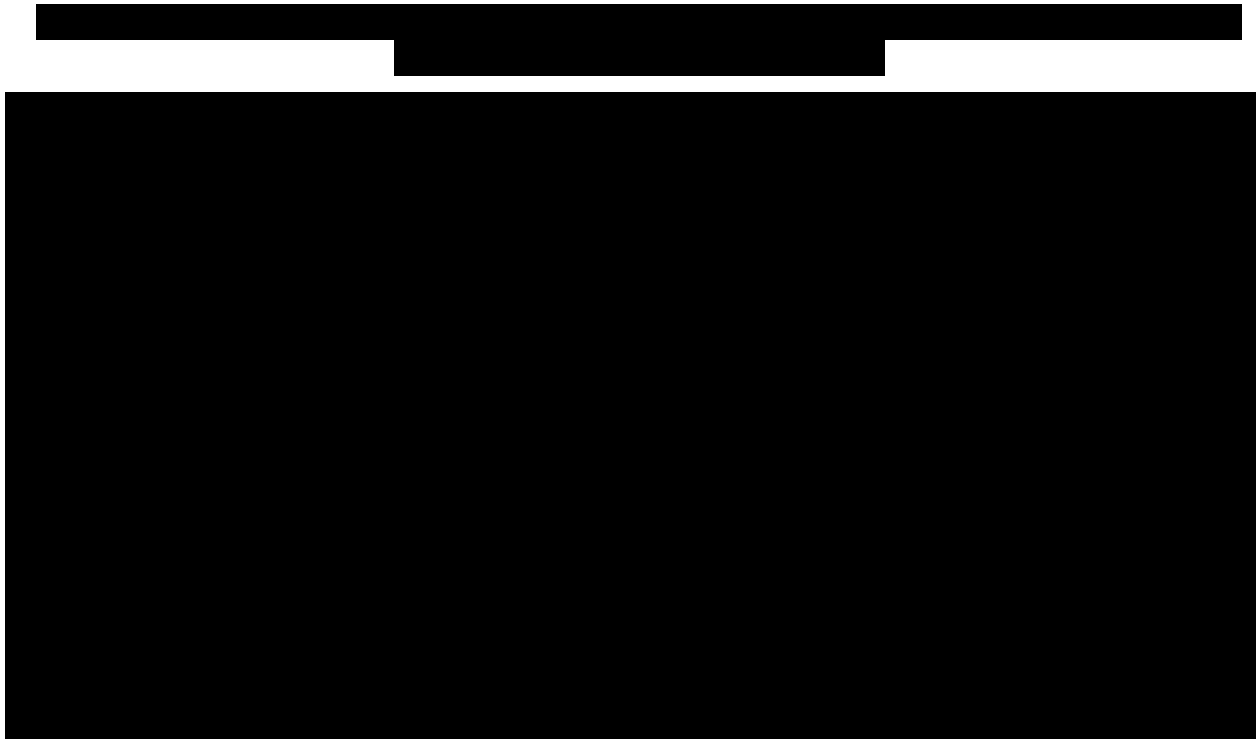
b. Limitation 1.a (“a motion sensor;”)

The WDI Products each include the claimed “motion sensor” in the form of an accelerometer specific. JDX-0004C, ¶¶112-32, 827-30, 933-35; CX-0148C, SAMS382-0012471;

[REDACTED]

CX-0160C, SAMS382-0013172; CX-0179C, SAMS382-0014881; CX-0150C, SAMS382-0012801; CX-0151C, SAMS382-0012809; CX-0152C, SAMS382-0012815; CX-0156C, SAMS382-0013147; CX-0157C, SAMS382-0013153; CX-0158C, SAMS382-0013161; CX-0164C, SAMS382-0013288; CX-0165C, SAMS382-0013296; CX-0159C, SAMS382-0013169; CX-0176C, SAMS382-0013620; CX-0177C, SAMS382-0013626; CX-1640C, SAMS382-0011270; CX-0992C, SAMS406-0074870; CX-1643 *see* Section IV.B.1.b (identifying models for accelerometers). For example, the accelerometers for the Galaxy Ring and Galaxy Watch6 are annotated with a red box in each of the block diagrams below.

[REDACTED]



c. Limitation 1.b (“one or more optical sensors includes a light emitting unit and a light receiving unit; and”)

Each of the WDI Products includes a SpO2 sensor (“one or more optical sensors”), with green, red, and IR LEDs (“light emitting unit[s]”) and photodetectors (“light receiving unit[s]”) as depicted below (annotated in green), and, thus, satisfy limitation 1.b. JDX-0004C, ¶¶112-32, 831-41, 936-38; CX-0261C, SAMS382-0029425-426; CX-1640C, SAMS382-0011274; *see* Section IV.B.1.d (identifying models for SpO2 sensor).

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

d. Limitation 1.c (“at least one processor configured to:”)

The WDI Products include the claimed “at least one processor,” [REDACTED]

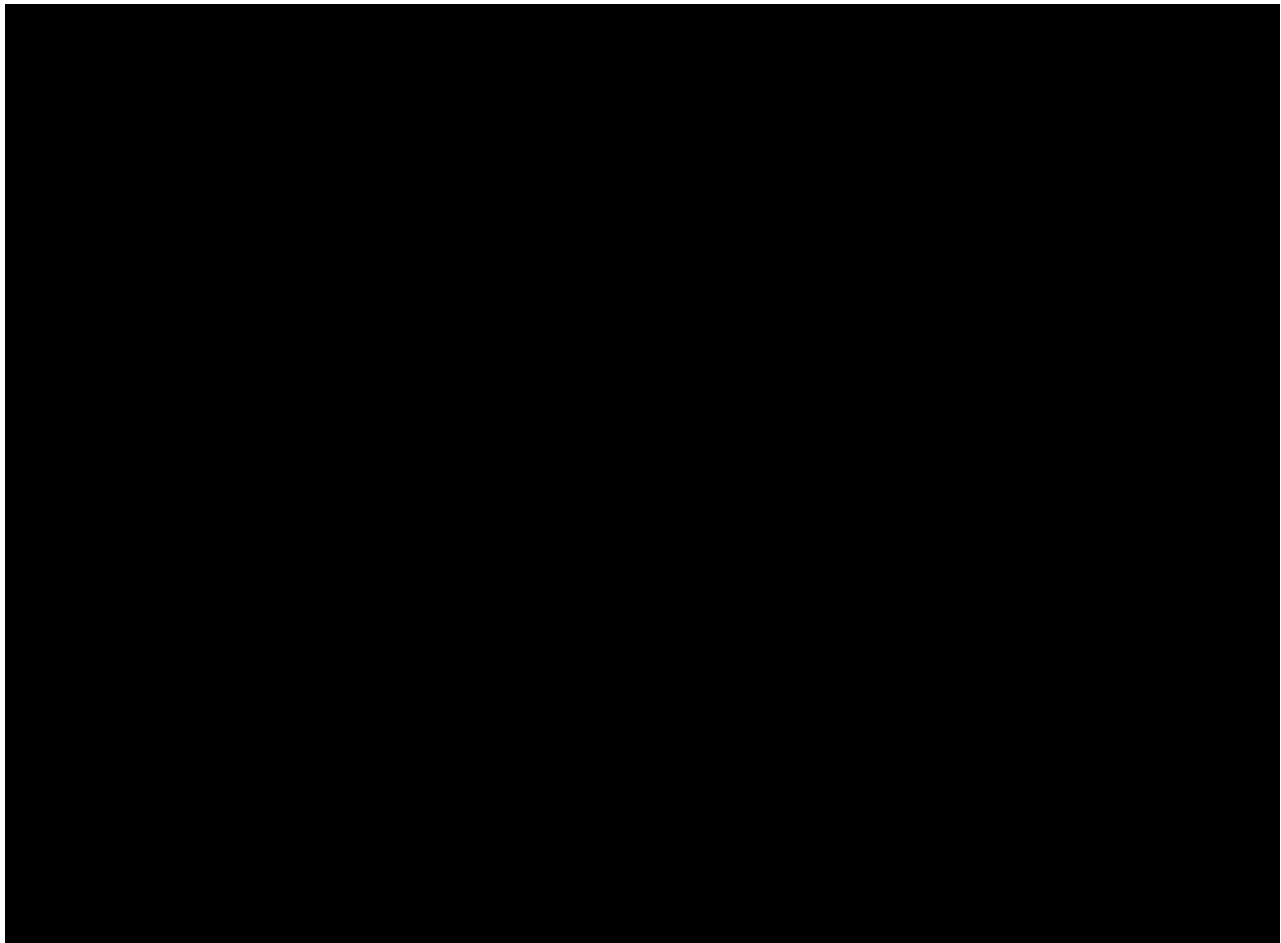
[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶ 112-32, 842-47, 939-41; CX-0129C, SAMS382-0009940; CX-0132C, SAMS382-0010123; CX-0135C, SAMS382-0010444; CX-0138C, SAMS382-0010728; CX-0126C; SAMS382-0009691; CX-0992C, SAMS406-0074870.

As described with respect to the remaining limitations of Claim 1, the processor is coupled to the accelerometer (“motion sensor”) and SpO2 sensor (“one or more optical sensor”) such that the processor is configured to perform each subsequent claimed step through the sensors.

The diagrams below depict representative processors (annotated with a blue box) coupled to the accelerometer (annotated with a red box) and SpO2 sensor (annotated with a green box).



[REDACTED]

[REDACTED]

[REDACTED]

- e. **Limitation 1.d (“obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,”)**

The accelerometers (i.e., “motion sensors”) of the WDI Products are used to “obtain” motion signals before the products measure an SpO2 signal (“biological signal”) because the SpO2 sensor is not active or emitting light until after motion signals are used to determine whether the user is asleep. JDX-0004C, ¶¶158-96, 842-62, 939-55; CX-0829C, SAMS406-0071843; CX-0829C, SAMS406-0071844-845; CX-0142, SAMS382-0011184; CX-1639, SAMS406-0001878; CX-1685C, SAMS406-SC-0000007, 0323-0325, 0331-0332; CX-1499C, SAMS382-SC-0000017-18, 0027-0029, 0057-0058, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300, 0955-0958; JDX-0022C, 46:16-48:25, 76:14-78:4, 78:18-79:5; JDX-0012C, 49:16-24, 50:15-52:4, 53:25-54:13, 55:7-14, 56:2-57:3, 62:5-64:17, 71:15-72:16; JDX-0013C, 50:3-19 [REDACTED]

[REDACTED]

[REDACTED]), 145:11-148:10.

[REDACTED]

Oura argues that this requirement is not met because Samsung has failed to show “obtain[ing]” “prior to a biological signal being obtained.” JDX-0006C, ¶¶143-148. However, as shown for limitation 1.e next, use of the SpO2 sensor to obtain a biological only happens after obtaining a motion signal that satisfies a specified condition.

f. Limitation 1.e (“emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition.”)

The WDI Products “emit light” using LEDs comprising the SpO2 sensor (“light emitting unit”) based on a determination that the user is asleep, i.e., the claimed “specified condition.” *Id.*; *see also* JDX-0013C, 145:11-148:10. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*

g. Limitation 1.f (“obtain the biological signal corresponding to a user using the light receiving unit based at least in part on reflected light which the light is reflected on the user.”)

The WDI Products obtain an SpO2 signal (“biological signal”) by emitting light from LEDs comprising the SpO2 sensor that is reflected by a user’s skin before being detected by photodetectors and, thus, satisfy limitation 1.f. JDX-0004C, ¶¶158- 96, 863-70, 956-59; CX-0142, SAMS382-0011184; CX-1639, SAMS406-0001878; CX-1685C, SAMS406-SC-0000327-0328, 0331-0332; CX-1499C, SAMS382-SC-0000011-0012, 0017-0018, 0027-0029, 0117, 0124, 0133, 0161, 0197, 0216, 0241, 0260-0261, 0284-0285, 0299-0300.

h. Limitation 1.g (“obtain, using the motion sensor, a second motion signal corresponding to the wearable electronic device during the obtaining of the biological signal, and”)

The accelerometer of the WDI Products “obtain” accelerometer data (“a second motion

[REDACTED]

signal”) “during the obtaining of the biological signal.” JDX-0004C, ¶¶158- 96, 871-73. As described below with respect to limitation 1.h, [REDACTED]

[REDACTED]. *Id.*

Oura argues that the WDI Products do not practice this limitation because Samsung fails to identify a “biological signal” or that to the extent Samsung does, Samsung calls biological information collected over a period of time a “biological signal” rather than identifying a clear “signal.” JDX-0006C, ¶¶148-151. As described with respect to limitation 1.f, an SpO2 signal is a “biological signal” and is obtained [REDACTED]

[REDACTED]. Regardless, the ’677 Patent does not preclude biological information collected over a period of time from being a “biological signal.”

i. Limitation 1.h (“refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition.”)

The WDI Products “refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition” as limitation 1.h requires. JDX-0004C, ¶¶158- 96, 874-85, 960-71; JDX-0022C, 76:14-78:4, 78:18-29:5; CX-0142, SAMS382-0011184; CX-1499C, SAMS382-SC-0000679-0680, 0955-0958; CX-1685C, SAMS406-SC-0000007, 0327-0328, 0331-0332. Specifically, [REDACTED] [REDACTED] [REDACTED] [REDACTED]

[REDACTED] *Id.* [REDACTED]

[REDACTED] *Id.*

[REDACTED]

To the extent that Oura argues that this claim element is not practiced because a PPG sensor continues to emit light for purposes other than SpO2 detection, such as HR detection, and after SpO2 sensing has stopped, this is wrong for at least two reasons. *First*, it is the SpO2 sensor that Samsung and Dr. Welch identify the claimed “one or more optical sensors;” not the PPG sensor. *See* Section V.B.1.c. Other LEDs, like the green LED used to measure HR, may be part of a PPG sensor but they are not part of the SpO2 sensor and are thus irrelevant with respect to whether the SpO2 sensor practices the Claims. *Second*, limitation 1.h requires “refrain[ing] from emitting of *the* light” where “the light” refers back to the specific light emitted “based at least in part on a determination that the first motion signal satisfies a specified condition” in 1.e. Even if the PPG sensor emits other light for other purposes, the specific light emitted and accused for 1.e is the light used for measuring SpO2 levels (*See* Section V.B.1.f). [REDACTED]

[REDACTED]

[REDACTED]

2. TDI Is Met for Claims 2-3

a. *Claim 2*

Limitation 2.pre (“The wearable electronic device of claim 1, wherein the at least one processor is further configured to:”)

Limitation 2.a (“determine whether the wearable electronic device is attached to the user using the one or more optical sensors, and:”)

Limitation 2.b (“perform the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user.”)

The WDI Products satisfy the added limitation of 2.a, i.e., they “determine whether the wearable electronic device is attached to the user using the one or more sensors” [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. JDX-0004C, ¶¶136- 96, 886-899; CX-1641C, WELCH1478-0000006; CX-1642C, WELCH1478-0000002; CX-0829C, SAMS406-0071485; CX-1499C, SAMS382-SC-0000048, 0051, 0054, 1011-1012, JDX-0022C, 63:10-64:1; JDX-0016C, 60:25-61:22; JDX-0012C, 68:13-16.

The WDI Products also perform the added step of limitation 2.b, i.e., “emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user” because [REDACTED]

[REDACTED]

[REDACTED]

b. *Claim 3*

Limitation 3.pre (“The wearable electronic device of claim 2, wherein the at least one processor is further configured to:”)

Limitation 3.a (“perform the determining based at least in part on a determination that the first motion signal satisfies a specified condition.”)

The WDI Products perform the added step of claim 3 of “determining based at least in part on a determination that the first motion signal satisfies a specified condition” because the WDI Products use a [REDACTED]

[REDACTED]

[REDACTED]. JDX-0004C, ¶¶136-96, 900-914, WELCH1478-0000006; CX-1642C, WELCH1478-0000002; JDX-0016C, 50:17-52:12, 72:10-73:15, 74:16-76:23.

3. TDI Is Met for Claim 9

a. Limitation 9.pre (“A wearable electronic device comprising:”)

9.pre is practiced for the same reason as 1.pre. *See* Section V.B.1.a.

b. Limitation 9.a (“a first sensor;”)

The “motion sensor” described for 1.a is “a first sensor.” *See* Section V.B.1.b.

c. Limitation 9.b (“a second sensor; and”)

The “optical sensor” described for 1.b is “a second sensor.” *See* Section V.B.1.c.

d. Limitation 9.c (“at least one processor configured to:”)

9.c is practiced for the same reason as 1.c. *See* Section V.B.1.d.

e. Limitation 9.d (“obtain, using the first sensor, first information corresponding to movement of the wearable electronic device,”)

9.d is practiced for the same reason as 1.d and 1.g. *See* Section V.B.1.e, V.B.1.h. The “first information corresponding to movement” of 9.d is [REDACTED]

[REDACTED] as described for 1.d and 1.g.

f. Limitation 9.e (“activate the second sensor based at least in part of the movement,”)

The WDI Products “activate the second sensor based at least in part of the movement” [REDACTED]

g. Limitation 9.f (“obtain, using the activated second sensor, second information corresponding to biological signal related to a user during the obtaining of the first information, and”)

9.f is practiced for the same reasons as 1.f. *See* Section V.B.1.g. The “biological signal” corresponding to the “second information” is obtained by emitting light using the “activated” SpO2 sensor (“second sensor”).

h. Limitation 9.g (“refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range.”)

9.g is practiced as described for 1.h. *See* Section V.B.1.i. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

C. The Asserted Claims of the '677 Patent Are Valid

Samsung's expert Dr. Welch and the evidence will demonstrate that the asserted claims are valid. Oura incorrectly asserts that the asserted claims are invalid based on the following prior art grounds:

- Ground 1 – Claims 1-3 and 9 as anticipated by Ring 1
- Ground 2 – Claims 1-3 and 9 as anticipated by U.S. Patent App. Pub. No. 2014/0073486 (“Ahmed486”)
- Ground 3 – Claims 1 and 9 as anticipated by WO 2015/139980 (“Presura”)
- Ground 4 – Claims 2 and 3 as obvious based on Presura in view of U.S. Patent App. No. 2015/0371028 (“Patel”)

As for Ring 1, as discussed for the '675 Patent, Oura will not be able to show that it is prior art or that the documents and code to which it points actually describe the functionality of the prototype Oura alleges was demonstrated before the priority date. Nor did the Ring 1 activate and deactivate a HR sensor (i.e. the “optical sensor” or “second sensor”) based on motion.

Presura and Ahmed486 teach continuously obtaining biological information regardless of motion rather than deactivating a sensor based on motion.

1. Ground 1: Ring 1 Does Not Anticipate Claims 1-3 and 9

a. Ring 1 Is Not Prior Art

As discussed for the '675 Patent, Oura has failed to show that Ring 1 qualifies as prior art, predating the '677 Patent's August 12, 2015 priority date. *See* Section IV.C.1.a. In summary, it is undisputed that the Ring 1 did not go on sale until late 2015. While Oura contends that prototypes

[REDACTED]

of the product were demonstrated in June/July 2015, Oura cites no evidence to connect the source code it relies upon to describe Ring 1’s features—which dates from 2016—to the Ring 1 prototypes as allegedly demonstrated a year earlier.

b. Ring 1 Does Not Teach Activating/Deactivating a Sensor:

Limitation 1.e (“emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition”)

Limitation 1.h (“refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition”)

Limitation 9.e (“activate the second sensor based at least in part of the movement”)

Limitation 9.g (“refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range”)

The asserted claims of the ’677 Patent all require activating or emitting light from an optical sensor (limitations 1.e, 9.e) based on motion and later refraining from using or emitting light from the sensor (limitations 1.h, 9.g) based on motion. Ring 1 does not teach these limitations.

[REDACTED], Oura argues Ring 1 teaches limitations 1.e and 9.e because, [REDACTED]

[REDACTED]

[REDACTED] JDX-0009C, ¶303. Oura argues that Ring 1 teaches limitations 1.h and 9.g because the device [REDACTED]

[REDACTED] JDX-0009C, ¶306. Oura’s arguments fail for a number of reasons. JDX-0009C, ¶¶302-314, 400-408; CX-1376C, OURA1478_00001495.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

JDX-0009C, ¶¶302-314, 400-408; CX-1376C, OURA1478_00001495.

Third, Oura conflates motion information obtained to trigger activation of the biological sensor with motion information obtained as part of wear detection. Claims 2-3 distinguish wear detection from obtaining a biological signal as required by Claims 1 and 9. For example, Claim 1 requires “emit[ting] light . . . based at least in part on a determination that the first motion signal satisfies a specified condition” while Claim 2 further requires “perform[ing] the emitting of the light further based at least in part on a determination that the wearable electronic device is attached to the user,” clearly distinguishing motion and wear determination as separate requirements. JX-0002, Claims 1-2. Oura’s mapping improperly conflates motion used for wear detection with motion separately required to trigger a HR sensor.

JDX-0009C, ¶¶302-314, 400-408; CX-1376C, OURA1478_00001495.

2. Ground 2: Ahmed486 Does Not Anticipate Claims 1-3 and 9

Limitation 1.e (“emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition”)

Limitation 1.h (“refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition”)

Limitation 9.e (“activate the second sensor based at least in part of the movement”)

Limitation 9.g (“refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range”)

As set forth above, the asserted claims of the ’677 Patent all require activating or emitting light from an optical sensor (limitations 1.e, 9.e) based on motion and later refraining from using or emitting light from the sensor (limitations 1.h, 9.g) based on motion. Like Ring 1, Ahmed486

does not teach these limitations.

Oura argues that these limitations are disclosed because Ahmed486 teaches “automatically and selectively activ[ating] one or more [] light emitters to determine a heart rate of the user” based on motion and “adjust[ing] the duty cycle of one or more light emitters and the corresponding sampling rate of the one or more light detectors based on motion status.” JDX-0009C, ¶380. However, adjusting the sampling schedule of a sensor is not the same as the claimed deactivating or activating the sensor based on motion. JDX-0009C, ¶¶378-380.

Ahmed486, which is titled “Systems, Devices and Methods for *Continuous Heart Rate Monitoring*,” is directed to continuously sampling HR; and does not teach intermittently activating and deactivating the biological sensor. *Id.*; RX-0016, Title. Ahmed486 purports to address the problem of “not enabl[ing] continuous sensing of heart rate, thereby preventing continuous fitness monitoring and reliable analysis of physiological data” (c, [0002]-[0003]) by providing “devices and methods for *continuous health and fitness monitoring*.” JDX-0009C, ¶¶378-380; RX-0016, Abstract. Ahmed486 explains the “term ‘continuous,’” which appears over 50 times throughout the specification, “refers to collection of heart rate data at a sufficient frequency to enable detection of every heart beat and also refers to collection of heart rate data continuously throughout the day and night.” JDX-0009C, ¶¶378-380; RX-0016, [0055]. As such, Ahmed486 is explicitly directed to continuous HR detection and teaches away from activating or deactivating the biological sensor based on motion to save battery. JDX-0009C, ¶¶378-380.

3. Ground 3: Presura Does Not Anticipate Claims 1 and 9

a. Presura Does Not Teach the Following Limitations:

Limitation 1.e (“emit light using the light emitting unit based at least in part on a determination that the first motion signal satisfies a specified condition”)

Limitation 1.h (“refrain from the emitting of the light, based at least in part on a determination that the second motion signal does not satisfy the specified condition”)

Limitation 9.e (“activate the second sensor based at least in part of the movement”)

Limitation 9.g (“refrain from the obtaining of the second information using the activated second sensor based at least in part on a determination that the first information satisfies a specified range”)

To reiterate, the asserted claims of the '677 Patent all require activating or emitting light from an optical sensor (limitations 1.e, 9.e) based on motion and later refraining from using or emitting light from the sensor (limitations 1.h, 9.g) also based on motion. Like Ring 1 and Ahmed486, Presura does not teach these limitations.

Oura argues that these limitations are met because Presura teaches activating/deactivating a biological sensor in response to motion. JDX-0009C, ¶¶350-351; RX-0015, 6:8-28. In doing so, Oura ignores Presura’s explanation that it is possible to estimate heartrate based on past heartrate measurements if motion remains relatively stable. *Id.* Thus, a biological sensor can be “deactivated at least *for a duration* during which the heart rate can be estimated accurately enough” to replace real measurements. *Id.* The deactivation disclosed by Presura is only temporary, it represents a change in the sampling schedule rather than the claimed activation/deactivation. *Id.* Presura’s sensor always re-activates, regardless of motion, once past HR measurements may no longer serve as an accurate proxy for current biological measurements.

[REDACTED]

Id. Accordingly, Presura does not disclose these limitations.

Oura also argues that Presura teaches a “specified range” or “specified condition” triggering the claimed activation/deactivation because Presura discloses that whether “the user is in constant motion or is not moving” or whether there “is a change in the motion or motion pattern of the user” may inform sensor activation/deactivation. JDX-0009C, ¶¶352-353; RX-0015, 6:8-28. This argument fails at least because these teachings focus on generalized motion characteristics rather than any *specified* range or condition. *Id.* Considering changes to whether “the user is in constant motion or is not moving” or whether there “is a change in the motion or motion pattern of the user” does not disclose any specific degree or range of change that should trigger a modification of sensor activity. *Id.*

b. Presura Does Not Teach the Claimed Sequence

Limitation 1.d (“obtain, using the motion sensor, a first motion signal corresponding to the wearable electronic device, prior to a biological signal being obtained using the one or more optical sensors,”)

Limitation 1.d requires “obtain[ing] . . . a motion signal . . . *prior* to a biological signal.” Oura argues a motion signal in Presura is obtained before a biological signal because it teaches that a motion signal may trigger biological sensing. JDX-0009C, ¶¶354-358. But Presura teaches modifying a biological sensor’s sampling schedule based on motion only *after* an original biological signal is obtained and used as a baseline for estimating heartrate when the sensor is temporarily inactive. JDX-0009C, ¶¶354-358; RX-0015, 6:8-28, 8:7-12. Thus, Presura discloses the exact opposite of the claimed order in limitation 1.d.

a. No Motivation To Combine

To the extent wear detection was desirable and could result in decreasing light-emission time for the optical sensor, Patel admits that “using proximity alone” rather than using an optical sensor was sufficient to enable wear detection. RX-0014, [0012]; JDX-0009C, ¶¶364-66. Oura admits that a “heart rate sensor consumes substantially more power than the motion sensor” and provides no explanation for why a POSITA would incorporate Patel’s HR sensor based wear detection into Presura’s system over non-optical wear detection. JDX-0001C, ¶1744. The only discernable explanation is that Oura needs to make this argument because Claim 2 and 3 require “determin[ing] whether the wearable electronic device is attached to the user *using the one or*

[REDACTED]

more optical sensors.” JX-0002, Claims 2-3.

D. The Asserted Claims Are Directed to Patent Eligible Subject Matter Under 35 U.S.C. § 101

Oura contends that the asserted claims are directed to the abstract idea of “using motion to activate sensing of a biological signal” and that of the ’677 Patent are directed to well-understood, routine and conventional technology based on disclosures similar to its invalidity grounds. JDX-0043, 230; JDX-0001C, ¶¶2337-2343. These arguments fail for the same reasons as discussed for the ’675 Patent, which shares a specification with the ’677 Patent. *See* Section IV.D. The ’677 Patent is not directed to an abstract idea and discloses a novel application of its multi-sensor system—selectively activating or deactivating a biological sensor based on motion information from a motion sensor—to produce improvements in battery life. JDX-0009C, ¶¶505-15, 584-93; JX-0002, 1:47-65, 19:65-20:29, 31:24-58, 32:28-33, Claims 1-3, 9; *See* Section V.C.

VI. U.S. PATENT NO. 10,642,359

A. The Accused Products Infringe the ’359 Patent⁵

The Accused Products infringe claims 1 and 4 of the ’359 Patent as detailed below. The only limitation Oura contests in claim 1 is 1.f, i.e., of “perform[ing] the specified function using a select portion of the biological signal.” Oura also contests that the Accused Products meet the additional limitations of claim 4.

1. The Accused Products Infringe Claim 1

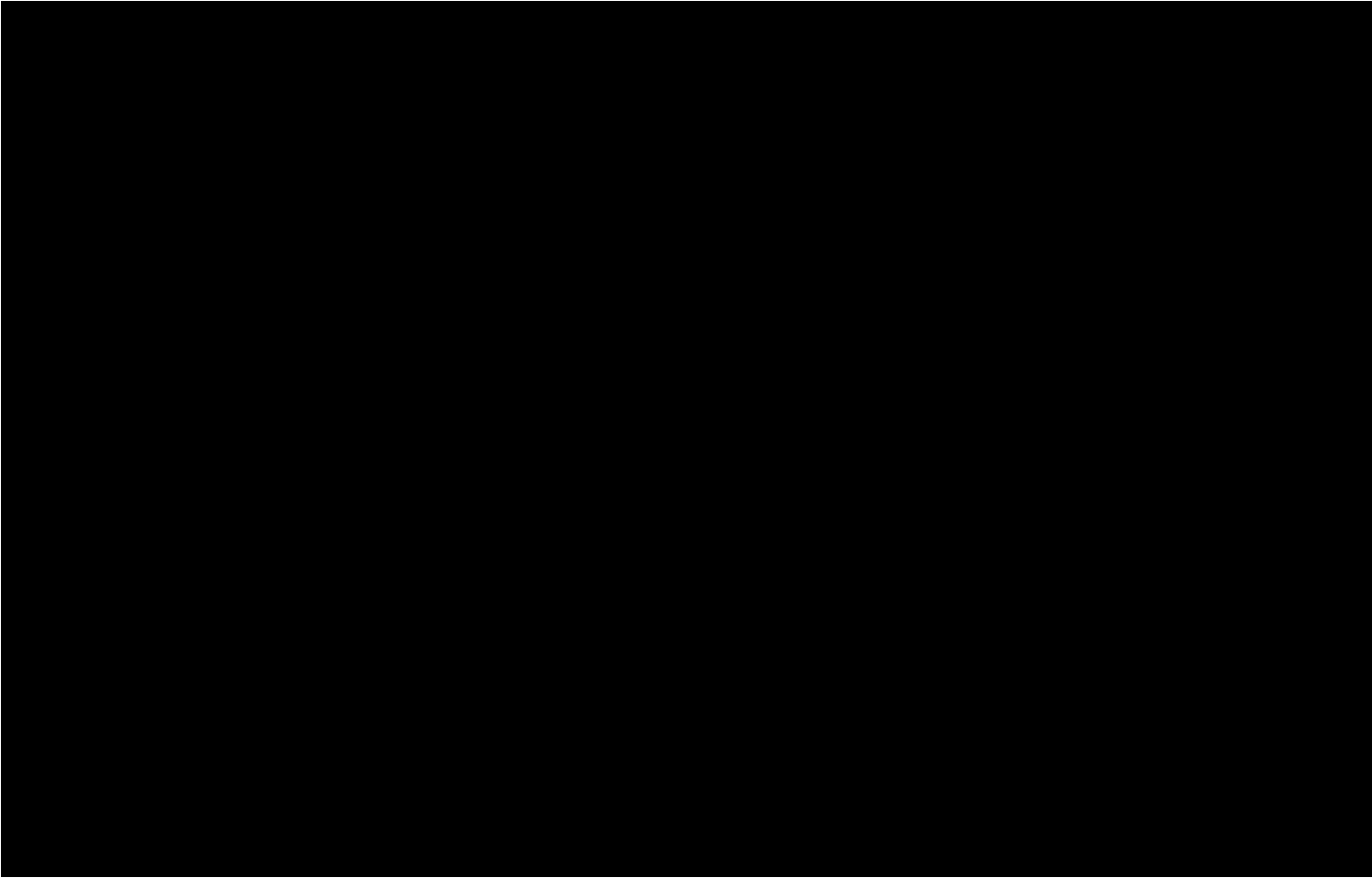
a. Limitation 1.pre (“A wearable device comprising:”)

The Accused Products, i.e., rings, are the claimed “wearable device[s].” JDX-0004C, ¶¶72-77, 202-204.

⁵ The analysis below focuses on Oura Ring 4. But the parties agree that Ring 4 is representative of Ring 5 and [REDACTED] with no material differences. JDX-0007C, ¶¶129-132.

[REDACTED]

[REDACTED]



b. Limitation 1.a (“a motion sensor;”)

The Accused Products have the claimed “motion sensor”—an accelerometer, such as a

[REDACTED], as shown below, which measures motion.

JDX-0004C, ¶¶72-77, 205-207.

[REDACTED]

[REDACTED]

c. Limitation 1.b (“a biological sensor; and”)

The Accused Products include the claimed “biological sensor,” i.e., their SpO2 sensor.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

d. Limitation 1.c (“a controller configured to:”)

Limitation 1.d (“obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor,”)

The Accused Products comprise a “controller,” [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

signal”).

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

f. **Limitation 1.f (“and perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition.”)**

The Accused Products satisfy limitation 1.f, which is the only one Oura contests in claim

1. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶88-111, 222-257; CX-1500C, OURA_1478_SC_0000080-81; *see also* CX-1500C, OURA_1478_SC_0000186, 188-189, 178, 192, 185 [REDACTED]

[REDACTED] JDX-0026C, 237:6-10

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶88-111, 222-257; CX-1500C, OURA_1468_SC_0000168-169 [REDACTED]

[REDACTED]

Oura’s dispute with this limitation focuses on “perform[ing] the specified function using a select *portion* of the biological signal.” JDX-0007C, ¶¶71-86. [REDACTED]

[REDACTED]

[REDACTED] *Id.*, ¶¶76, 82-83. That is, Oura argues that Samsung “provides no evidence that the average” discussed “is obtained by using a subset of data (select portion of the biological signal).” *Id.*, ¶83.

As Dr. Welch explains, the Accused Products use [REDACTED]

[REDACTED]. JDX-0004C, ¶¶88-111, 222-257. More specifically, he shows that the Accused Products use [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]

[REDACTED]

[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] *Id.*; CX-1500C, OURA_1468_SC_0000168-169

[REDACTED] [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *See id.* [REDACTED]

[REDACTED] *See id.*

2. **The Accused Products Infringe Claim 4 (“The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in**

[REDACTED]

response to the magnitude of the biological signal not satisfying a second threshold value.”)

The Accused Products’ [REDACTED] to determine SpO2 levels (i.e., the “biological signal”) to be i [REDACTED]

[REDACTED]. JDX-0004C, ¶¶275-285; CX-1500C, OURA_1478_SC_0000441-442. [REDACTED]

[REDACTED] JDX-0004C, ¶¶275-285; CX-1500C, OURA_1478_SC_0000441-442; CX-1671; CX-1646; CX-1219C, 54:23-55:21. [REDACTED]

[REDACTED] JDX-0004C, ¶¶275-285; CX-1500C, OURA_1478_SC_0000441-442.

[REDACTED]

As Dr. Welch explains, Oura's "[REDACTED]" platform involves an [REDACTED] determination [REDACTED] JDX-0004C, ¶¶275-285; CX-1500C, OURA_1478_SC_0000441-442; CX-1671; CX-1646; CX-1219C, 54:23-55:21.

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶275-285; CX-1500C, OURA_1478_SC_0000441-442. Thus, the Accused Products "determine the biological signal to be invalid ... in response to the magnitude of the biological signal not satisfying a second threshold value," as the claim requires; the claim does not require any further action based on the invalidity determination, such as "choos[ing] to store or not store that data," as Oura argues. *See id.*, JDX-0007C, ¶¶92, 97.

B. TDI Is Met for the '359 Patent

The WDI Products practice claims 1 and 4 of the '359 Patent. As with infringement, Oura only disputes limitation 1.f of claim 1 and claim 4.

1. TDI Is Met for Claim 1

a. Limitation 1.pre ("A wearable device comprising:")

The WDI Products are rings or watches and are, thus, "wearable device[s]." JDX-0004C, ¶¶112-132, 177, 651-654.

Limitation 1.a ("a motion sensor;")

The WDI Products each include a "motion sensor," specifically an accelerometer that measures motion, such as:

- [REDACTED] in Galaxy Watch4, Watch5, Watch6, and Watch7
- [REDACTED] in Watch Ultra;
- [REDACTED] in Watch8;
- [REDACTED] in the Galaxy Ring.

[REDACTED]

JDX-0004C, ¶¶112-132, 655-664; CX-0148C, SAMS382-0012471; CX-0160C, SAMS382-0013172; CX-0179C, SAMS382-0014881; CX-0150C, SAMS382-0012801; CX-0151C, SAMS382-0012809; CX-0152C, SAMS382-0012815; CX-0156C, SAMS382-0013147; CX-0157C, SAMS382-0013153; CX-0158C, SAMS382-0013161; CX-0164C, SAMS382-0013288; CX-0165C, SAMS382-0013296; CX-0159C, SAMS382-0013169; CX-0176C, SAMS382-0013620; CX-0177C, SAMS382-0013626; CX-1640C, SAMS382-0011270; CX-0992C, SAMS406-0074870; CX-1643.

b. Limitation 1.b (“a biological sensor; and”)

The WDI Products each include a “biological sensor” in the form of one or more optical SpO2 sensors that includes red and IR LEDs and corresponding photodetectors: [REDACTED]

[REDACTED]

JDX-0004C, ¶¶112-132, 177-198, 665-681; CX-0261C, SAMS382-0029425-426; CX-1640C, SAMS382-0011274.

c. Limitation 1.c (“a controller configured to:”)

Limitation 1.d (“obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor,”)

The WDI Products each include “a controller”:

- The Galaxy Watch4 and Watch5 include a [REDACTED]
- The Galaxy Watch6 includes a [REDACTED]
- The Galaxy Watch7, Watch Ultra, and Watch8 include a [REDACTED]
- The Galaxy Ring includes an [REDACTED]

JDX-0004C ¶¶112-132, 177-198, 655-687; CX-0129C, SAMS382-0009940; CX-0132C,

[REDACTED]

SAMS382-0010123; CX-0135C, SAMS382-0010444; CX-0138C, SAMS382-0010728; CX-0126C; SAMS382-0009691; CX-0992C, SAMS406-0074870. Each of these controllers is configured to obtain a movement signal of an object using the accelerometer (“motion sensor”) and to obtain a biological signal of the object using the SpO2 sensor (“biological sensor”). JDX-0004C ¶¶112-132, 177-198, 655-687; CX-0142; CX-0227; CX-0237; CX-0232; CX-0226; CX-0247; CX-1685C, SAMS406-SC-0000323-335, 327, 328, 331-332; CX-1499C, SAMS382-SC-00000011-12, 17-18, 28-29, 117, 124, 133, 161, 197, 216, 241, 260-261, 216, 284, 285, 299-300. The diagrams below depict representative controllers (annotated with a blue box) coupled to the accelerometer (annotated with a red box) and SpO2 sensor (annotated with a green box).

[REDACTED]

[REDACTED]

d. Limitation 1.e (“determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal,”)

The WDI Products use data from the accelerometer to determine whether the movement signal indicates that the user is sleeping (i.e., to “determine, based on an analysis of the movement signal”), [REDACTED]

[REDACTED]. JDX-0004C, ¶¶112-132, 158-198, 655-708; CX-0829C; CX-0262C; CX-1499C, SAMS382-SC-0000039, 48, 57-58, 599-600, 758, 761, 776. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶112-132, 158-198, 655-708; CX-0829C; CX-0262C; CX-1499C, SAMS382-SC-0000039, 48, 57-58, 599-600, 758, 761, 776. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶112-132, 158-198, 655-708; CX-0142, SAMS382-0011184; CX-0066,

[REDACTED]
[REDACTED]
SAMS406-0001878; CX-0234; CX-0991C ([REDACTED]); CX-0792C ([REDACTED]);
CX-0260C ([REDACTED]); CX-1499C, SAMS382-SC-0000117 ([REDACTED]
[REDACTED], 124-126 ([REDACTED]
[REDACTED]).

e. Limitation 1.f (“and perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition.”)

When a score from processed accelerometer data is lower than a threshold for determining that the user is asleep, the WDI Products determine that the user is sleeping, activate the SpO2 sensor, and obtain SpO2 levels (i.e., “biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition”). JDX-0004C, ¶¶112-132, 158-198, 655-708; CX-0829C; CX-0262C; CX-1499C, SAMS382-SC-0000039, 48, 57-58, 599-600, 758, 761, 776. [REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

JDX-0004C, ¶¶112-132, 158-198, 655-720; CX-1499C, SAMS382-SC-0000117
[REDACTED]
[REDACTED]

Oura disputes that the WDI Products “perform the specified function using a *select portion* of the biological signal.” JDX-0006C, ¶¶92-106. [REDACTED]
[REDACTED]
[REDACTED]

[REDACTED]

As Dr. Welch explains, the WDI Products use [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] See JDX-0004C, ¶¶ 112-132, 158-198, 655-720; CX-1499C, SAMS382-SC-0000117 [REDACTED]

[REDACTED] The WDI Products [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] See *id.* Thus, the WDI Products [REDACTED]

[REDACTED] (i.e., “perform[ing] the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition”). See *id.*

2. TDI Is Met for Claim 4 (“The wearable device of claim 1, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.”)

The WDI Products’ controller is configured to determine the signal used to determine SpO2 levels (i.e., the “biological signal”) to be invalid, irrespective of the movement signal, when its magnitude does not satisfy a threshold value, such as being within a range indicating an invalid

[REDACTED]

signal and/or not being within a range indicating a valid signal. JDX-0004C, ¶¶729-749; CX-0991C; CX-0260C, SAMS406-0029420; JX-0023, 77:10-20 (confirming that the WDI Products

[REDACTED]), 78:16-79:2, 84:1-4, 81:19-23; JX-0014, 127:22-129:21, 133:6-134:11.

The WDI Products determine whether the signal for measuring SpO2 is invalid by [REDACTED]. JDX-0004C, ¶¶729-749; CX-0991C; CX-0260C, SAMS406-0029420 [REDACTED]

[REDACTED]; JX-0023, 77:10-20, 78:16-79:2 [REDACTED]

[REDACTED], 84:1-4 (same), 81:19-23; JX-0014, 127:22-129:21, 133:6-134:11.

[REDACTED] JX-0014, 133:6-134:11.

The [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶729-749; CX-1499C, SAMS382-SC-0000122 [REDACTED]

[REDACTED]; CX-0991C; CX-0260C, SAMS406-0029420; JX-0023, 77:10-20, 78:16-79:2, 84:1-4, 81:19-23; JX-0014, 127:22-129:21. If the

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0004C, ¶¶729-749; CX-1499C, SAMS382-SC-0000122, 117; CX-0991C; CX-0260C, SAMS406-0029420; JX-0014, 127:22-129:21 [REDACTED]

[REDACTED]

[REDACTED]

Oura argues that the WDI Products do not “determine the biological signal to be invalid ... in response to the magnitude of the biological signal not satisfying a second threshold value.”

JDX-0006C, ¶¶109-114. Oura argues that, in the WDI Products, [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*, ¶¶111-112. Oura further argues that [REDACTED]

[REDACTED] *Id.*, ¶112.

As Dr. Welch explains, the WDI Products perform [REDACTED]

[REDACTED] JDX-0004C, ¶¶729-749; JX-0014, 127:22-129:21, 133:6-134:11. He confirms that the [REDACTED]

[REDACTED], as Oura argues. JDX-0004C, ¶¶729-749; CX-1499C, SAMS382-SC-0000122, 117; CX-0991C; CX-0260C, SAMS406-0029420; JX-0014, 127:22-129:21 ([REDACTED])

[REDACTED], 133:6-134:11. [REDACTED]

[REDACTED]. *Id.* Dr. Welch explains that Samsung’s [REDACTED]

[REDACTED].

[REDACTED]

See JDX-0004C, ¶¶729-749; CX-1499C, SAMS382-SC-0000122, 117; CX-0991C; CX-0260C, SAMS406-0029420 [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JX-0014, 127:22-129:21, 133:6-134:11 [REDACTED]

[REDACTED]

[REDACTED].

C. The Asserted Claims of the '359 Patent Are Valid

Oura incorrectly asserts that the asserted claims are invalid based on the following prior art grounds:

- Ground 1 – Claim 1 as anticipated by U.S. Patent App. Pub. No. 2008/0139952 (“Kuroda II”)
- Ground 2 – Claim 4 as obvious based on Kuroda II in view of U.S. Patent App. Pub. No. 2012/0078066 (“Ahmed066”)
- Ground 3 – Claim 4 as obvious based on Kuroda II in view of Compressed Sensing System Considerations for ECG and EMG Wireless Biosensors (“Dixon”)
- Ground 4 – Claim 1 as anticipated by U.S. Patent No. 7,351,206 (“Suzuki”)

Kuroda II and Suzuki fail to disclose both a specified function using a select portion based on motion-based determination, as required by limitation 1.e, and a movement-defined “select period” with a defined beginning and end, as required by limitation 1.f. Oura’s primary references also fail to disclose the invalidity check required by Claim 4, and Oura’s secondary references cannot cure these shortcomings.

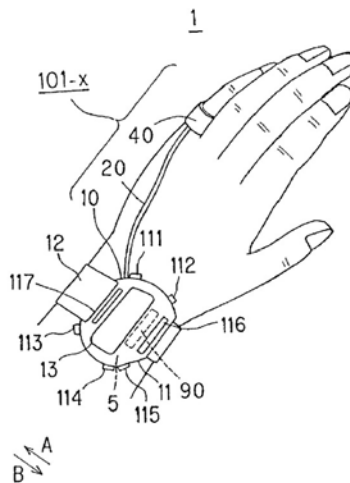
1. Ground 1: Kuroda II Does Not Anticipate Claim 1

a. Limitation 1.d (“obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor”)

Limitation 1.d requires the motion sensor and biological sensor to sense the same object, as required by the antecedent basis of “the object.” JDX-0009C, ¶75.

Dr. Mortazavi maps Kuroda II’s “‘activity signal’ or ‘activity data’ to ... the claimed movement signal obtained using ‘activity sensor 302,’ and ... the ‘pulse wave signal,’ ‘heart rate HR,’ or ‘pulse data’ to ... the claimed biological signal obtained using ‘pulse sensor 30,’” claiming that “a POSITA would understand the ‘activity sensor 302’ to correspond to the claimed motion sensor, and the ‘pulse sensor 30’ to correspond to the claimed biological sensor.” JDX-0001C, ¶141.

However, Dr. Welch notes that Kuroda II does not disclose sensing *the same object* with the motion sensor and the biological sensor. The activity sensor of Kuroda II (e.g., an acceleration detector) is housed in the watch case worn on the wrist, whereas the pulse sensor is worn on the finger and connects to the main unit via a cable, as illustrated in Kuroda II, Figure 1 reproduced below. JDX-0009C, ¶76; RX-0003, [0056]-[0059], [0064], [0080], Fig. 1.

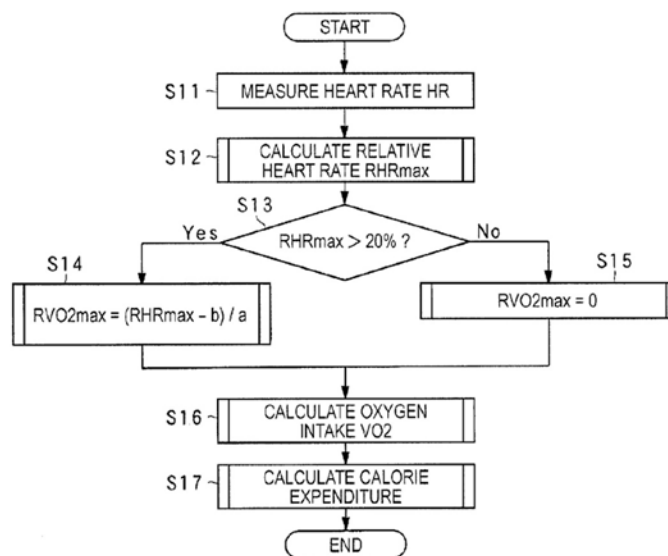


Kuroda II, Fig. 1

Dr. Welch explains that the wrist and the finger are different objects with distinct dynamics; they do not move in unison and they can exhibit different motion profiles during activities such as walking, running, or gesturing. JDX-0009C, ¶76 Dr. Mortazavi agrees with Dr. Welch: “Q: A user’s finger can move without the wrist moving, right? A: It can.” JDX-0051C, 239:11-13. The ’359 Patent consistently refers to “a target” as a single body part to which the wearable device is attached. *Id.*; JX-0001, 5:26-27.

b. Limitation 1.e (“determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal”)

Limitation 1.e requires a binary decision—whether to perform or not perform a specified function—gated by the movement signal analysis. Dr. Mortazavi identifies Kuroda II’s “activity signal” as the “movement signal” and identifies “the ‘pulse wave signal,’ ‘heart rate HR,’ or ‘pulse data’” as the “biological signal.” JDX-0001C, ¶140. Dr. Mortazavi further identifies “the determination to calculate heart rate, RVO2max, VO2, and/or calorie expenditure” as the specified function, but he does not identify which single operation’s performance is determined “based on an analysis of the movement signal,” nor does he explain how the movement-signal analysis controls that decision. *Id.*, ¶147; JDX-0009C, ¶78. Dr. Mortazavi also points to the “RHRmax>20%?” step, Step S13, of Kuroda II Figure 8 (below) to argue that “a POSITA would also understand that the comparison of the activity data to a prescribed threshold corresponds to the claimed ‘analysis of the movement signal.’” JDX-0001C, ¶152.



Kuroda II, Fig. 8

However, as Dr. Welch concludes, the “RHRmax > 20%” decision is not a check of movement but an anaerobic threshold determination, classifying the user’s physiological state as either “aerobic or anaerobic.” JDX-0009C, ¶78; RX-0003, [0175]-[0176]. This comparison is based on *relative HR*, a value derived from the purported “biological signal” rather than the “movement signal.” Further, it occurs *after* HR has already been determined, which is a fundamentally different operation from the movement signal-based binary decision that limitation 1.e requires. JDX-0009C, ¶78.

Kuroda II also fails to disclose the step of determining “whether to perform a specified function” using a “select portion” of the biological signal based on an analysis of the movement signal. JDX-0009C, ¶79. Limitation 1.e requires a conditional determination and the analysis of the movement signal must be the basis for deciding whether the specified function will be performed using the biological signal. *Id.* Kuroda II does not disclose whether to perform a specified function; instead, it discloses using different operations to calculate calorie expenditure based on an analysis of the movement signal, such as when the subject is exercising or resting. *Id.*

[REDACTED]

In Kuroda II, the calorie expenditure calculation is performed both “when the subject is exercising and when the subject is resting.” RX-0003, Fig. 8, [0031], [0034]. Kuroda II therefore does not disclose determining whether or not to perform the “specified function” identified by Dr. Mortazavi using a “select portion” of the biological signal.

Additionally, Kuroda II’s entire process flow, which is directed to calculating calorie expenditure from the user’s HR and oxygen intake, does not determine *whether to perform* a “specified function.” JDX-0009C, ¶80. Rather, Kuroda II’s activity-component threshold check determines whether the MPU 308 *can* extract a usable HR value. *Id.*; RX-0003, [0136]-[0139]. Thus, Kuroda II never makes an affirmative decision to perform or not perform a specified function of the wearable device and, therefore, does not satisfy limitation 1.e. JDX-0009C, ¶80.

c. Limitation 1.f (“perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition”)

Kuroda II does not meet limitation 1.f because it does not disclose (1) a “select period, of the movement signal,” i.e., a period of time with a defined beginning and ending, and (2) the use of a “select portion of the biological signal.” Apart from Dr. Mortazavi’s own usage of the term “the determined period” to describe “when activity data is less than or equal to a prescribed threshold value” in Kuroda II, Dr. Mortazavi fails to identify an actual “select period” of time in Kuroda II with a defined beginning and ending. JDX-0001C, ¶157. Dr. Mortazavi also broadly maps Kuroda II’s calculation of “heart rate HR, relative heart rate RHRmax, RVO2max, VO2, and/or calorie expenditure by MPU308” using “pulse data during the period that the activity data is less than or equal to a prescribed threshold value” to “perform[ing] the specified function using a select portion of the biological signal” required by limitation 1.f. *Id.*, ¶155.

However, as Dr. Welch explains, Kuroda II fails to disclose “perform[ing] the specified

[REDACTED]

function using a select portion of the biological signal obtained during a select period, of the movement signal.” JDX-0009C, ¶¶82-85. Apart from Dr. Mortazavi qualitatively describing “when activity data is less than or equal to a prescribed threshold value” as a “select period,” neither Kuroda II nor Dr. Mortazavi define temporal boundaries for any such period. JDX-0001C, ¶157; JDX-0009C, ¶¶81, 85. On the contrary, Kuroda II discloses a continuous fast Fourier transformation (FFT) processing of a continuous stream of acceleration data through sequential FFT operations. JDX-0009C, ¶¶84-85; RX-0003, [0134]-[0136]. There is no concept of a bounded “period.” Rather, the signal processing is ongoing and does not identify distinct temporal intervals. JDX-0009C, ¶84.

Nor does Kuroda II identify a “select portion” of the biological signal corresponding to such a period. *Id.*, ¶85. Without a “select period” of the movement signal determined by the system that has a defined beginning and a defined end, there can be no corresponding “select portion” of the biological signal obtained during that period. *Id.* The select-period/select-portion architecture of limitation 1.f requires both elements to be present, but Kuroda II has neither. *Id.*

2. Ground 2: Kuroda II In View Of Ahmed066 Does Not Render Obvious Claim 4

Oura’s position that Kuroda II combined with Ahmed066 renders Claim 4 obvious similarly fails. As explained below, (a) a POSITA would not have been motivated to combine Kuroda II and Ahmed066, and (b) even if combined, Ahmed066’s disclosure of a bandpass filter and peak-detection process is not the equivalent of the claimed invalidity determination that Kuroda II lacks. JDX-0009C, ¶¶118-131.

a. No Motivation to Combine

As Dr. Welch explains, Kuroda II is directed to a wrist- and finger-worn combination device that calculates calorie expenditure using HR zones during physical exercise, whereas

Ahmed066 is directed to an ear cuff that addresses signal noise reduction in optical HR monitoring systems. JDX-0009C, ¶118; RX-0003, Abstract; RX-0008, [0002]-[0003]. In addition to their different form factors, combining Kuroda II and Ahmed066 would have been impractical, such as due to the references' redundancy, and would not have yielded the claimed invention. JDX-0009C, ¶118. Kuroda II's motion-related processing is limited to using an activity sensor and FFT to subtract the activity component from the pulse signal so that a clean HR can be extracted for calorie computation; it does not use the motion signal to determine the validity of a biological signal. *Id.*; RX-0003, [0134]-[0136]. Motion sensing by Ahmed066's pressure sensor, on the other hand, is used to reduce noise in the HR signal obtained from the light sensor. JDX-0009C, ¶118; RX-0008, [0023]. A POSITA would not have been motivated to combine a wrist- and finger-worn combination device with an ear cuff that does virtually the same thing—to do so would be redundant.

b. Claim 4 (“determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value”)

Claim 4 requires a determination of whether the biological signal is invalid based on “the magnitude of the biological signal not satisfying a second threshold value,” “irrespective of the movement signal.” JDX-0009C ¶100. Dr. Mortazavi maps Kuroda II's “RHRmax > 20%” decision and Ahmed066's bandpass filtering and peak detection to the invalidity determination required by Claim 4. *Id.*, ¶¶101, 107.

However, even if combined, Kuroda II and Ahmed066 do not teach or disclose Claim 4. *Id.*, ¶¶117-31. As explained above in Section VI.C.1.b, Kuroda II's “RHRmax > 20%” decision is not a determination of whether the signal is valid or invalid—it is an anaerobic threshold determination. *Id.*, ¶101; RX-0003, [0175]-[0176]. Dr. Mortazavi's citation of Kuroda II at

[REDACTED]

[0137]-[0142] is similarly unpersuasive; the thresholding in Kuroda II that Dr. Mortazavi cites is of the activity component (i.e., the movement signal) rather than the HR signal (i.e., the biological signal). JDX-0009C, ¶102; JDX-0001C, ¶388. In contrast, Claim 4 requires a validity determination “irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.” *Id.*; JX-0001, 14:26-30.

As Dr. Welch explains, Ahmed066 does not cure the deficiencies of Kuroda II. JDX-0009C, ¶¶122-126. Ahmed066 does not teach an invalidity determination when “the magnitude of the biological signal [does] not satisfy[] a second threshold value,” as required by Claim 4. *Id.*, ¶124. Ahmed066 discloses only a bandpass filter that processes all incoming signals equally without determining the validity of the biological signal. *Id.*; RX-0008, [0026]. The subsequent peak-detection process similarly does not label peaks as unreliable or invalid. JDX-0009C, ¶¶124-25; RX-0008, [0028]-[0030]. The peak detection disclosed in Ahmed066 merely determines the number of peaks counted by a downstream counter—it does not “determine the biological signal to be invalid.” JDX-0009C, ¶125. Without an invalidity determination, Ahmed066 cannot disclose Claim 4.

Because neither reference discloses an invalidity determination based on the magnitude of the biological signal, the combination of Kuroda II and Ahmed066 fails to render Claim 4 obvious. JDX-0009C, ¶¶127-31. In addition, the combination fails to disclose the limitations of the independent claim not disclosed by Kuroda II individually (see Section VI.C.1) and not addressed by Oura’s proposed combination.

3. Ground 3: Kuroda II In View Of Dixon Does Not Render Obvious Claim 4

Oura’s third ground, the combination of Kuroda II and Dixon, also fails because (a) a POSITA would not have been motivated to combine Kuroda II and Dixon, and (b) even if

combined, neither reference discloses or suggests a validity determination based on the magnitude of the biological signal as Claim 4 requires. JDX-0009C, ¶¶97-116.

a. No Motivation to Combine

Dr. Mortazavi first argues that a POSITA would have been motivated to combine Dixon with Kuroda II because of their “technical similarities.” JDX-0001C, ¶326; JDX-0009C, ¶98. However, Kuroda II is directed to exercise physiology, specifically, measuring calorie expenditure using HR zones during physical exercise. JDX-0009C, ¶98. Dixon, by contrast, is directed to reducing data rates in wireless biosensor networks through compressed sensing techniques—it has nothing to do with movement sensing. *Id.* These are unrelated fields with different goals and architectures: Kuroda II seeks to quantify energy expenditure during exercise, while Dixon seeks to minimize power consumption in a wireless sensor node by reducing the amount of data that must be transmitted. *Id.*

Dr. Mortazavi’s fallback theory—that both references disclose “performance of a specified function using the biological [signal] based on the motion sensor detecting movement in reference to a specified condition”—conflates Dixon’s “activity detection” with detection of physical movement. JDX-0009C, ¶99 (quoting JDX-0001C ¶327). In his deposition, Dr. Mortazavi admitted that Dixon’s “activity” refers to electrocardiogram (ECG) activity within the biosignal waveform itself, not to physical movement measured by a motion sensor. JDX-0051C, 258:2-21; JDX-0009C, ¶99; RX-0006, 156, 159. The language of Dixon confirms this, as Dixon clearly uses “activity” exclusively in the context of ECG signals, for example, stating that “a typical ECG signal” includes “a high activity QRS complex followed by a low activity region between complexes.” RX-0006, 156; JDX-0009C, ¶107. Dixon has no use for, and does not disclose, a motion sensor at all. JDX-0009C, ¶99. It would have made little sense to a POSITA to combine

[REDACTED]

Dixon, which does not detect physical movement, with Kuroda II's motion sensor.

b. Claim 4 (“determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value”)

Even combined, Dixon does not cure the deficiencies of Kuroda II. JDX-0009C, ¶¶104-16. Dr. Mortazavi relies on Dixon's “dynamic thresholding of an EMG/ECG signal” to argue that Dixon discloses Claim 4's invalidity determination. JDX-0001C, ¶¶340-342. However, as Dixon itself explains, its dynamic thresholding is a sparsity and compression technique, RX-0006, 157, 160, not a means of determining whether a biological signal is invalid. JDX-0009C, ¶108. Dixon deliberately sets small samples to zero to make the biosignal more sparse so that compressed sensing can reduce data rate and energy consumption; it does not discard those samples or otherwise determine that those samples are physiologically invalid or unreliable for a downstream function. *Id.*, ¶¶106-10.

Thus, even if Dixon were combined with Kuroda II, the combination would still not disclose or suggest Claim 4 given their respective failures to disclose an invalidity determination and Dixon's inability to remedy Kuroda II's gaps. *Id.*, ¶115. Dixon makes clear that its purpose is to reduce data rate and transmitter duty cycle through compressed sensing, not to determine whether a biological signal can be relied upon for a specified function. RX-0006, Abstract; JDX-0009C, ¶110. Its dynamic thresholding is, therefore, merely a mathematical transformation applied to the biosignal to increase sparsity, not a validity determination regarding whether the biological signal can be relied upon for the relevant purpose. RX-0006, 157, 160; JDX-0009C, ¶110. In addition, the combination fails to disclose the limitations of the independent claim not disclosed by Kuroda II individually (see Section VI.C.1) and not addressed by Oura's proposed combination.

4. Ground 4: Suzuki Does Not Anticipate Claim 1

a. Limitation 1.e (“determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal”)

Limitation 1.e requires a binary decision—whether to perform or not perform a specified function—gated by the movement-signal analysis. Dr. Mortazavi maps Suzuki’s “determination of the body movement data as above or equal to/less than a predetermined threshold” to the “analysis of the movement signal” that allegedly determines “whether to perform a specified function.” JDX-0001C, ¶462.

However, as Dr. Welch explains, Suzuki does not disclose a motion-based determination of “whether to perform a specified function of the wearable device using the biological signal;” it merely filters which data to keep for a processing step that always occurs. JDX-0009C, ¶135; RX-0005, Abstract, 4:51-5:12, 6:12-53, 10:57-11:09, Fig. 11. Specifically, Suzuki’s “body movement determining unit” checks whether the body movement data exceeds a threshold for large movement. JDX-0009C, ¶135. If such movement occurs during sleep, the system flags the concurrently measured pulse interval data as a “waveform abnormality” and removes it. *Id.* The remaining pulse interval data is then processed, and sleep state is determined. *Id.* Thus, Suzuki’s threshold determination of body movement data excludes data from *inevitable* sleep-state processing that always occurs--it is not a “determin[ation], based on an analysis of the movement signal, *whether to perform* a specified function of the wearable device using the biological signal,” as required by limitation 1.e. JDX-0009C, ¶135.

b. Limitation 1.f (“perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition”)

Limitation 1.f requires the use of “a select portion of the biological signal” to perform the

[REDACTED]

“specified function,” and a “select period,” i.e., a period of time with a defined beginning and ending. JDX-0009C ¶133. Dr. Mortazavi’s mapping of these limitations to Suzuki rests on two alternate identifications of the claimed “specified function”: (1) “at least the detection and removal of abnormal pulse wave data where a sleep state is detected,” or (2) “the subsequent determination of the sleep state of the subject using pulse interval data.” JDX-0001C, ¶462; JDX-0009C, ¶136. Suzuki fails under either theory. JDX-0009C, ¶¶133-40.

Under Oura’s first alternate theory, if the alleged “specified function” is “the detection and removal of abnormal pulse wave data where a sleep state is detected,” Suzuki does not disclose performing that function “using a select portion of the biological signal obtained during a select period,” as limitation 1.f requires. JDX-0009C, ¶137. As explained above, Suzuki discloses a system that detects and removes abnormal pulse interval data. *See supra* VI.C.4.a; JDX-0009C, ¶137; RX-0005, 6:12-53, 10:47-51. This is motion-based data exclusion followed by downstream sleep-state processing, not a movement-based decision on *whether* to perform a “specified function using a select portion of the biological signal.” JDX-0009C, ¶137. Furthermore, Suzuki’s “removal” of “pulse interval data detected as the waveform abnormality” is effected by fitting a high-order polynomial curve through all of the raw data and generating equidistant pulse interval data for subsequent processing. *Id.*; RX-0005, 10:47-51. Thus, Suzuki’s “detection and removal of abnormal pulse wave data where a sleep state is detected” is performed using *all* of the biological signal obtained during the alleged select period, not just a “select portion.” JDX-0009C, ¶137.

Under Oura’s second alternate theory, if the alleged “specified function” is instead the subsequent determination of sleep state using pulse-interval data, Suzuki still fails to disclose limitation 1.f for the same reason explained above for limitation 1.e. The sleep-state determination

[REDACTED]

occurs *regardless* of movement analysis, and movement affects only which pulse interval data are removed, but it does not determine *whether* the sleep-state function is performed. JDX-0009C ¶¶138-39; RX-0005, Abstract, 4:51-5:12, 6:12-53, 9:42-10:7, 10:57-11:09. That is, Suzuki’s sleep-state determination at step S1119 is triggered by powering on the device at step S1101, not by the movement-based determination at step S1107, and both the “Yes” and “No” branches of step S1107 lead to step S1119. JDX-0009C, ¶139; RX-0005, Fig. 11.

Additionally, Suzuki does not disclose the “select period, of the movement signal, determined to correspond to a specified condition” that limitation 1.f requires. JDX-0009C ¶¶142-44. The alleged “select period” Dr. Mortazavi points to in Suzuki is a continuously-updated one-minute statistical analysis window used to count body-movement occurrence frequency; it is not a “select period, of the movement signal” during which the biological signal is selected for use in performing the specified function. JDX-0009C, ¶¶142-43; JDX-0001C, ¶462; RX-0005, 9:42-61, 4:51-5:12, Fig. 11. Further, as Dr. Welch explains, this one-minute window is chosen arbitrarily and could be replaced with another timeframe; the window does not align with the subject’s moving-but-asleep or moving-and-awakening state. JDX-0009C, ¶143; RX-0005, 9:42-61.

D. The Asserted Claims Are Directed to Patent Eligible Subject Matter Under 35 U.S.C. § 101

Oura contends that the asserted claims are directed to the abstract idea of “performing a function based on an amount of movement” and that of the ’359 Patent are directed to well-understood, routine and conventional technology based on disclosures similar to its invalidity grounds. JDX-0043, 91; JDX-0001C, ¶¶1281-1288. These arguments fail for the similar reasons as discussed for the ’675 Patent. *See* Section IV.D. The ’359 Patent is not directed to an abstract idea and discloses a novel application of its multi-sensor system—cross validating biological information taken using the biological sensor with motion information taken using the motion

[REDACTED]

sensor—to improving the accuracy of gesture recognition and the user experience. JDX-0009C, ¶¶505-15, 572-83; JX-0001, 1:34-45, 4:20-25, 7:34-37, Claims 1, 4; JX-0039, 112:2-14; *see generally* CX-1365; *see* Section VI.C. For example, Claim 1 recites “based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal.” The ’359 Patent clarifies, and Oura admits, that the purpose of this cross-validation is to improve gesture recognition because prior systems that only used one sensor for gesture recognition were prone to error. *Id.* Thus, the cross-validation provided by the asserted claims of the ’359 Patent address a long-felt but unmet need in pre-existing sensor systems. *See* Section VIII.A.1.

VII. U.S. PATENT NO. 10,978,789

A. The Accused Products Infringe the ’789 Patent

As detailed below on a claim element-by-element basis, the Accused Products infringe Claims 7 and 9 of the ’789 Patent. The analysis below focuses on Ring 4, which is representative of Ring 5 and [REDACTED]⁶ [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Independent Claim 7 recites over 15 claim elements, from [7.pre] to [7.c][vi][e]. Of these elements, Oura and its expert Dr. Mortazavi dispute only two elements for Ring 4, [7.c][v][c] and

⁶ As explained in Samsung’s Opposition to Oura’s MSD Non-Infringement by Oura’s Alternative Products, Oura’s noninfringement theories specific to Ring 5 and [REDACTED] were never disclosed during fact discovery, and instead were raised for the first time in Dr. Mortazavi’s rebuttal report. EDIS ID 889193. It is Samsung’s position that those untimely theories should be excluded, and neither product should be adjudicated. *Id.* Samsung addresses those theories here, to the extent the ALJ decides to consider them.

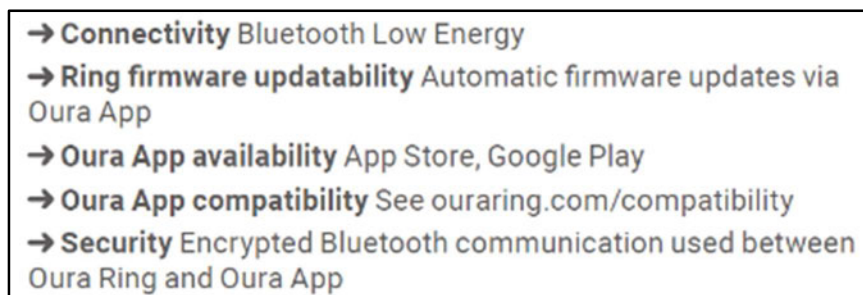
[7.c][vi][c], on the same flawed basis: that Samsung’s expert, Dr. Warren, improperly grouped together and mapped multiple insulating materials to a single “insulating layer” in the claim. JDX-0007C ¶¶597-600, 617-620. For dependent Claim 9, Oura raises the same objection with grouping multiple insulating materials and calling it a single “insulating layer.” *Id.* ¶¶639-641. For every other limitation of Claims 7 and 9, Dr. Mortazavi offers no noninfringement opinion whatsoever; his discussion of those limitations is confined to an “improper practicing the prior art defense” by comparing Ring 4 to Ring 1. *See id.* ¶¶559-596, 601-616, 621-629. As discussed for the ’675 Patent, the Federal Circuit and ALJ have rejected “practicing the prior art” as a defense to infringement.

1. The Accused Products Satisfy Each Limitation of Claim 7

a. Limitation [7.pre] (“A portable communication device comprising:”)

The Accused Products are portable communication devices in the form of finger-worn wearable devices that include a Bluetooth Low Energy (BLE) transceiver for communicating with a paired device, such as a smartphone running the Oura mobile application. JDX-0003C ¶¶76-78.

For instance, the Oura Ring 4 User Manual states that “[e]ncrypted Bluetooth communication [is] used between Oura Ring and Oura App” for “[a]utomatic firmware updates”:



CX-0219, SAMS382-0027564. Bluetooth communication also occurs “when the ring syncs with the app,” as explained by the Oura blog post, “Explore the Technology in Oura Ring 4”:

Bluetooth Connectivity

The bluetooth technology in Oura Ring is only active during a small portion (well below 1%) of each day. Data is transmitted more continuously when the ring syncs with the app and during firmware updates. [Airplane mode](#) disables all radio transmission from Oura Ring.

The limit for SAR (Specific Absorption Rate) for a device such as Oura Ring is 2.0 W/kg for head and body, and Oura's SAR level is 0.0003 W/kg. In other words, the electromagnetic field (EMF) levels of Oura Ring are about *3,000 times* lower than a typical smartphone.

CX-0224, SAMS382-0027621; *see also* CX-0241, SAMS382-0027804; CX-0994C, OURA1468_00029593; JDX-0025C, 71:13-72:1; JDX-0026C, 164:8-24.

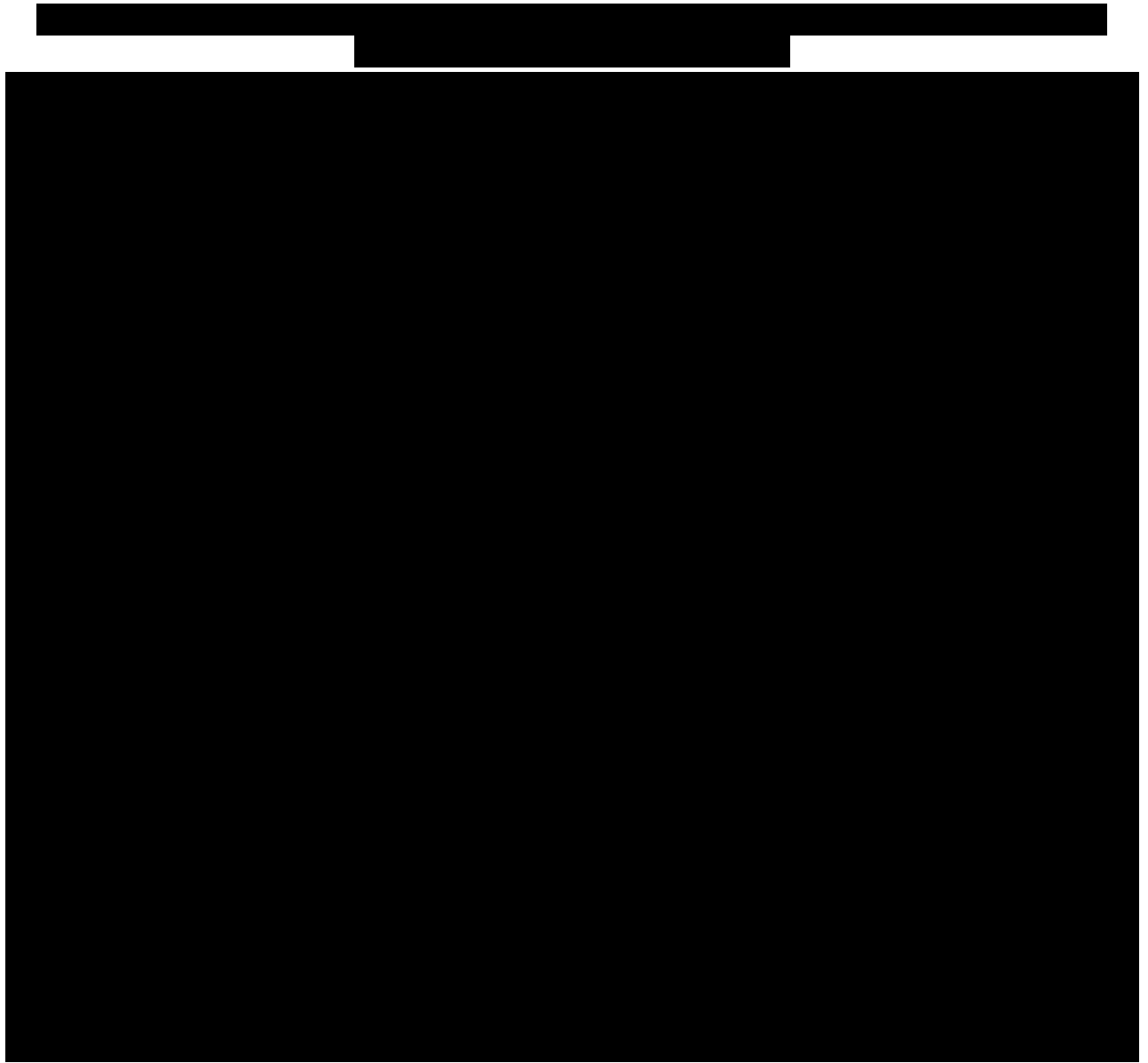
b. Limitations [7.a] (“a first electronic component;”) and [7.b] (“a second electronic component; and”)

The Accused Products include the claimed “first” and “second” electronic components. JDX-0003C ¶¶79-102. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



[REDACTED]

CX-1361C [REDACTED] [REDACTED] *See also* CX-0994C, OURA1468_00029597, 29605, 29608; CX-1355C, p.44; CX-0219, SAMS382-0027564; CX-0243, SAMS382-0027818; CX-1362C; CX-1354; CX-0994C, OURA1468_00029593, 29599, 29607, 29610, 29612-13; CX-0246, SAMS382-0027836-37; JDX-0025C, 68:24-70:25.

Any pair of these components satisfies the “first electronic component” and “second electronic component” limitations. JDX-0003C ¶¶79-102.

c. Limitation [7.c] (“a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:”)

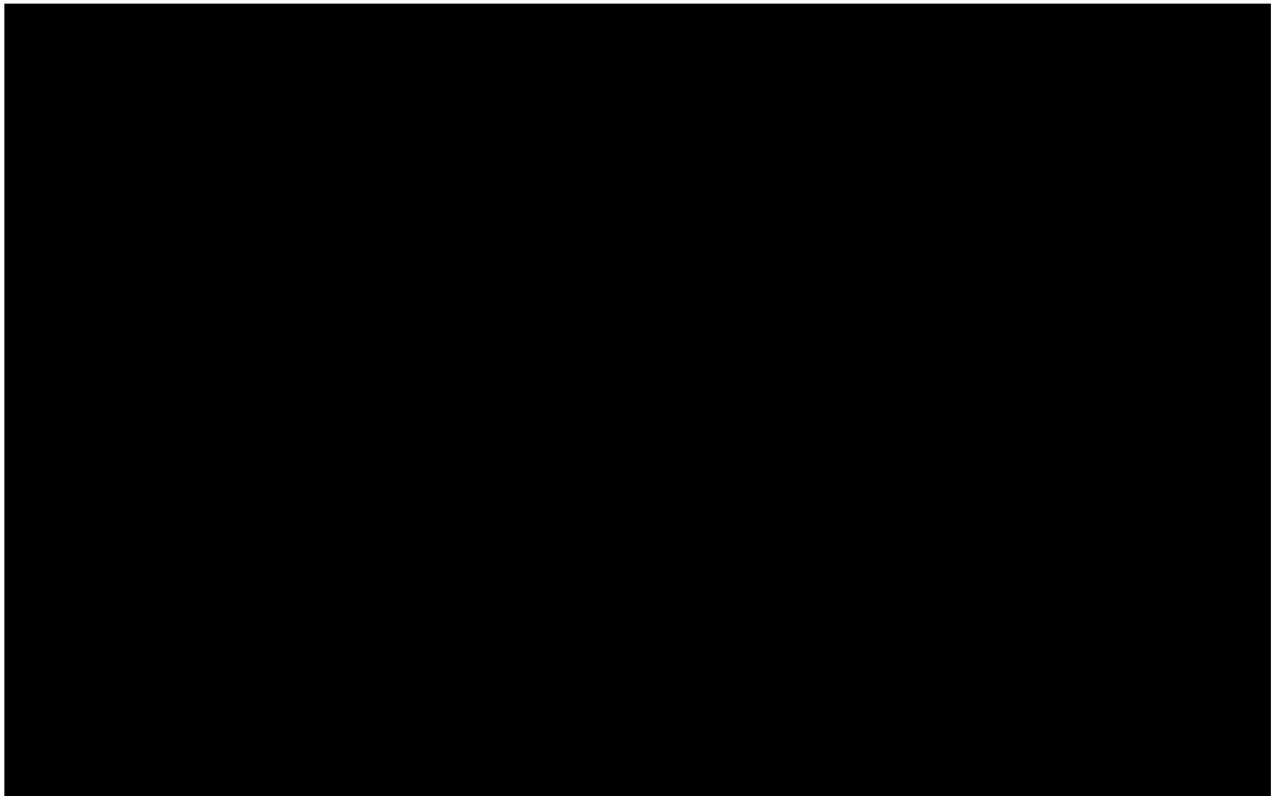
The Accused Products include a PCB that is electrically connected with the first and second electronic components. JDX-0003C ¶¶103-109. The PCB is illustrated in the below exploded

⁷ For purposes of the internal PCB structure, [REDACTED]. *Compare* CX-1357C (Ring 4 Ceramic PCB stack-up) *with* CX-1359C (Ring 4 PCB stack-up).

[REDACTED]

[REDACTED]

[REDACTED]



CX-0994C, OURA1468_00029616; *see also* CX-1355C, p.44.

As the CAD images below illustrate, the PCB in each Accused Product serves as the

[REDACTED]:

[REDACTED]

[REDACTED]



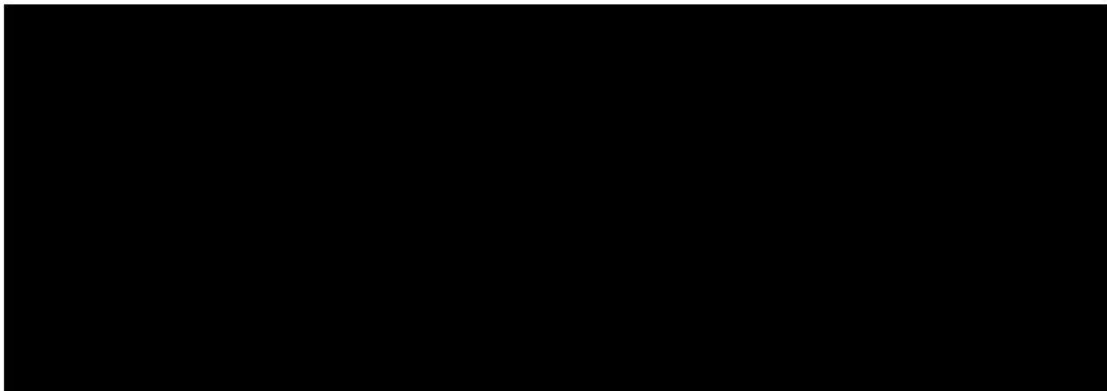
CX-1361C ([REDACTED]); *see also* CX-0246, SAMS382-0027836-37; CX-1502; CX-1362C.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED], are illustrated in Oura's schematic of the PCB, below:



CX-1358C; *see also* JDX-0025C, 72:7-22.

[REDACTED]

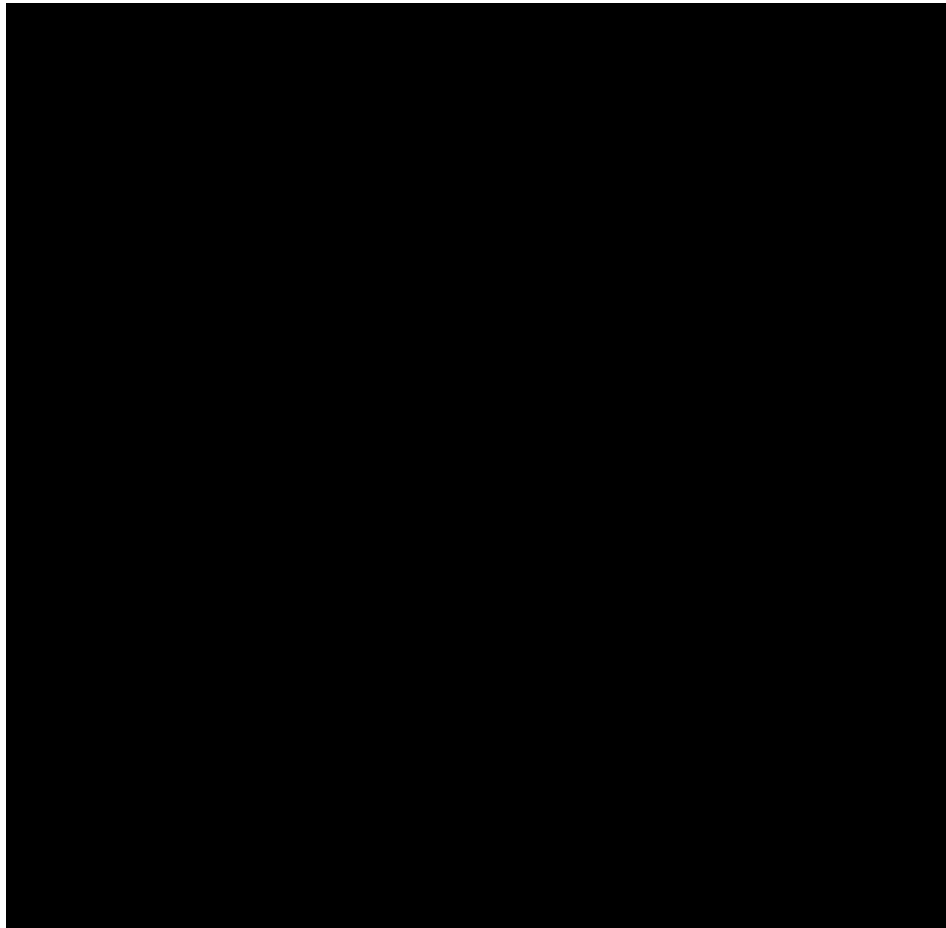
The PCB also provides the electrical interconnections between the ring's electronic components. [REDACTED]

[REDACTED] JDX-0003C ¶112. As Dr. Warren explained, [REDACTED]

[REDACTED]

[REDACTED] *Id.* Oura's corporate representative, Mr. Christophy, confirmed this understanding. JDX-0025C, 81:14-19 [REDACTED]

[REDACTED]. The [REDACTED] for Ring 4 is reproduced below:



CX-1360C, OURA1468_00219859 (annotated); *see also* CX-1359C; CX-1357C; JDX-0025C, 93:19-94:2, 96:23-17, 99:6-100:20; JDX-0003C ¶107.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

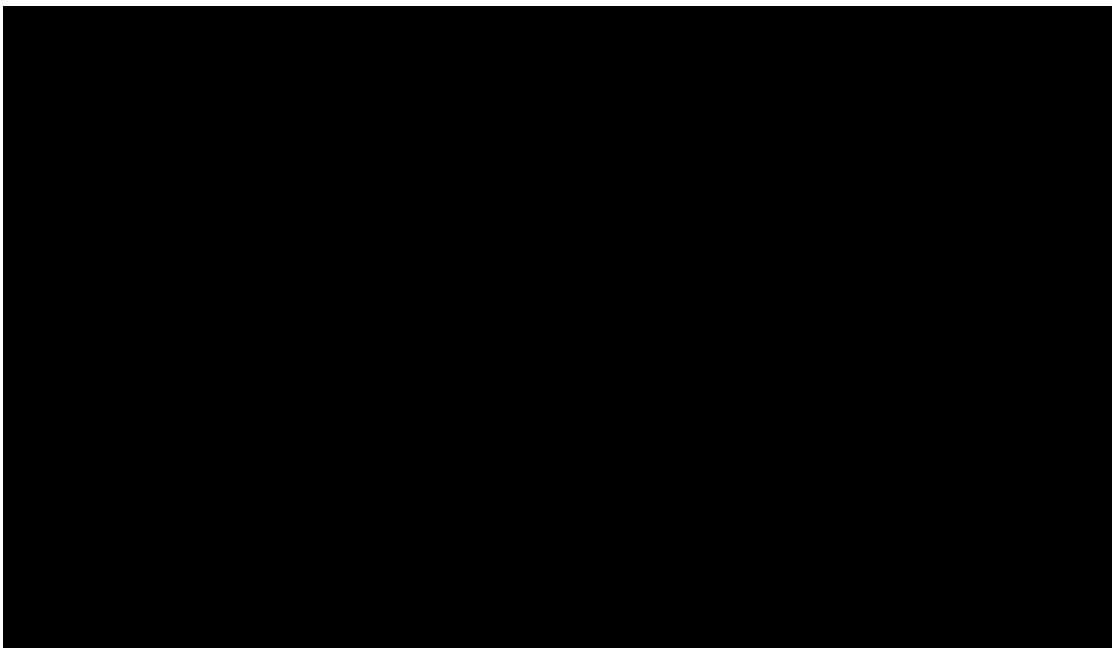
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]⁸ JDX-0003C ¶¶107-108; JDX-0025C, 116:19-25. [REDACTED]

[REDACTED]. The specific electrical connections required can be seen, for example, in the below circuit diagram:



⁸ [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0025C, 82:14-19, 110:10-13. [REDACTED]

[REDACTED]

[REDACTED] Id., 160:7-12.

[REDACTED]

CX-0994C, OURA1468_00029596 (annotated); *see also* [[OURA1468_00096107]], JDX-0025C, 70:14-25.

- d. **Limitations [7.c][i]-[iv] (“a first rigid PCB portion; a first flexible PCB portion in contact with the first rigid PCB portion; a second rigid PCB portion in contact with the first flexible PCB portion; and a second flexible PCB portion in contact with the second rigid PCB portion”)**

The PCB in the Accused Products includes the alternating rigid-flex-rigid-flex PCB structure required by Claim 7. JDX-0003C ¶¶110-135.

[REDACTED]

[REDACTED]

[REDACTED]:

[REDACTED]

[REDACTED]

0003C ¶¶110-111, 113-115, 121-122, 127-128, 131-132; JDX-0025C, 82:10-15, 83:21-25, 95:7-96:4.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-1360C, OURA1468_00219859 (annotated); *see also* JDX-0003C ¶112 [REDACTED]

[REDACTED]

[REDACTED]; JDX-0025C, 81:14-19 (similar).

[REDACTED]

[REDACTED] JDX-0025C, 95:7-96:4. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

e. ***Structural Requirements of the Rigid PCB Portions —***
Limitation [7.c][v] (“wherein each of the first rigid PCB
portion and the second rigid PCB portion includes”¹⁰

- (1) **Limitation [7.c][v][a] (“a first insulating layer including**
a first material and forming a first exterior surface of
each of the first rigid PCB portion and the second rigid
PCB portion;”)

Each of the rigid portions of the Accused Products’ PCBs—including the “first rigid PCB portion” and the “second rigid PCB portion”—includes a first insulating layer of a first material forming a first exterior surface. JDX-0003C ¶¶136-141. [REDACTED]

[REDACTED]

[REDACTED]

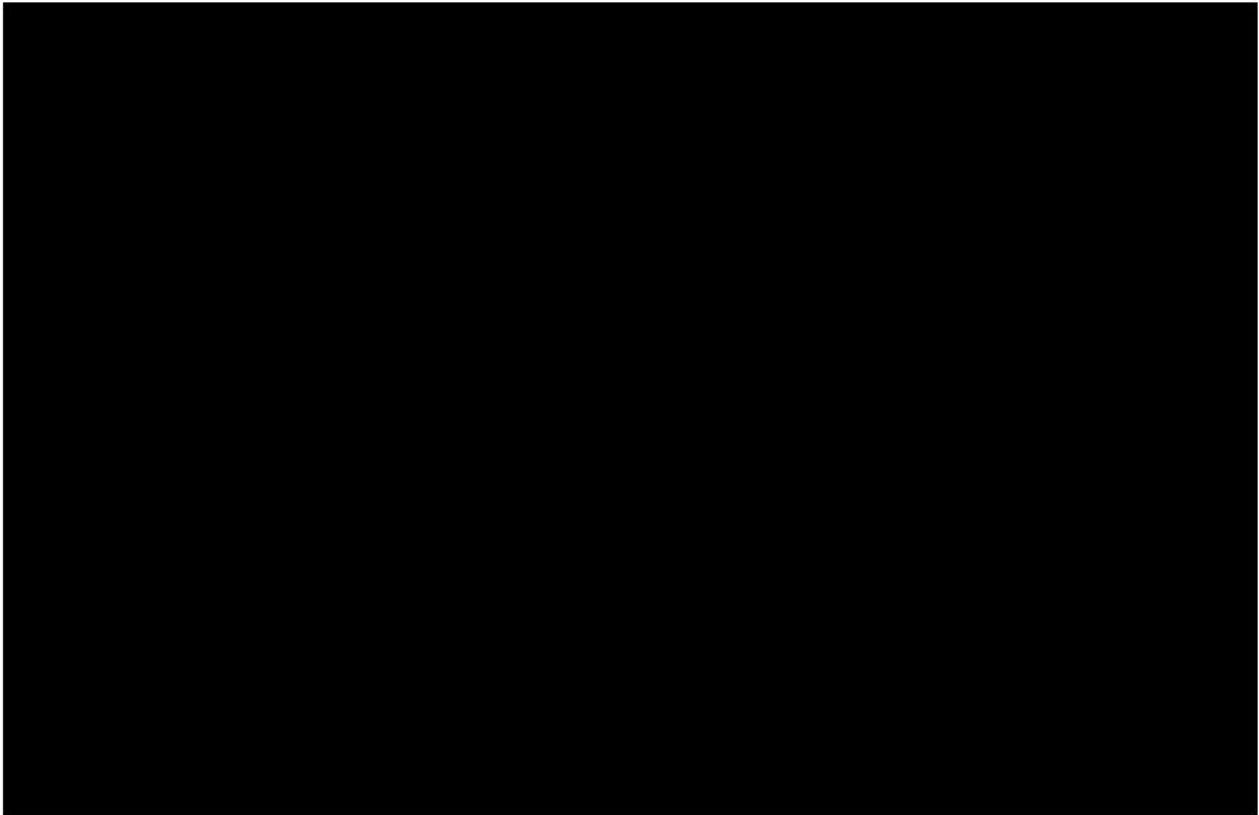
[REDACTED]

Specifically-Identified “First Rigid PCB Portion”:

¹⁰ [REDACTED]
[REDACTED] The Accused Products also meet Limitations [7.c][v][a]-[e] under [REDACTED], as Dr. Warren explains in his report. JDX-0003C ¶¶135-211.

[REDACTED]

[REDACTED]



CX-0349C (annotated).



CX-0331C (annotated).

Specifically-Identified “Second Rigid PCB Portion”:

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-1360C, OURA1468_00219859 (annotated).

[REDACTED]

Id. (annotated). Both Oura's expert, Dr. Mortazavi, and Oura's corporate representative, Mr.

[REDACTED]

Christophy, [REDACTED]. JDX-0052C, 392:3-16, 395:18-21, 396:18-21; JDX-0025C, 88:23-89:1, 89:18-23, 104:4-6.

(2) Limitation [7.c][v][b] (“a first conduction layer located on or above the first insulating layer;”)

Each of the rigid portions of the Accused Products’ PCBs includes a first conduction layer located on or above the first insulating layer. JDX-0003C ¶¶142-158. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Mortazavi, and Oura's corporate representative, Mr. Christophy, [REDACTED]
[REDACTED]. JDX-0025C, 88:14-16; JDX-0052C, 390:1-3, 396:1-2. Mr.
Christophy further confirmed that [REDACTED] JDX-0025C,
90:24-25.

[REDACTED]

[REDACTED]

[REDACTED]

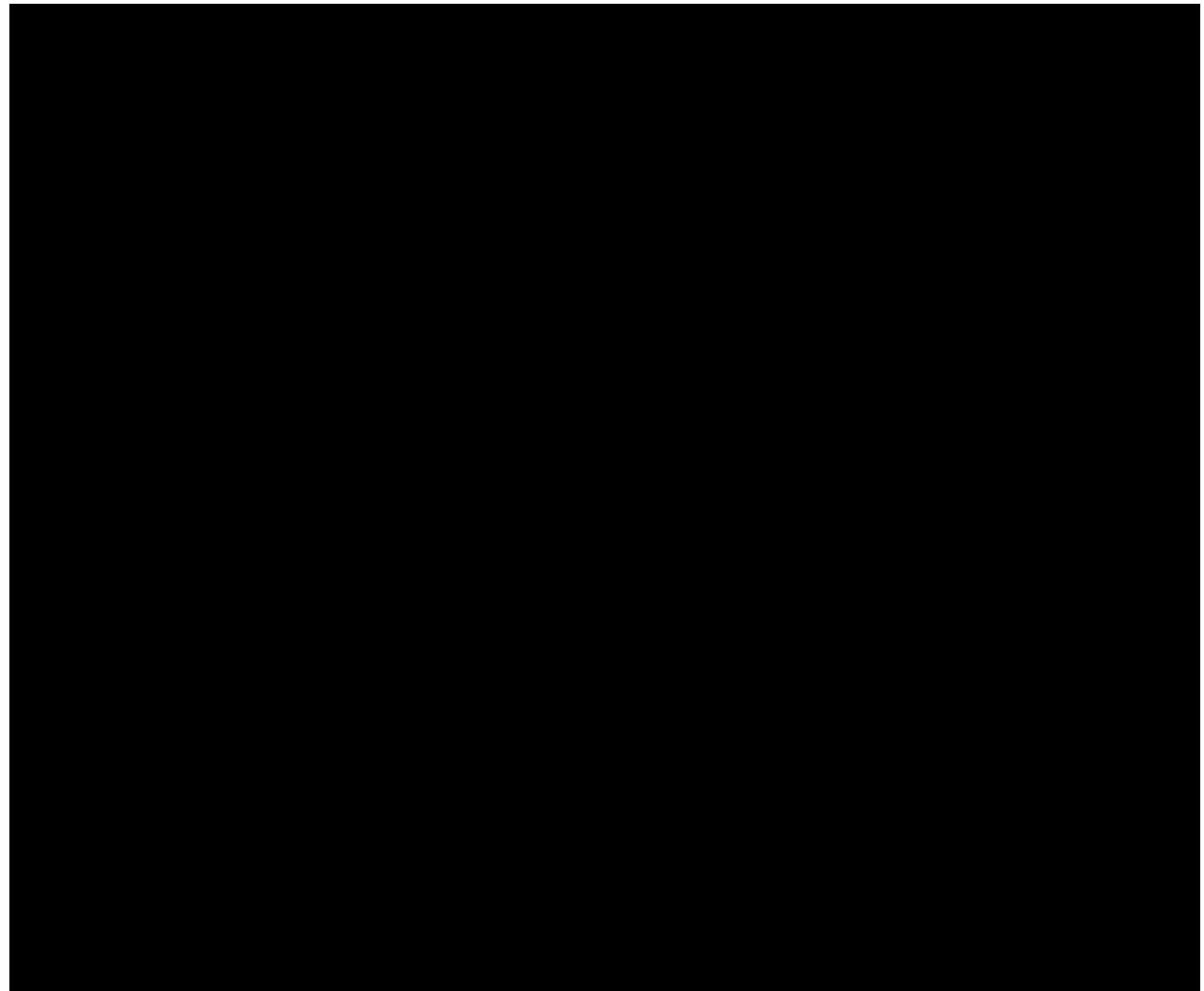
Specifically-Identified "First Rigid PCB Portion":

[REDACTED]

[REDACTED]

CX-0349C (annotated).

Specifically-Identified “Second Rigid PCB Portion”:



CX-0349C (annotated).

- (3) **Limitation [7.c][v][c] (“a second insulating layer located on or above the first conduction layer and having a first thickness;”)**

Each of the rigid portions of the Accused Products’ PCBs includes a second insulating layer located above the first conduction layer and having a first thickness. JDX-0003C ¶¶159-188.



[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Specifically-Identified “First Rigid PCB Portion”:

[REDACTED]

CX-0295C (annotated).

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0331C (annotated). [REDACTED]

[REDACTED]

[REDACTED] JDX-0003C ¶186.

Specifically-Identified “Second Rigid PCB Portion”:

[REDACTED]

CX-0293C (annotated).

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0333C (annotated). [REDACTED]

[REDACTED]

[REDACTED] JDX-0003C ¶¶187-188.

Oura's expert, Dr. Mortazavi, disputes Samsung's mapping for this limitation, contending that [REDACTED] JDX-0007C ¶¶597-599, 604, 617-623, 639-640, 643. That position is unsupported, and contradicted by the intrinsic evidence and Dr. Mortazavi's own analysis. First, the specification explicitly teaches that a single insulating layer may be formed from multiple constituent materials, such as "polyimide (PI) and an adhesive" (JX-0004, 8:8-9), or "a resin and a glass fabric" (*id.*, 7:21-24). During his deposition, Oura's expert, Dr. Mortazavi acknowledged that the specification contemplates that a single "insulating layer" may include multiple materials. JDX-0052C, 382:19-385:13; *see also* JDX-0050C, 45:13-19. And, as Dr. Warren explained at his deposition, the claim language itself—"including a first material" in limitation [7.c][v][c]—"does not discount a composite layer structure"; it simply requires that the insulating layer include the first material (e.g., as part of the composite structure). JDX-0054C, 103:20-104:7.

Second, Dr. Mortazavi himself acknowledged during his deposition that the Oura Ring

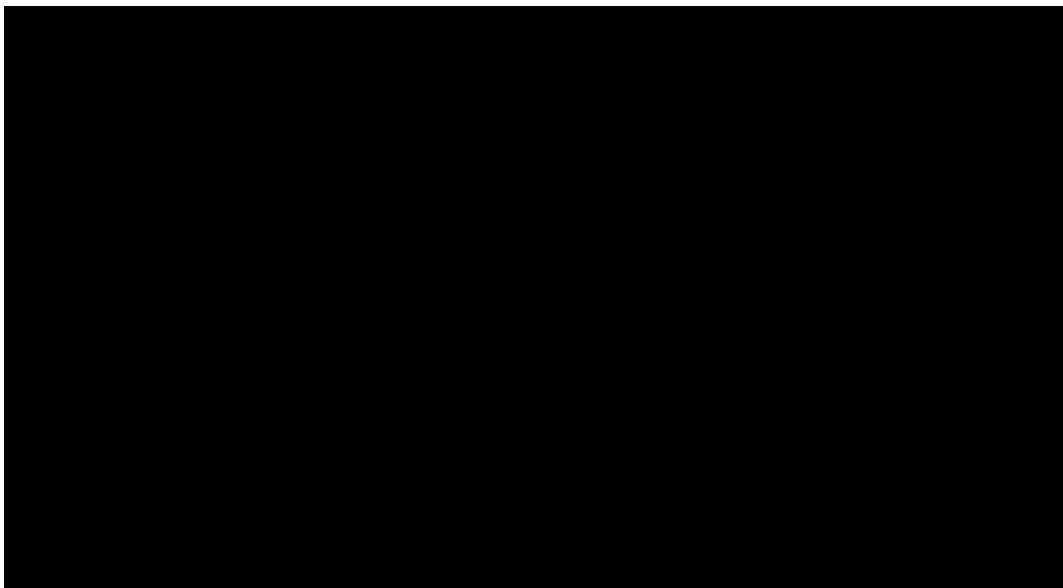
[REDACTED]

Gen. 4 [REDACTED]

[REDACTED]. JDX-0052C, 390:20-391:11, 395:4-16. As Dr. Warren explained in both his report and at deposition, a POSITA would view the insulating material located between two adjacent conductive layers—whether formed from a single material or multiple materials bonded together—as a single insulating layer because its function is to provide electrical isolation between those two conductive layers. JDX-0003C ¶162; JDX-0054C, 93:12-16, 100:14-106:6; JDX-0050C, 44:17-45:1. A POSITA designing or analyzing a PCB stack-up identifies insulating layers by their position between conductive layers and by their dielectric function, not by whether the insulating material was formed from one sub-layer or multiple sub-layers that were laminated together. JDX-0008C ¶131.

Third, Oura’s own invalidity theory undermines Dr. Mortazavi’s noninfringement position. In his opening invalidity report, Dr. Mortazavi maps the [REDACTED] to the same “second insulating layer” limitation. JDX-0001C ¶3432, Appx. B. But, as illustrated below, and as Dr. Mortazavi agreed during his deposition, [REDACTED]

[REDACTED]:



[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

(4) Limitation [7.c][v][d] (“a second conduction layer located on or above the second insulating layer; and”)

Each of the rigid portions of the Accused Products’ PCBs includes a second conduction layer located on or above the second insulating layer. JDX-0003C ¶¶200-204. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Specifically-Identified “First Rigid PCB Portion”:



CX-0349C (annotated).

Specifically-Identified “Second Rigid PCB Portion”:



- (5) **Limitation [7.c][v][e] (“a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion,”)**

Each of the rigid portions of the Accused Products’ PCBs includes a third insulating layer including the first material [REDACTED] located on or above the second conduction layer, and forming a second exterior surface of each of the rigid PCB portions. JDX-0003C ¶¶205-211. As illustrated

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]:

[REDACTED]

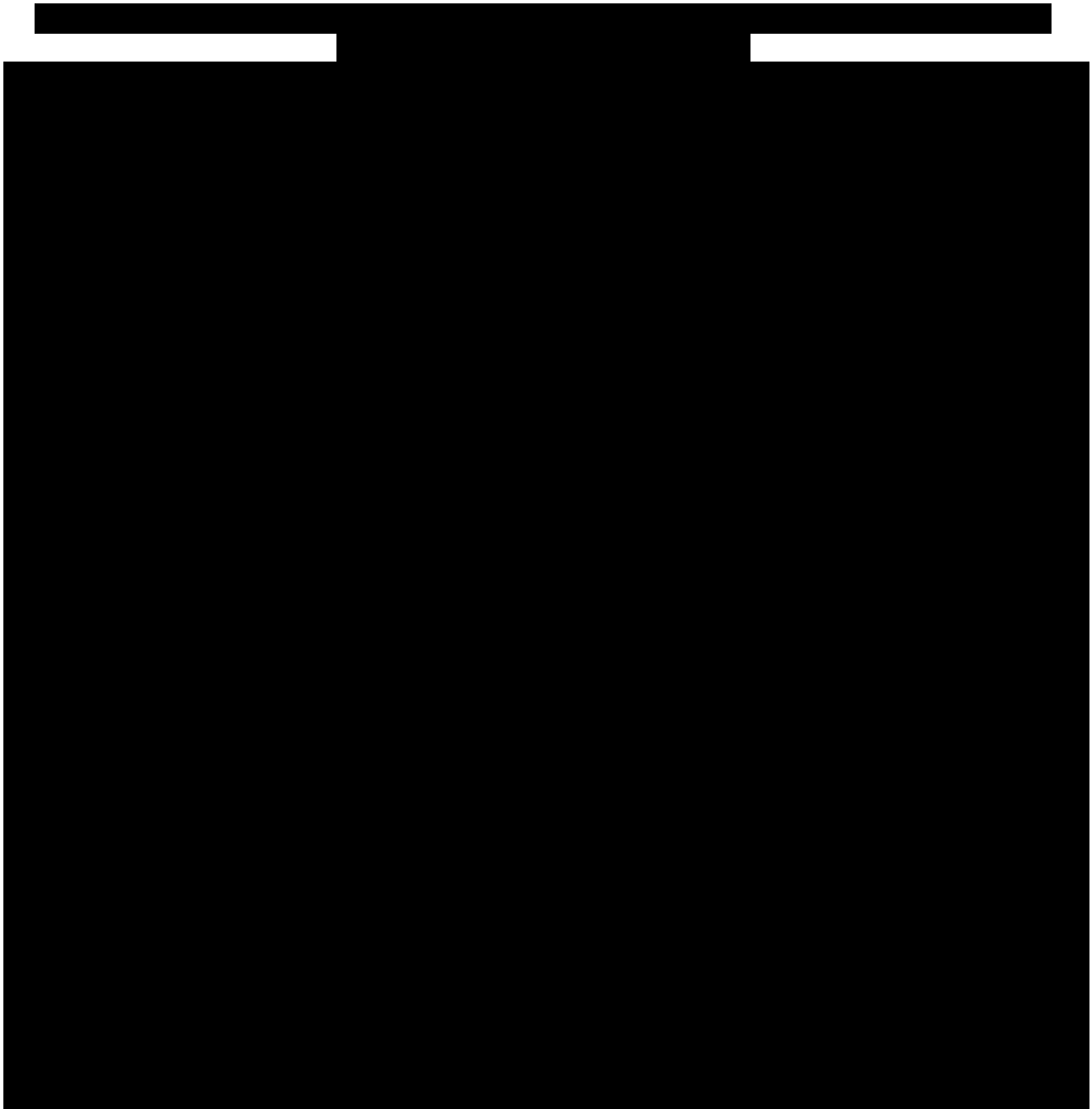
CX-1360C, OURA1468_00219859 (annotated).

[REDACTED]

[REDACTED]

[REDACTED]

Specifically-Identified “First Rigid PCB Portion”:



[REDACTED]

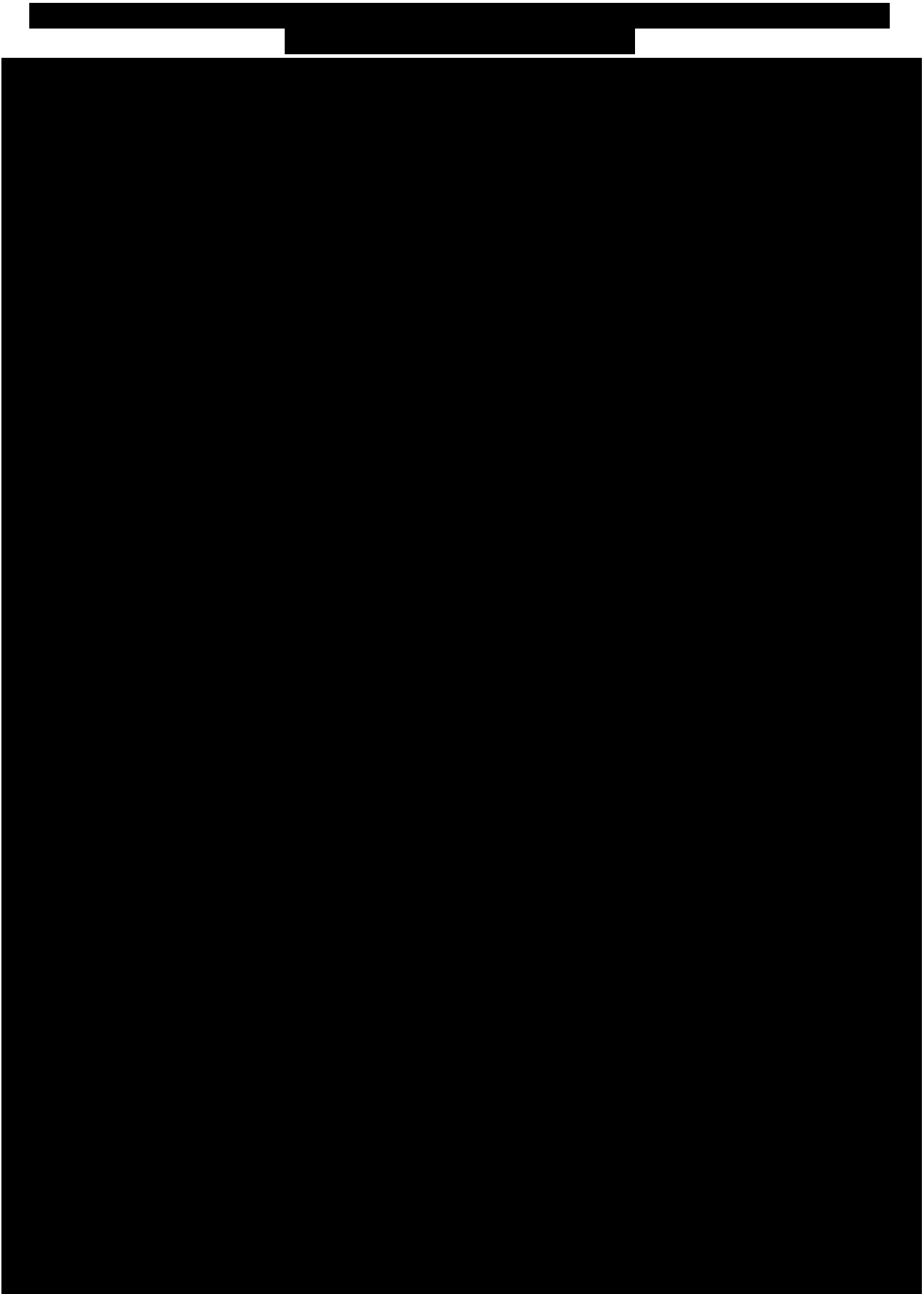
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

f. ***Structural Requirements of the Flexible PCB Portions — Limitation [7.c][vi] (“wherein each of the first flexible PCB portion and the second flexible PCB portion includes”***¹¹

- (1) **Limitation [7.c][vi][a] (“a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;”)**

Each of the flexible portions of the Accused Products’ PCBs—including the “first flexible PCB portion” and the “second flexible PCB portion”—includes a first insulating layer of a second material forming a first exterior surface. JDX-0003C ¶¶212-219. [REDACTED]

[REDACTED]

[REDACTED]

Specifically-Identified “First Rigid PCB Portion”:

[REDACTED]

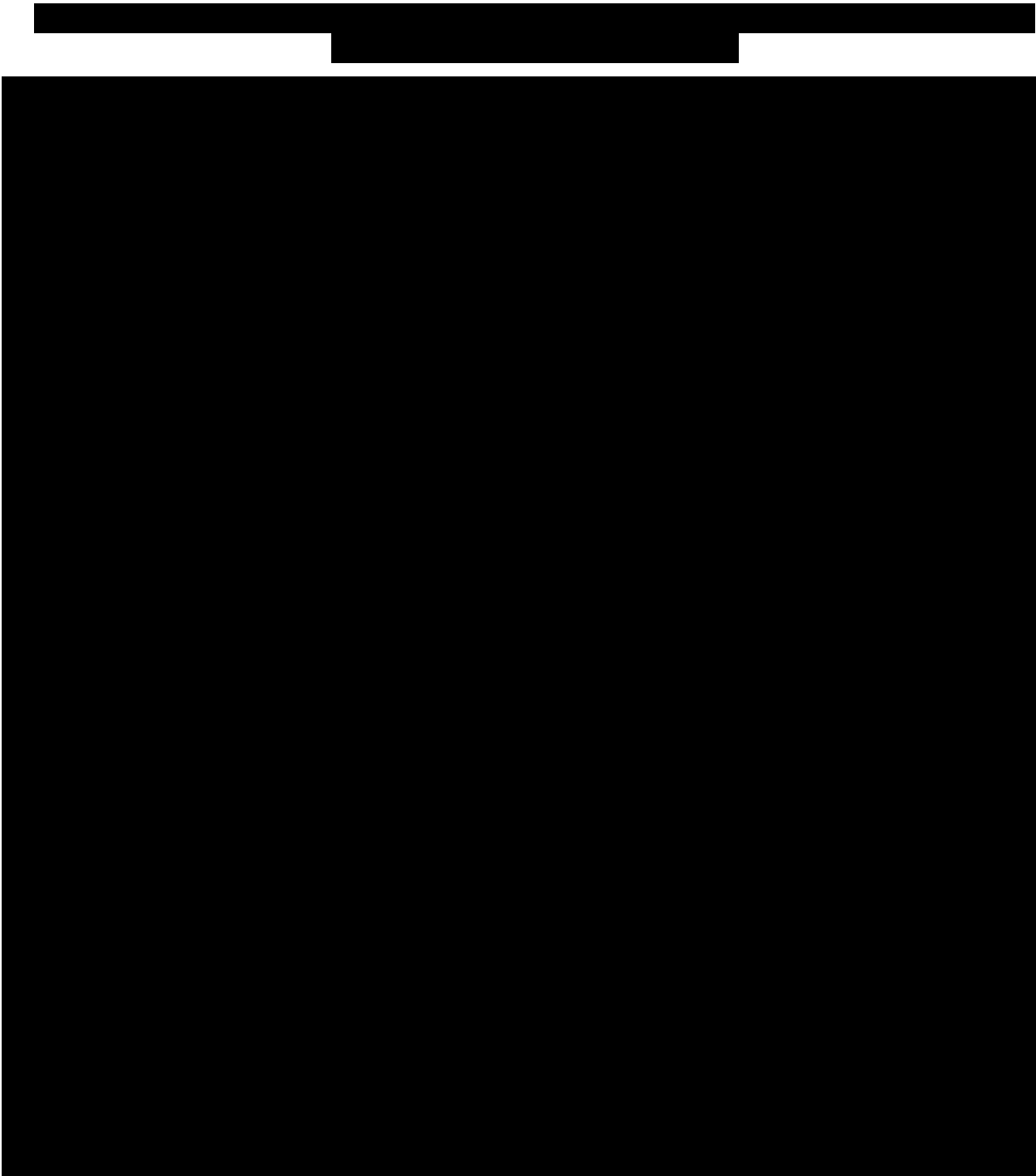
CX-0295C (annotated).

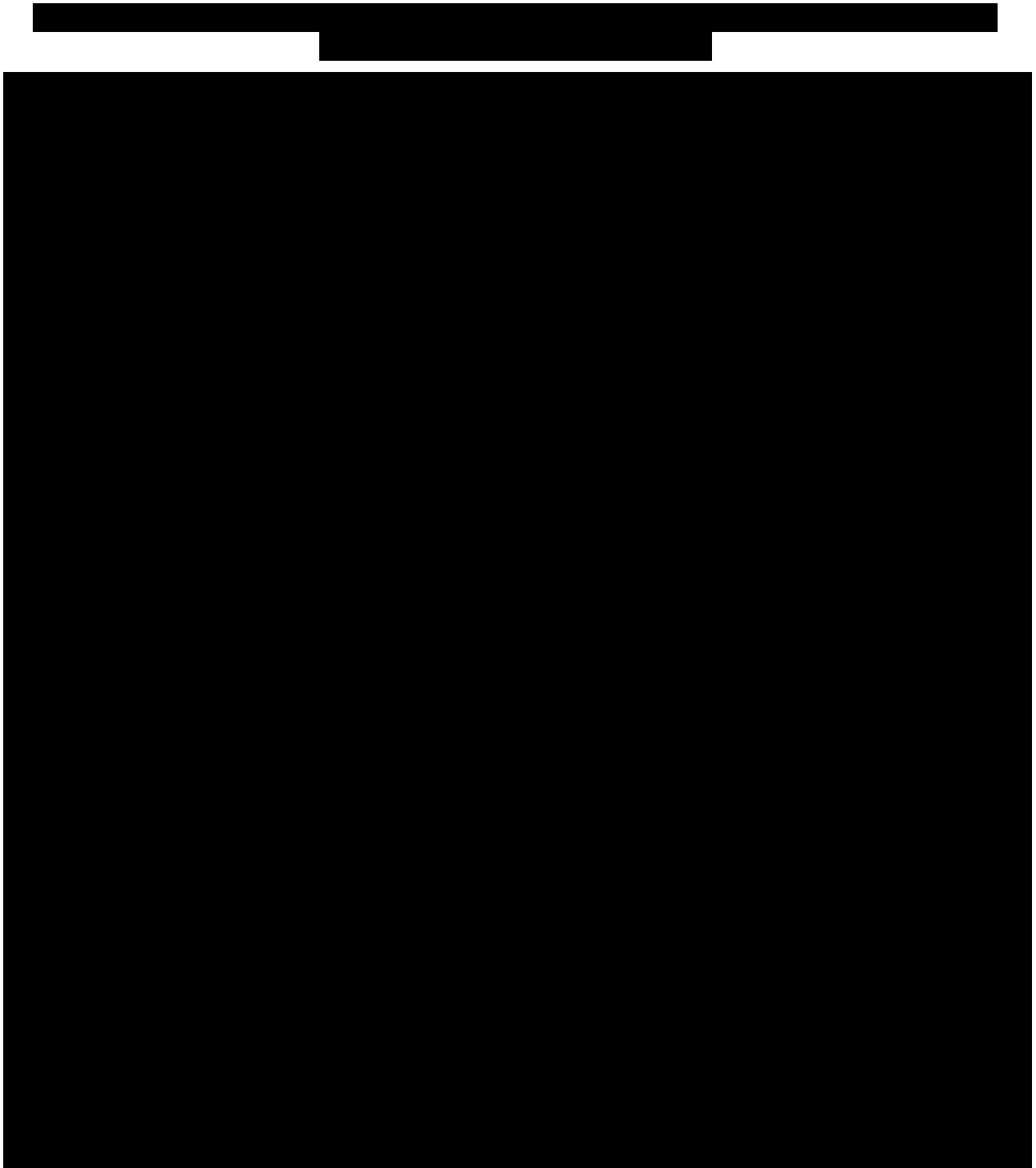
¹¹ The analysis below presents one mapping of the claimed “insulating” and “conduction” layers to the layers of the Accused Products’ PCBs. The Accused Products also meet Limitations [7.c][vi][a]-[e] under alternative mappings, as Dr. Warren explains in his report. JDX-0003C ¶¶212-300.

[REDACTED]

[REDACTED]

[REDACTED]





[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Id. (annotated). Both Oura's expert, Dr. Mortazavi, and Oura's corporate representative, Mr. Christophy, [REDACTED]. JDX-0052C, 397:19-398:11; JDX-0025C, 92:23-9:3. [REDACTED]

[REDACTED] JDX-0025C, 93:12-18, 100:24-101:3.

[REDACTED] JDX-0003C

¶217.

[REDACTED]

[REDACTED]

[REDACTED]

Id., SAMS382-0034586.

Ring 5 – Response to Oura’s Rebuttal Argument:

While Oura’s expert, Dr. Mortazavi, does not dispute that Ring 4 and [REDACTED] meet this

[REDACTED]
[REDACTED]
[REDACTED]
limitation, [REDACTED]
[REDACTED]
[REDACTED]. JDX-0007C ¶¶655-

660.

[REDACTED]

Id., ¶658. Specifically, Dr. Mortazavi contends that [REDACTED]
[REDACTED]
[REDACTED] *Id.*, ¶659.

Oura's own teardown data, however, contradicts Dr. Mortazavi's opinion that the two "first insulating layers" (of the rigid and flexible portions) are made from the [REDACTED]

[REDACTED]¹² The SEM image below, identified by Dr. Mortazavi within a set of "[REDACTED] testing

¹² As detailed in Samsung's Opposition to Oura's MSD Non-Infringement by Oura's Alternative Products, Oura produced its own teardown, SEM, and EDS images and analysis for [REDACTED] on only the last day of fact discovery on May 29. EDIS ID 889193.

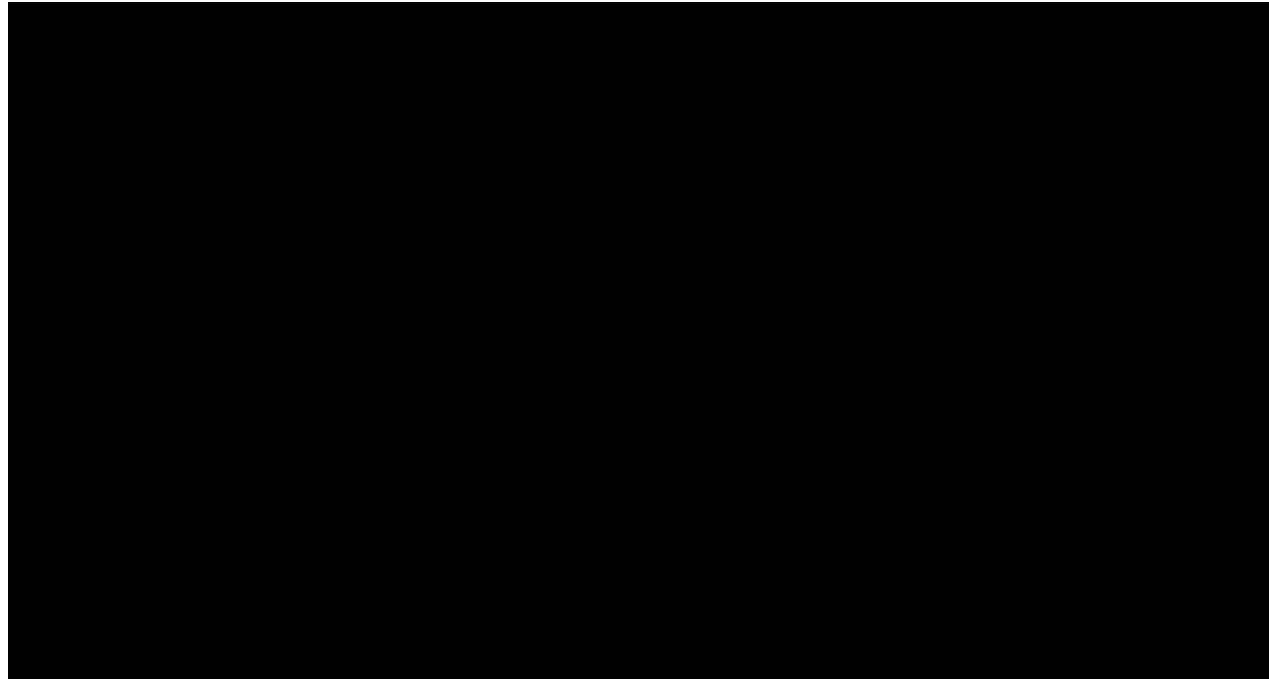
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



CX-1689 (contrast adjusted); *see also* CX-1688; JDX-0054C, 205:21-206:4 (Dr. Warren testifying that, as depicted in this image, [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

- [REDACTED]
- [REDACTED]
- (2) **Limitation [7.c][vi][b] (“a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of [the] first rigid PCB portion and the first conduction layer of the second rigid PCB portion;”)**

Each of the flexible portions of the Accused Products’ PCBs includes a first conduction layer located on or above the first insulating layer. JDX-0003C ¶¶220-243. [REDACTED]

[REDACTED]

[REDACTED]

Specifically-Identified “First Flexible PCB Portion”:



CX-0349C (annotated).

Specifically-Identified “Second Flexible PCB Portion”:

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

(3) Limitation [7.c][vi][c] (“a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;”)

Each of the flexible portions of the Accused Products’ PCBs includes a second insulating layer located above the first conduction layer and having a second thickness thinner than the first thickness. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-1360C, OURA1468_219859 (annotated). [REDACTED] JDX-0003C

¶¶246; JDX-0025C, 148:4-15; JDX-0052C, 397:14-16. [REDACTED]

[REDACTED]

[REDACTED] CX-1360C, OURA1468_219859; JDX-0003C, ¶246.

[REDACTED]

[REDACTED] JDX-0003C ¶¶247-265. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

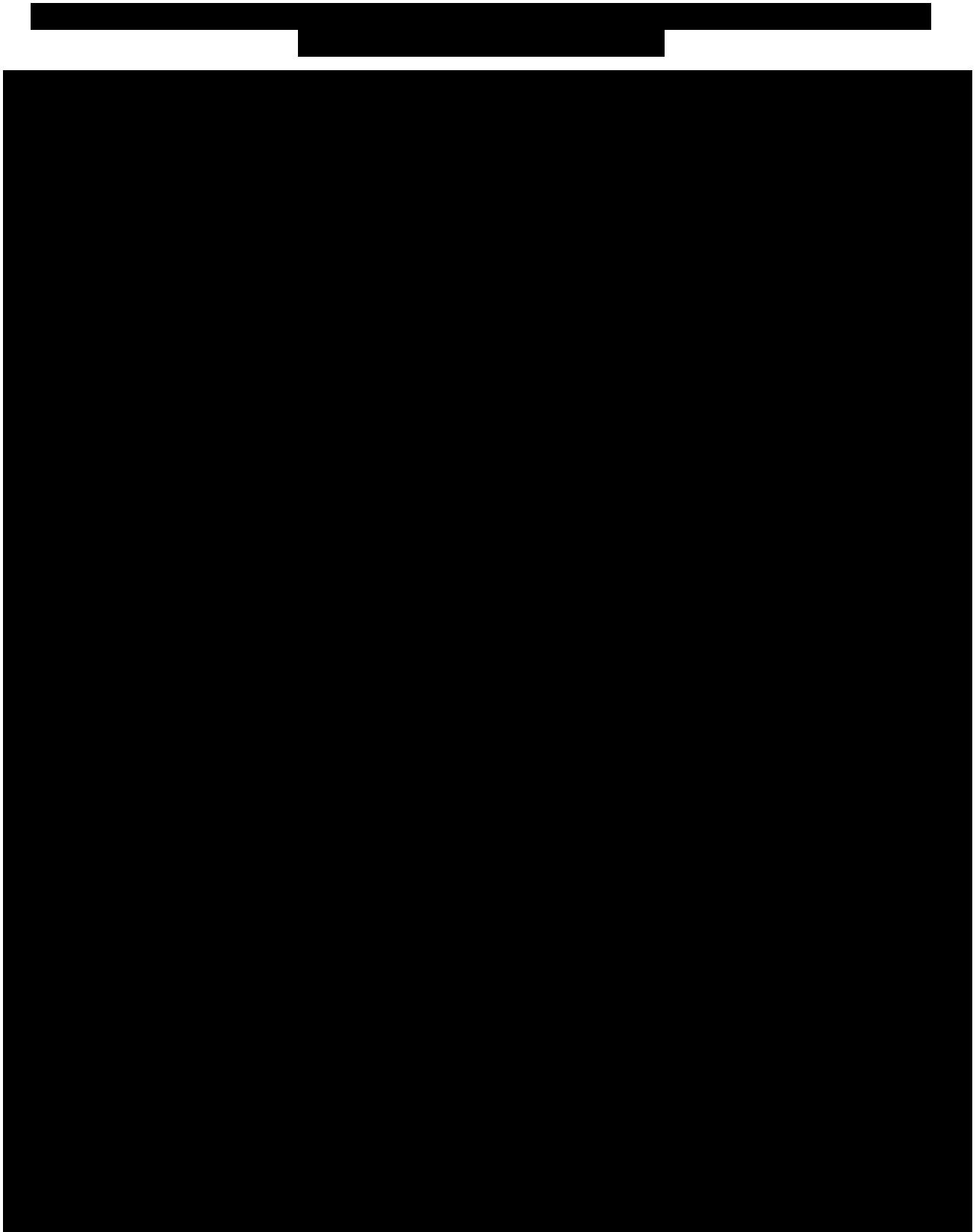
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

- (4) **Limitation [7.c][vi][d] (“a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and”)**

Each of the flexible portions of the Accused Products’ PCBs includes a second conduction layer located on or above the second insulating layer. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

- [REDACTED]
- [REDACTED]
- (5) **Limitation [7.c][vi][e] (“a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.”)**

Each of the flexible portions of the Accused Products’ PCBs includes a third insulating layer including the second material, which is located on or above the second conduction layer, and forming a second exterior surface of each of the flexible PCB portions. [REDACTED]

[REDACTED]

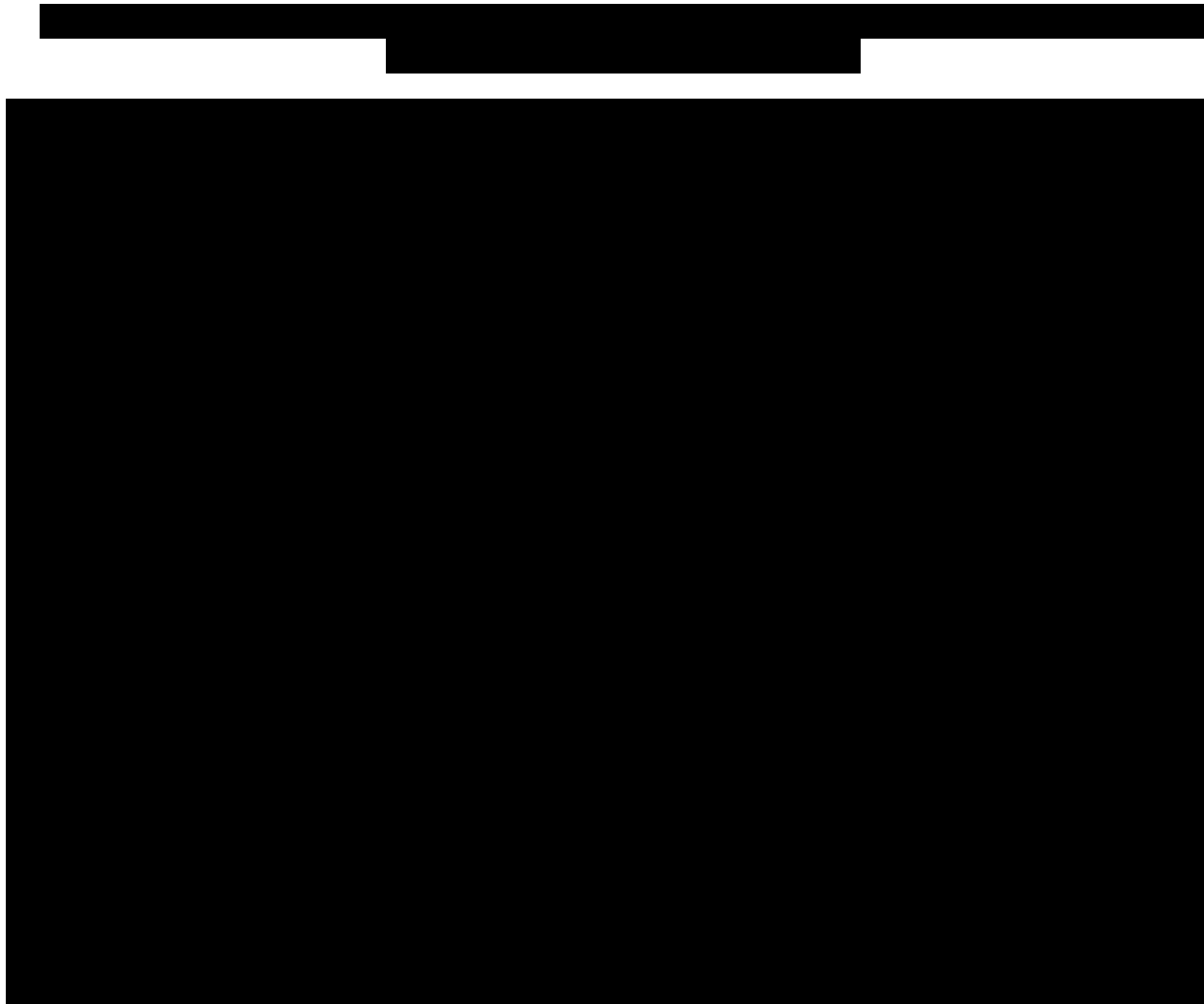
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



2. The Accused Products Infringe Claim 9 of the '789 Patent

- (1) Limitation [9.a] (“wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer”)**

Each of the flexible portions of the Accused Products’ PCBs—including the “first flexible PCB portion” and the “second flexible PCB portion”—includes a fourth insulating layer located between the second conduction layer and the third insulating layer. JDX-0003C ¶¶301-306. 







Specifically-Identified “First Rigid PCB Portion” and “Second Rigid PCB Portion”:

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-1360C, OURA1468_00219859 (annotated).

While Oura's expert, Dr. Mortazavi, disputes Samsung's mapping for this limitation,

[REDACTED]

[REDACTED] (JDX-0007C ¶¶638-641), that position is unsupported, and contradicted by the intrinsic evidence and Dr. Mortazavi's own analysis, as explained for Limitation [7.c][v][c]. In particular, the specification expressly contemplates that a single "insulating layer" may comprise multiple insulating materials. JX-0004, 7:21-24, 8:8-9; JDX-0054C, 103:20-104:7; *see also* JDX-0052C, 390:20-391:11, 395:4-16; JDX-0003C ¶162; JDX-0054C, 93:12-16, 100:14-106:6; JDX-0008C ¶131; JDX-0050C, 45:13-19. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

(2) Limitation [9.b] (“and a third conduction layer located between the third insulating layer and the fourth insulating layer.”)

Each of the flexible portions of the Accused Products’ PCBs includes a third conduction layer located between the third insulating layer and the fourth insulating layer. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

B. TDI Is Met for the '789 Patent

The technical prong of the domestic industry requirement is satisfied because the PDI Products practice Claim 6 of the '789 Patent, as detailed below.

The analysis that follows uses the Samsung S25 Basic, S24 Ultra, S23 Ultra, and S22 Ultra as representative of their respective series.¹³ JDX-0003C, ¶¶314-316 (Dr. Warren explaining representativeness and common design rules across each Galaxy series); JDX-0014C, 44:11-16, 61:19-62:2, 62:20-63:7.

1. The PDI Products Satisfy Each Limitation of Claim 1, From Which Asserted Claim 6 Depends

Notably, Oura does not contest that the PDI Products satisfy limitations [1.pre], [1.a], and [1.b]. JDX-0006C ¶¶163-185. Oura's disputes are limited to two issues: (1) Limitation [1.c], specifically whether the antenna FPCB-type RF cable ("FRC") in the DI Products constitutes a "printed circuit board" or PCB, and (2) Limitation [1.c][v][a], specifically whether the traces thereon are "ground lines." As explained below, the evidence clearly establishes both and Oura's

¹³ S25 Series: S25, S25+, S25 Edge; S24 Series: S24, S24+, S24 Ultra; S23 Series: S23, S23+, S23 Ultra; S22 Series: S22, S22+, S22 Ultra.

rebuttal arguments misread Samsung’s documentation and disregard the intrinsic evidence.

a. Limitation [1.pre] (“A portable communication device comprising:”)

Each PDI Product is a portable communication device—a portable, battery-powered smartphone with cellular and other wireless communication capabilities, including GSM, CDMA, HSPA, LTE, and 5G. JDX-0003C ¶¶317-319.



The image is a screenshot of the Samsung Galaxy S25 product page. It features a blue smartphone on the left. To the right, key specifications are listed: Released 2025, February 03; 162g, 7.2mm thickness; Android 15, up to 7 major upgrades, One UI 7; 12GB/256GB/512GB storage, no card slot. Below these, a 6.2" display (1080x2340 pixels), 50MP camera (430g), 12GB RAM (Snapdragon 8 Elite), and 4000mAh battery (25W fast charging, 45W wireless) are shown. The bottom section contains a table with details on network, launch, body, and display.

NETWORK	Technology	GSM / CDMA / HSPA / EVDO / LTE / 5G
LAUNCH	Announced	2025, January 22
	Status	Available. Released 2025, February 03
BODY	Dimensions	146.9 x 70.5 x 7.2 mm (5.78 x 2.78 x 0.28 in)
	Weight	162 g (5.71 oz)
	Build	Glass front (Gorilla Glass Victus 2), glass back (Corilla Glass Victus 2), aluminum frame
	SIM	- Nano-SIM + Nano-SIM + eSIM + eSIM (max 2 at a time) - INT - Nano-SIM + eSIM + eSIM (max 2 at a time) - USA - Nano-SIM + Nano-SIM - CN
		IP68 dust tight and water resistant (immersion up to 1.5m for 30 min) Armor aluminum 2 frame
DISPLAY	Type	Dynamic LTPO AMOLED 2X, 120Hz, 480Hz PWM, HDR10+, 2600 nits (peak)
	Size	6.2 inches, 94.4 cm ² (~91.1% screen-to-body ratio)
	Resolution	1080 x 2340 pixels, 19.5:9 ratio (~416 ppi density)
	Protection	Coming Corilla Glass Victus 2, Mohs level 5

CX-0121; *see also* CX-0125; CX-0123; CX-0116; CX-0118; CX-0113; CX-0110; CX-0108; CX-0107; CX-0102; CX-0100; CX-0104.

b. Limitation [1.a] (“a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;”)

Each PDI Product includes a housing with a first conduction portion forming a first portion of a side surface and a second conduction portion forming a second portion of the side surface,

[REDACTED]

where the first conduction portion operates as a first antenna and the second conduction portion operates as a second antenna. JDX-0003C ¶¶320-324. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0674C, SAMS382-0045580 (annotations in original); *see also* CX-0119C, SAMS382-007934; CX-0124C, SAMS382-009316, 9321; CX-0114C, SAMS382-007064; CX-0105C, SAMS382-004669; CX-0101C, SAMS382-004118.

[REDACTED]

[REDACTED]:

[REDACTED]

[REDACTED]

[REDACTED]

CX-0673C, SAMS382-0045575, 45578 (annotated); *see also* CX-1625C, SAMS382_0029238;
CX-0203C, SAMS382_0021186-89; CX-0257C, SAMS382_0029236-38.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0119C, SAMS382-0008015 (annotated); *see also* CX-0124C, SAMS382-0009394, 9395; CX-0114C, SAMS382-0007141; CX-0117C, SAMS382-0007837; CX-0109C, SAMS382-0005672; CX-0105C, SAMS382-0004766; CX-0101C, SAMS382-0004194; CX-0103, SAMS382-0004468.

c. Limitation [1.b] (“a wireless communication circuitry”)

Each PDI Product includes wireless communication circuitry that transmits, receives, and processes the [REDACTED]. JDX-0003C ¶¶325-332. [REDACTED]

[REDACTED]

[REDACTED]:

[REDACTED]

CX-0119C, SAMS382-0008125 (annotated), 8133; CX-0098C, 3814, 3905; CX-0101C, SAMS382-0004119; CX-0103, SAMS382-0004396, 4469; CX-0105C, SAMS382-0004767; CX-0109C, SAMS382-0005595; CX-0111C, SAMS382-0005905; CX-0114C, SAMS382-0007167;

[REDACTED]

[REDACTED]

[REDACTED]

d. Limitation [1.c] (“a first printed circuit board (PCB) including”)

Each PDI Product includes a first PCB in the form of an antenna FRC (“FPCB-type RF cable”). [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0119C, SAMS382-0007934 (annotated), 7939; *see also* CX-0098C, SAMS382-0003812; CX-0101C, SAMS382-0004118; CX-0103, SAMS382-0004395; CX-0105C, SAMS382-0004669; CX-0109C, SAMS382-0005592-93; CX-0111C, SAMS382-0005903; CX-0114C, SAMS382-0007165; CX-0117C, SAMS382-000 7597; CX-0122C, SAMS382-0008428; CX-0124C, SAMS382-0009321.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0215C, SAMS382-0026614; *see also* CX-0637C, SAMS382-0043794; CX-0214C, SAMS382-0026506; CX-0208C, SAMS382-0021320; CX-0206C, SAMS382-0021227; CX-0255C; CX-1625C; CX-0258C; CX-0259C; CX-0382C CX-0382C-CX-0390C; JDX-0014C, 62:20-65:16 ([REDACTED]

[REDACTED]).

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0006C ¶166. [REDACTED]

[REDACTED]

[REDACTED] JDX-0050C, 24:4-16.

Moreover, his deposition testimony reveals he cannot maintain the distinction.

First, [REDACTED]

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] [REDACTED]

[REDACTED]

■■■■■■■■■■

- [REDACTED]
- e. **Limitations [1.c][i]-[iv] (“a first PCB portion; a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion; a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion; a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion”)**

The FRC in each PDI Product is constructed in a rigid-flex-rigid-flex-rigid configuration.

JDX-0003C ¶¶335-336. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED].

[REDACTED]

CX-0638C, SAMS382-003948 (annotated); *see also* CX-0637C, SAMS382-0043842; CX-0214C, SAMS382-0026556; CX-0208C, SAMS382-0021368-69; CX-0255C; CX-1625C; CX-0258C; CX-0259C.

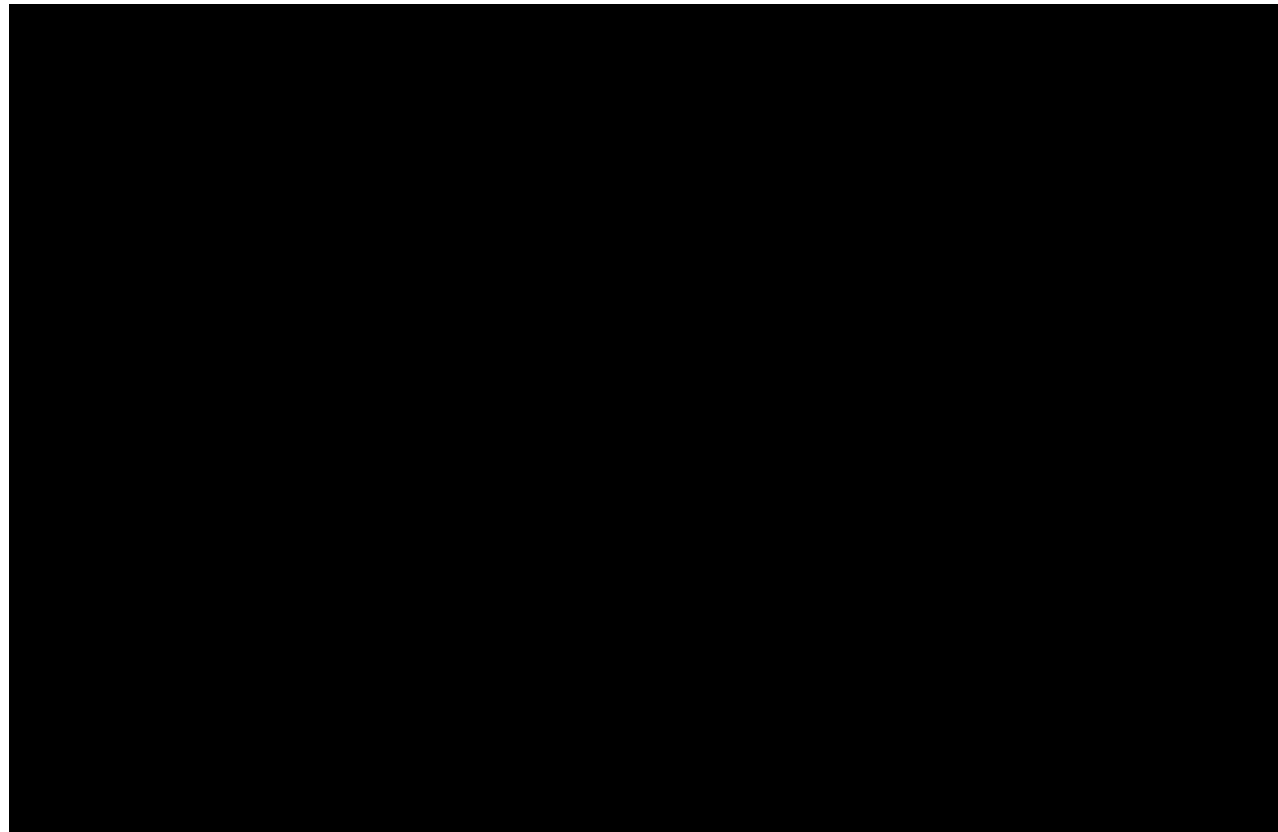
[REDACTED]

[REDACTED]

[REDACTED] JDX-0003C ¶¶336-343.



CX-0215C, SAMS382-0026614 (annotated); *see also* CX-0637C, SAMS382-0043794; CX-0214C, SAMS382-0026506; CX-0208C, SAMS382-0021320; CX-0206C, SAMS382-0021227; CX-0255C; CX-1625C; CX-0258C; CX-0259C; CX-0382C CX-0382C-CX-0390C.



CX-0215C, SAMS382-0026615 (annotated); *see also* CX-0638C at SAMS382-0043901; CX-0637C, SAMS382-0043795; CX-0214C, SAMS382-0026507; CX-0208C, SAMS382-0021321; CX-0206C, SAMS382-0021228; CX-0255C; CX-1625C; CX-0258C; CX-0259C; CX-0382C.

- f. Limitation [1.c][v][a] (“a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line,”)**

The FRC in each PDI Product carries a plurality of lines running from the first PCB portion to the fourth PCB portion through the second and third PCB portions, including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the two ground lines, and a second signal line located between the first signal line

[REDACTED]

and the second ground line. JDX-0003C ¶¶344-355.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0661C–CX-0671C ([REDACTED]); *see also* CX-0642C–CX-0651C ([REDACTED]);
CX-1352C ([REDACTED]); CX-1350C ([REDACTED]); CX-1351C ([REDACTED]
[REDACTED]).

As Dr. Warren explains, [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED].

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Antenna 1 Antenna 2

[REDACTED]

CX-0661C–CX-0671C (S25 design files); *see also* CX-0642C–CX-0651C (S25+ design files);
see also CX-1484C. Dr. [REDACTED]

[REDACTED]. JDX-
0003C ¶¶354-355; CX-1352C; CX-1350C; CX-1351C.

[REDACTED]

[REDACTED] JDX-0006C ¶¶174-177. But this ignores a basic principle of PCB design—
ground lines, ground planes, and ground vias are all commonly interconnected and part of a single
ground network. JDX-0003C ¶357; JDX-0014C, 85:15-86:12. [REDACTED]

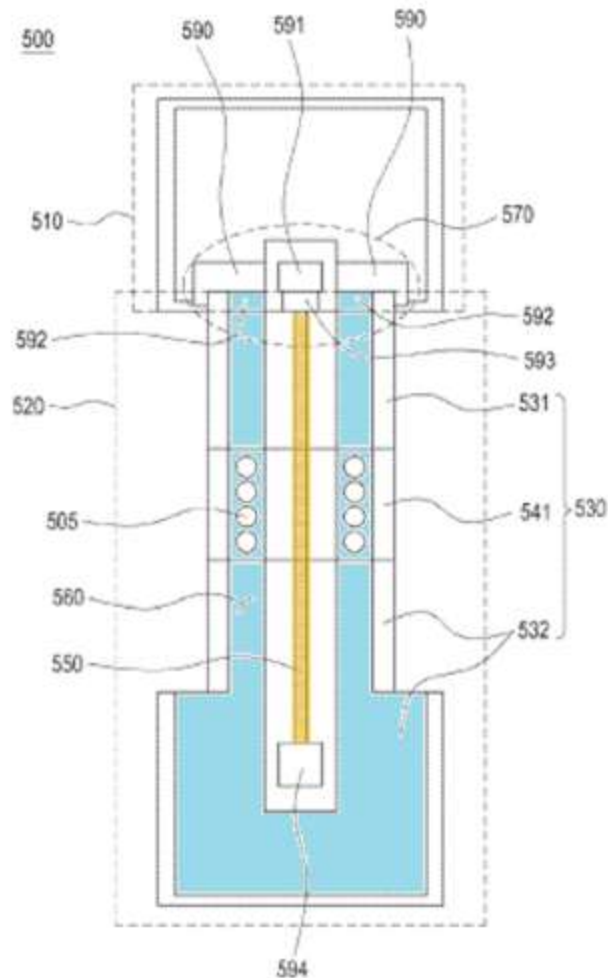
[REDACTED]

[REDACTED] JDX-0003C ¶¶344-
355, 357-358.

Indeed, the specification teaches the same configuration in its Figure 5 annotated below,

[REDACTED]

[REDACTED]



JX-0004, FIG. 5 (colored; illustrating ground lines 560 that are connected at the lower end of the PCB). Shown Figure 5, Dr. Mortazavi acknowledged that “the patent says a ground line 560,” even while describing 560 as “a single ground area that wraps around 550.” JDX-0052C, 369:8-370:2, 370:14-15, 372:3-5. He likewise agreed, as to the specification’s teaching that “the entire ground line 360 may be connected to the all-ground layer through the conductive vias 305,” that “the 360 ground lines are distinct lines that are connected to a ground plane somewhere,” and that he has “seen ground lines connected to singular ground sources.” *Id.*, 374:9-375:3, 376:13-16; JX-0004, 9:19-24. Thus, the ’789 Patent refutes Oura and Dr. Mercier’s distinction between a “ground line” versus a “ground area/portion.” Indeed, the ’789 Patent’s configuration in its Figure

[REDACTED]

5 is exactly identical to the PDI Products, [REDACTED].

JDX-0003C ¶¶344-355, 357; JDX-0014C, 85:15-86:12.

- g. **Limitation [1.c][v][b] (“the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.”)**

In each PDI Product, the first signal line is electrically connected with the first antenna [REDACTED] and the wireless communication circuitry, and the second signal line is electrically connected with the second antenna [REDACTED] and the wireless communication circuitry.

[REDACTED]

[REDACTED]

[REDACTED] JDX-0003C ¶¶356-362. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0014C, 70:15–24.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0661C–CX-0671C (S25 design files); *see also* CX-0642C–CX-0651C (S25+ design files); CX-1352C (S22 Ultra design files); CX-1350C (S23 Ultra design files); CX-1351C (S24 Ultra design files).

[REDACTED]

[REDACTED] JDX-0003C ¶358.

[REDACTED]

[REDACTED] JDX-0003C ¶¶358-359. [REDACTED]

[REDACTED] *Id.*; JDX-0014C, 63:6-19, 64:1-17.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

CX-0638C, SAMS382-0043856-57 (annotated); *see also* CX-0637C, SAMS382-0043750-51.

[REDACTED]

CX-0661C–CX-0671C (S25 design files); *see also* CX-0642C–CX-0651C (S25+ design files).

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0003C ¶¶361-362; CX-0206C, 21202-03 (S22); CX-1352C (S22 Ultra design files); CX-0208C, 21279-80 (S23); CX-1350C (S23 Ultra design files); CX-0214C, 26466-67 (S24); CX-1351C (S24 Ultra design files).

2. The PDI Products Practice Claim 6 of the '789 Patent

The PDI Products also practice dependent Claim 6, which incorporates the limitations of Claim 1 established above.

a. Limitation [6.a] (“further comprising: a second PCB on which the wireless communication circuitry is located”)

Each PDI Product includes a second PCB—the main PBA—on which the wireless communication circuitry is located. JDX-0003C ¶¶387-390.

[REDACTED]

[REDACTED]

[REDACTED]

CX-0119C, SAMS382-0007939 (annotated), 8015-16; CX-0124C, SAMS382-0009321; CX-0114C, SAMS382-0007141; CX-0109C, SAMS382-0005672; CX-0101C, SAMS382-0004194; JDX-0014C, 17:24–18:10 [REDACTED]

[REDACTED].

As explained for Limitation [1.b], [REDACTED]

[REDACTED]

[REDACTED]:

[REDACTED]

[REDACTED]

[REDACTED]

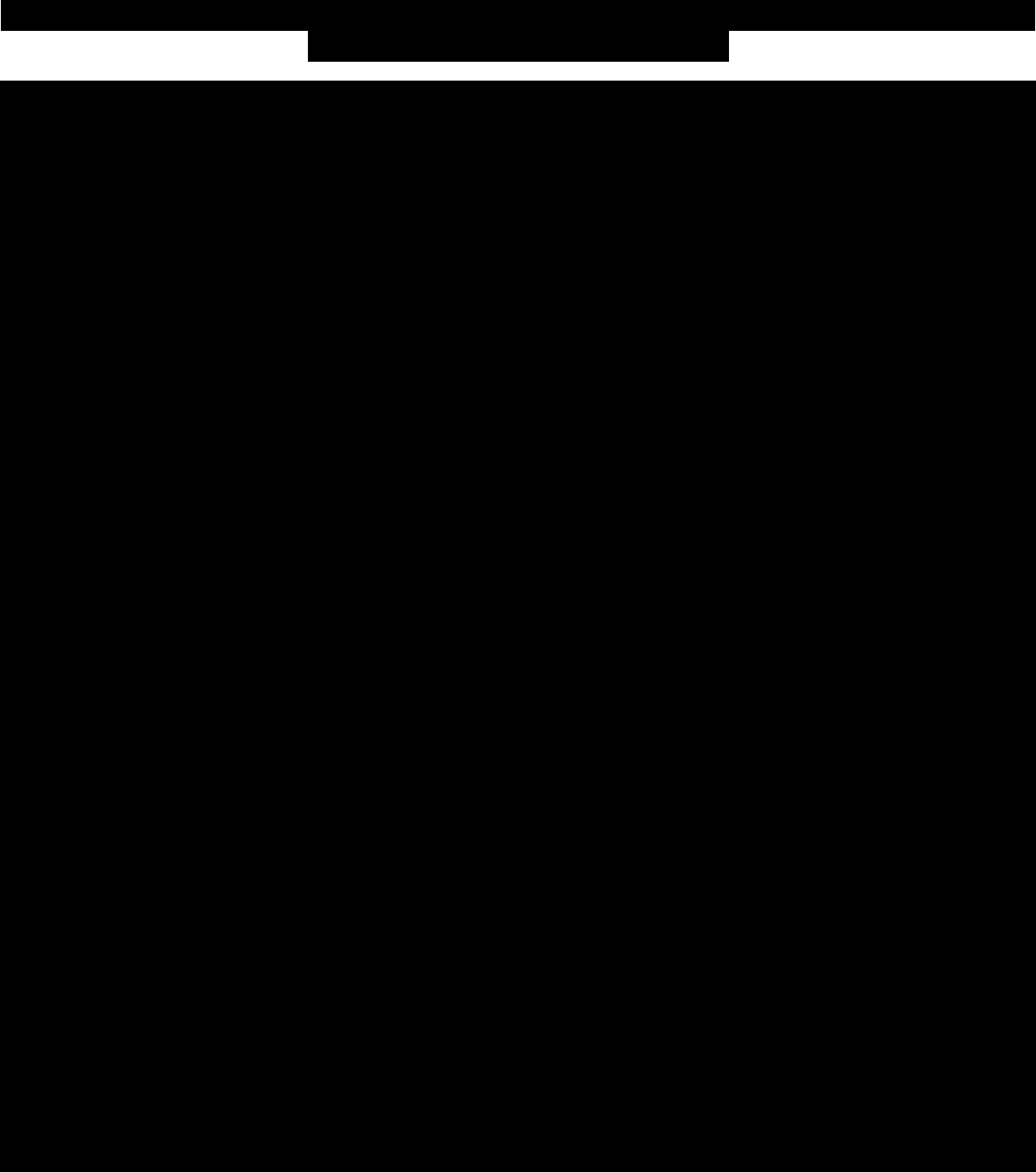
- b. Limitation [6.b] (“wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion”)**

The first and second ground lines are electrically connected with a ground member located in rigid section D (the first PCB portion), [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]



CX-0661C-CX-0671C (S25 Basic design files); *see also* CX-0642C–CX-0651C (S25+ design files); CX-0206C, 21202-03 (S22); CX-1352C (S22 Ultra design files); CX-0208C, 21279-80 (S23); CX-1350C (S23 Ultra design files); CX-0214C, 26466-67 (S24); CX-1351C (S24 Ultra design files).

C. The Asserted Claims of the '789 Patent Are Valid

Oura incorrectly asserts that the asserted claims are invalid based on the following prior art

grounds:

- Ground 1 – Claim 7 as anticipated by Oura Ring Gen 1 (“Ring 1”);
- Ground 2 – Claims 7 and 9 as obvious based on U.S. Patent No. 10,064,276 (“Williams”);
- Ground 3 – Claims 7 and 9 as obvious based on Williams in view of U.S. Patent No. 8,383,948 (“Aoyama”);
- Ground 4 – Claims 7 and 9 as obvious based on Williams in view of U.S. Patent App. Pub. No. 2012/0325524 A1 (“Naganuma”);
- Ground 5 – Claims 7 and 9 as obvious based on Aoyama;
- Ground 6 – Claims 7 and 9 as obvious based on Aoyama in view of Williams;
- Ground 7 – Claim 6 as obvious based on the combination of Samsung Galaxy S5 (“S5”), U.S. App. Patent Pub. No. 2015/0280771 (“Mow”), and U.S. Patent App. Pub. No. 2019/0045630 (“Kim”).

As explained below, Oura’s prior art grounds, however, fail to disclose one or more of the following key claim features: (i) Alternating rigid and flexible PCB portions (Limitations [7.c][i]-[iv]); (ii) Different exterior materials for the rigid versus flexible PCB portions (Limitations [7.c][v][a], [7.c][v][e], [7.c][vi][a], [7.c][vi][e]); (iii) Thinner interior insulating layer in the flexible versus rigid PCB portions (Limitations [7.c][v][c], [7.c][vi][c]); (iv) Conduction layers of the flexible PCB portions electrically connected to corresponding conduction layers of the rigid PCB portions (Limitations [7.c][vi][b], [7.c][vi][d]); and (v) Four insulating layers and three conduction layers in each flexible PCB portion (Limitations [9.a], [9.b]). JDX-0008C, ¶¶58-61.

1. Ground 1: Ring 1 Does Not Anticipate Claim 7

Ground 1 relies on Oura’s first-generation ring, Ring 1, and challenges only Claim 7, not Claim 9. To establish anticipation, Oura must show that the Ring 1 product actually sold or publicly used met every limitation of Claim 7, arranged as claimed. JDX-0008C, ¶¶62-80.

[REDACTED]

a. Oura Fails to Establish the Properties of Ring 1, as Sold

In arguing that Ring 1 anticipates Claim 7, Oura’s expert, Dr. Mortazavi, does not present any teardown analysis of an actual, as-manufactured ring. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

**(1) A Stack-up Drawing Reports Pre-Curing Dimensions,
Not As-Manufactured Ones**

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

- [REDACTED]
- b. **Oura Has Not Established that Ring 1 Uses a Different Insulating Material for the Exterior Layers of its Flexible PCB Portions, as Compared to its Rigid PCB Portions (Limitations [7.c][v][a], [7.c][v][e], [7.c][vi][a], [7.c][vi][e])**
- [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

**c. Oura Has Not Established that Ring 1 Provides the Required
Electrical Connections (Limitations [7.c][vi][b] and [7.c][vi][d])**

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

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[REDACTED]

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[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

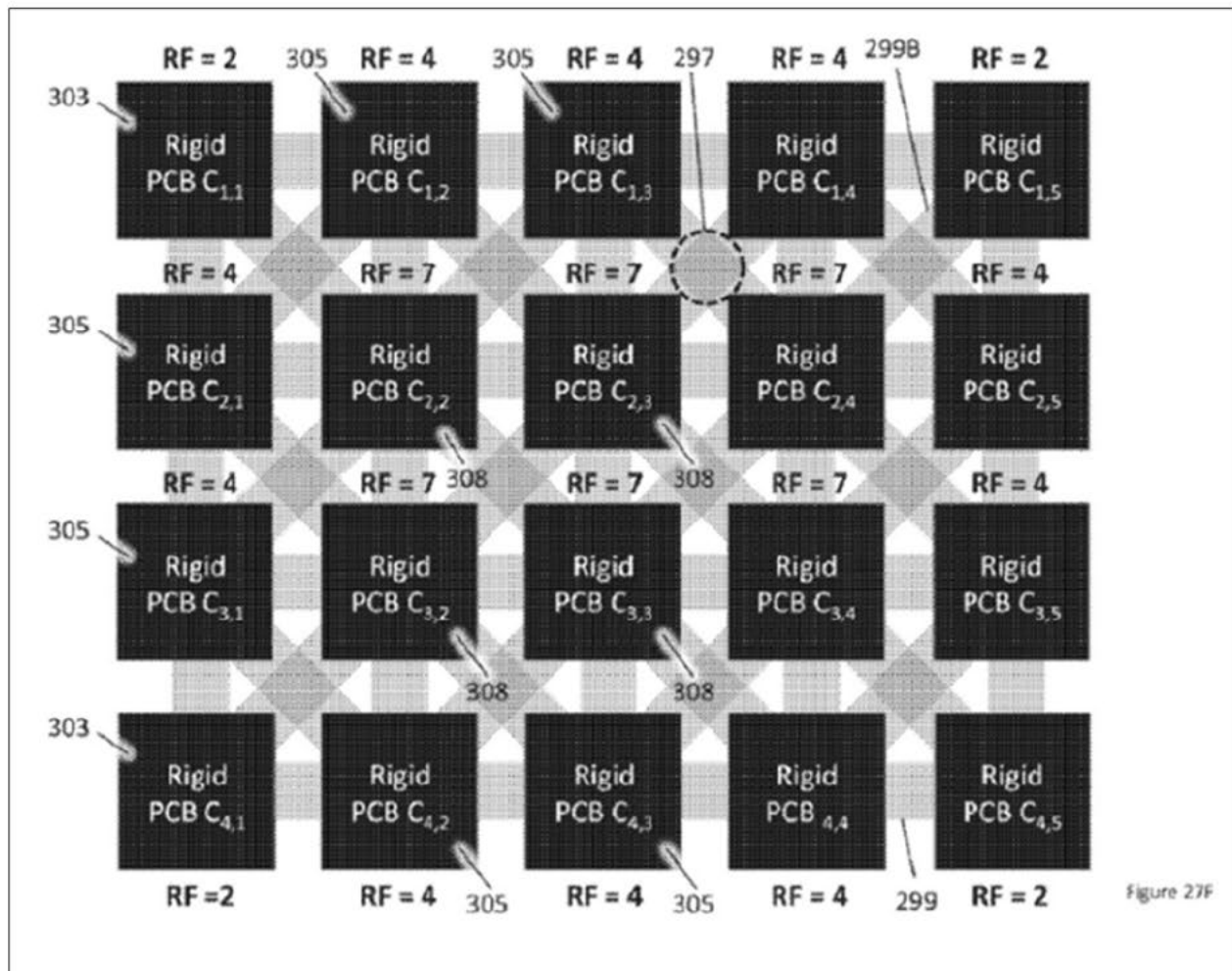
2. Ground 2: Williams Does Not Render Obvious Claims 7 or 9

Oura's Ground 2 asserts that Williams renders obvious Claims 7 and 9. It does not. JDX-0008C, ¶¶91, 104, 105, 113-15. Dr. Mortazavi's analysis is the product of impermissible hindsight, using the claims of the '789 Patent as a roadmap to piece together elements from disparate portions of Williams—the Background, distinct embodiments, and separate fabrication methodologies.. JDX-0008C, ¶¶81-83.

a. Williams Is Directed to Large-Area Distributed PCBs, Not a Portable Communication Device (Limitation [7.pre])

Claim 7's preamble recites "[a] portable communication device," and the body of the claim describes the compact rigid-flex transmission circuit that such a device requires—a single PCB, with two rigid and two flexible portions alternating, that routes high-frequency antenna signals through the narrow interior of the device in place of coaxial cables. JDX-0008C, ¶¶26, 58-61; JX-0004, 1:28-2:34. Williams is directed to the opposite problem.

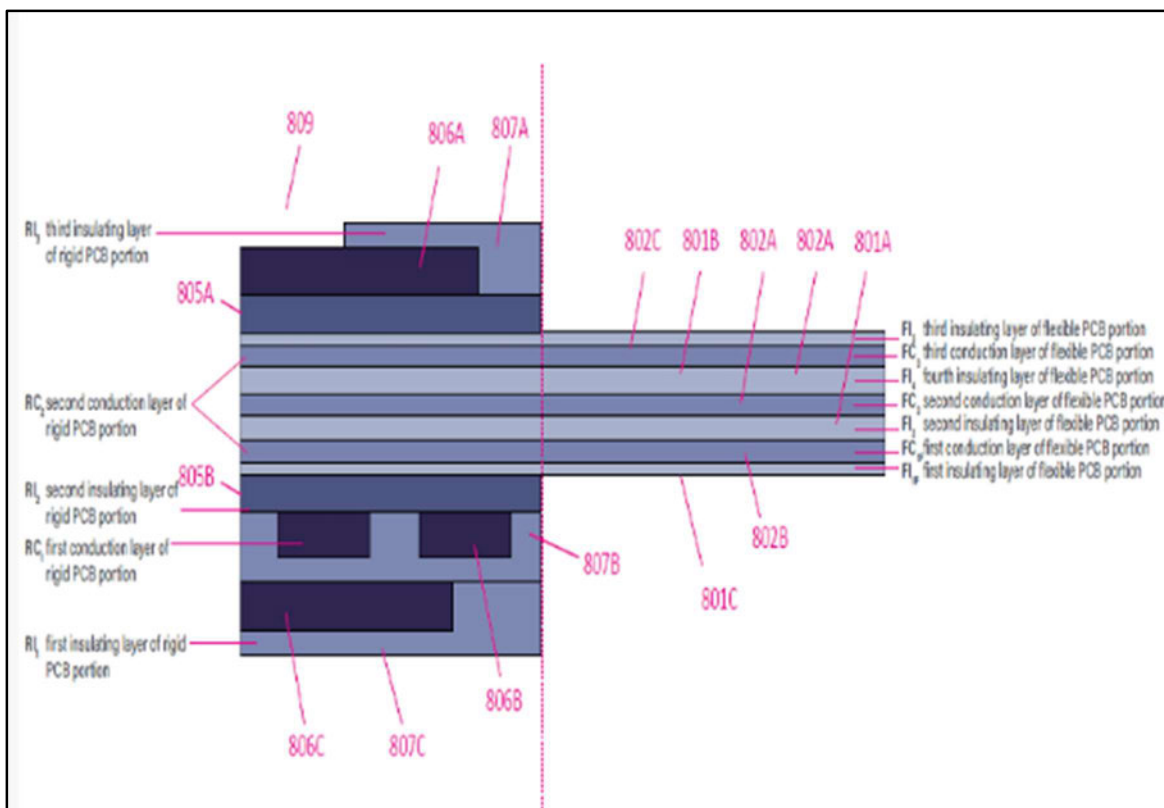
Williams discloses a "rigid-flex PCB fabrication" method for use in the "fabrication of PCBs occupying large areas"—such as the distributed rigid-flex PCB below—not for the compact, space-constrained architecture a portable communication device requires. JDX-0008C, ¶¶81-83.



RX-0076, Fig. 27F; *see also id.*, 68:3-31, 76:30-31, Fig. 99. Indeed, Williams emphasizes that existing flex/rigid-flex fabrication processes “are [] not directly applicable to PCBs covering large areas...but instead are limited to small PCBs, typically the size of cell phones and smaller.” RX-0076, 76:17-22.

b. Dr. Mortazavi Relies on a Mix-and-Match Approach

Dr. Mortazavi relies on disclosures from fundamentally different contexts within Williams that would not be readily combined. JDX-0008C ¶¶83-90. As a base disclosure, Dr. Mortazavi relies on the below cross-sectional schematic of Williams’s Figure 69E. JDX-0001C, ¶¶3613, 3646-47, 3663, 3671, 3685, 3720.



But, because Williams assigns no material composition to any layer, states no dimension for any layer, and depicts only a single rigid-to-flexible transition in Figure 69E, he is forced to combine this figure with unrelated and incompatible embodiments. JDX-0008C, ¶¶101, 104, 112.

[REDACTED]

For the material composition and comparative thicknesses requirements, Mortazavi points to both Williams's Background, and separate discussion of quasi-rigid-flex ("QRF") fabrication. JDX-0008C, ¶¶88-89; JDX-0001C, ¶¶3650, 3654, 3675, 3678. Neither is applicable Figure 69E's structure. The Background provides a summary of the prior art, which compares standalone flex boards to standalone rigid boards, not regions of a single rigid-flex board. JDX-0008C, ¶¶95-96; JDX-0052C, 493:1-11. And, QRF is a different fabrication technology altogether, which Williams describes as an expressly "alternative" and "new" method. JDX-0008C, ¶98; RX-0076, 12:54-13:3, 86:5-17. At his deposition, Dr. Mortazavi conceded that his mapping draws on "at least three things: the embodiment of 69E, the prior art embodiment and then this quasi-rig[id]-flex embodiment." JDX-0052C, 492:22-493:11. That defeats anticipation, which requires that a single embodiment disclose every limitation "arranged or combined in the same way as in the claim." *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1370-71 (Fed. Cir. 2008).

c. Williams Does Not Disclose Different Exterior Materials For the Rigid and Flexible Portions (Limitations[7.c][v][a], [7.c][v][e], [7.c][vi][a], [7.c][vi][e])

The claims require that each rigid portion's exterior surfaces be formed by insulating layers of a "first material," and each flexible portion's exterior surfaces be formed by insulating layers of a "second material different from the first material." JDX-0008C, ¶103.

Williams does not disclose these material requirements. Williams teaches that, after patterning the metal layers in the flexible portions, "protective caps comprising insulating material" are laminated onto the flexible PCB, but never specifies the composition of those cap layers beyond the generic label "insulating material." RX-0076, 81:55-60; JDX-0008C, ¶104. Williams's general listing of candidate materials (e.g., RX-0076, 6:42-49) does not disclose that a specific "second material" forms both exterior surfaces of each flexible portion, much less that a

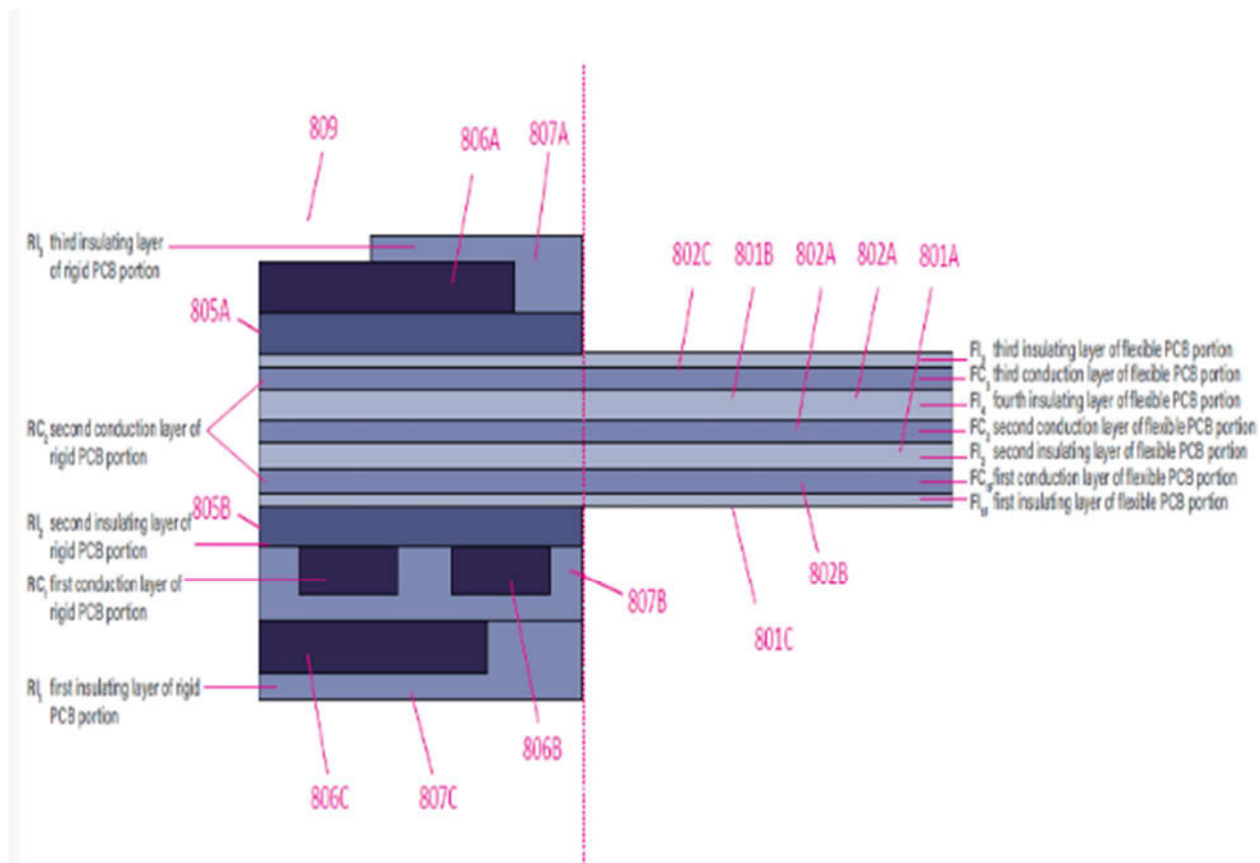
[REDACTED]

different “first material” forms both exterior surfaces of each rigid portion. JDX-0008C ¶¶103-104.

d. Williams Does Not Disclose An Interior Insulating Layer in the Flexible PCB Portions Thinner than a Corresponding Layer in the Rigid PCB Portions (Limitations [7.c][v][c] and [7.c][vi][c])

As claimed, the “second insulating layer” is an interior layer—it sits between two conduction layers—and its thickness in *each flexible portion* must be thinner than that of a corresponding interior layer in *each rigid portion*. JDX-0008C, ¶¶91-92. Meeting the limitation requires identifying a specific interior layer in the flexible portion, a specific interior layer in the rigid portion, and a thickness relationship between the two.

As illustrated below, Dr. Mortazavi maps the flexible portion’s interior layer to Figure 69E’s insulating layer 801A—the flexible insulating layer sandwiched between flex metal layers 802A and 802B—and the rigid portion’s interior layer to Figure 69E’s insulating layer 805B or 807B. JDX-0001C, ¶¶3613, 3671; RX-0076, 71:58-72:30.



JDX-0001C, ¶¶3613 (reproducing and annotating RX-0076, Fig. 69E (RX-0076)). But Dr. Mortazavi never establishes a thickness for either layer, because Figure 69E states none. JDX-0008C, ¶101. His theory instead rests on three substitutes, none of which supplies the comparison.

First, he cites Williams’s Background generalization that flex PCBs “generally utilize much thinner copper layers and thinner insulating substrates than rigid PCBs.” JDX-0008C, ¶¶89, 95; JDX-0001C, ¶3675; RX-0076, 6:42-49. That passage compares standalone flex boards to standalone rigid boards, not the layers within the rigid and flexible portions of one rigid-flex board. JDX-0008C, ¶¶89, 95-96. It is also inconsistent with the embodiment he maps: in Figure 69E, layer 801A is a single *continuous* flexible insulating layer running through both regions—one layer of one thickness, not two separately dimensioned structures. JDX-0008C, ¶97. *Second*, he relies on Williams’s QRF discussion, but QRF is a separate fabrication technology in which a movable

[REDACTED]

print head deposits a thicker polymeric layer on quasi-rigid areas and a thinner layer on flexible areas, calibrated so that etching removes the thinner layer to form vias. JDX-0008C, ¶98; RX-0076, 12:54-13:3, 86:5-17. Those are printed masking deposits, not interior insulating layers positioned between two conduction layers as Claim 7 requires. JDX-0008C, ¶98. *Third*, he relies on the apparent relative thickness of the layers as drawn in Figure 69E. JDX-0001C, ¶3674. But patent drawings do not define proportions absent a statement that they are to scale. *Hockerson-Halberstadt, Inc. v. Avia Grp. Int’l, Inc.*, 222 F.3d 951, 956 (Fed. Cir. 2000); JDX-0008C, ¶101; JDX-0052C, 503:7-504:3. Williams contains no such statement.

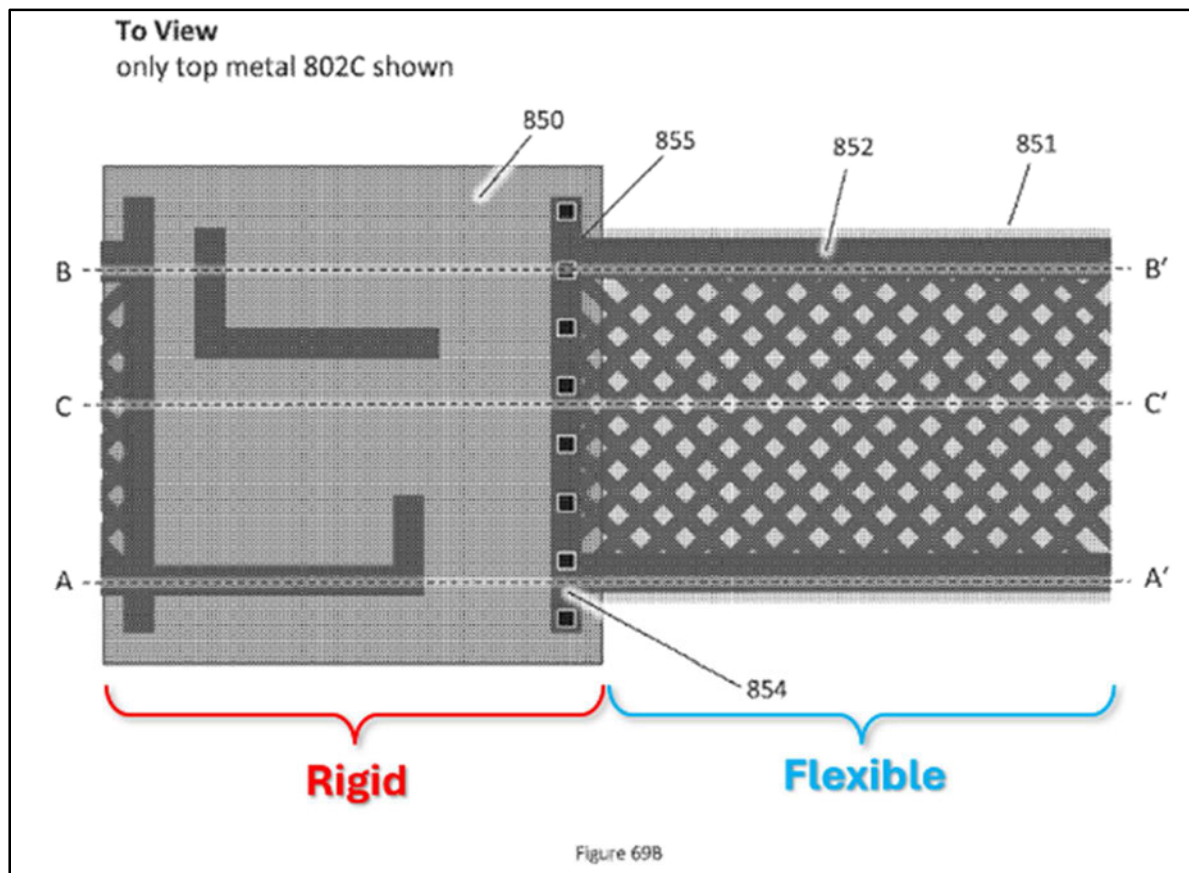
e. Williams Does Not Disclose First and Second Conduction Layers of the Flexible PCB Portions that are “Electrically Connected” with Corresponding Conduction Layers of the Rigid Portions (Limitations [7.c][vi][b] and [7.c][vi][d])

These limitations require proof of actual electrical paths: in each flexible portion, the first and second conduction layers must each connect to the corresponding conduction layer of *both* rigid portions. JDX-0008C, ¶¶74, 105. Dr. Mortazavi again maps to Figure 69E, identifying flex metal layers 802B and 802C as connecting to rigid metal layers 806C and 806B through Williams’s disclosed vias. JDX-0001C, ¶¶3661-63, 3685.

Dr. Mortazavi thus relies on two propositions: that certain flex metal layers (802B, 802C) are “continuous” through the rigid and flexible portions, and that Williams discloses vias that “can” connect various metal layers. JDX-0008C, ¶107; JDX-0001C, ¶¶3661-62. But neither establishes that any specific via *does* connect the specific layer pairings he maps. JDX-0008C, ¶112. Whether a given via at a given location joins flex layer 802B to rigid layer 806C depends on the circuit layout—the routing of traces, the placement of vias, and the signal assignment of each trace—and Williams provides no layout diagrams showing those connections. JDX-0008C, ¶112. This is the same defect Oura’s own corporate representative confirmed is fatal: the presence of vias in a

stack-up does not show where they are placed or which layers they join, and answering that question requires “open[ing] up a layout file and go[ing] through it trace by trace.” JDX-0008C, ¶111; JDX-0025C, 125:8-127:6.

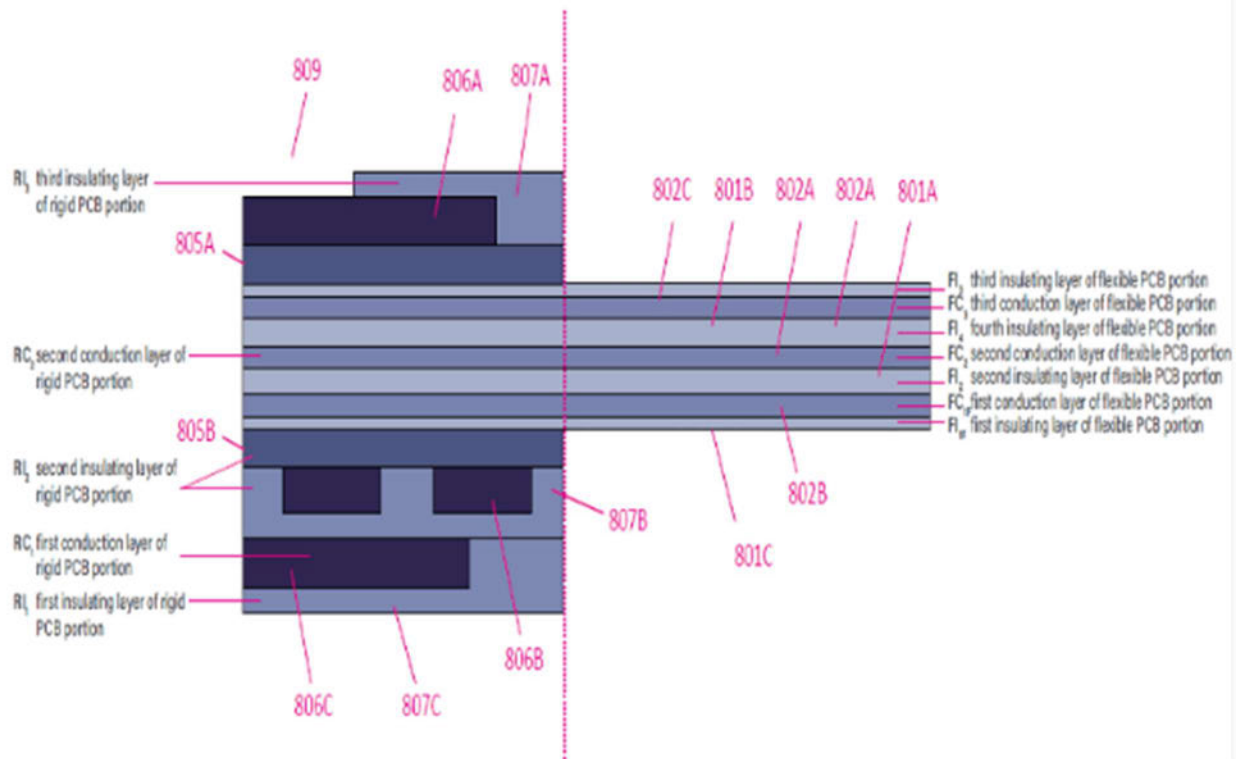
Williams’s own disclosure cuts against the mapping. In one example, Williams explains that flex layer 802C at the flex-to-rigid interface may serve as a “metallic mesh 852” providing “mechanical support” rather than carrying signal, as illustrated below. RX-0076, 74:24-44.



RX-0076, Fig. 68B (annotated). Williams expressly states that “the mesh 852 is not necessarily biased to any circuit potential” and in “[m]any embodiments ... is electrically floating.” RX-0076, 74:28-36. A section of a conduction layer that is “electrically floating” is by definition not “electrically connected with” any other conductive sections. JDX-0008C, ¶110.

Dr. Mortazavi’s annotated reproduction on Figure 69E shows one connection: the second

conduction layer of a first rigid portion joined to the second conduction layer of a first flexible portion. JDX-0008C, ¶112; JDX-0001C, ¶¶3663, 3685. Three of the four required connections are missing from the figure.



JDX-0001C, ¶¶3663 (reproducing and annotating RX-0076, Fig. 69E).

First, the figure does not depict the corresponding *first* conduction layer connection—only the second. JDX-0008C, ¶112. *Second*, the figure does not show a second rigid portion at all, so it cannot show either flexible-portion layer connecting to a second rigid portion, as the claims require. *Id.* *Third*, the figure does not show a second flexible portion, so it cannot show that portion's first or second conduction layers connecting to *either* rigid portion. *Id.*

Dr. Mortazavi's alternative mappings do not close those gaps. He offers dozens of alternative candidates for the rigid-portion layers and, separately, dozens for the flexible-portion layers. JDX-0008C, ¶90; JDX-0001C, ¶¶3517-3521. But he never pairs a specific rigid-layer

mapping with a specific flexible-layer mapping. JDX-0008C, ¶90.

f. Flexible PCB Portions Including Four Insulating Layers and Three Conduction Layers (Limitations [9.a] and [9.b])

Claim 9 adds two layers to each flexible PCB portion, in a specified order: a fourth insulating layer between the second conduction layer and the third insulating layer, and a third conduction layer between the third and fourth insulating layers. JDX-0008C, ¶113. The result is that *each* flexible portion must contain at least three conduction layers and four insulating layers in the recited sequence. *Id.* Dr. Mortazavi maps these to Figure 69E’s insulating layer 801B and metal layer 802C. JDX-0001C, ¶3720.

Two problems defeat that mapping. *First*, the layers are not consistently identified. Dr. Mortazavi relies on multiple alternative “mappings” that relabel and reorient the same Williams layers in different ways. JDX-0008C, ¶114; JDX-0001C, ¶¶3517-3521, 3715-3722; RX-0076, 74:24-44. *Second*, the additional layers must be present in *both* flexible portions. JDX-0008C, ¶115. Dr. Mortazavi identifies no single, consistent Williams embodiment in which each flexible portion includes the claimed additional layers in the claimed order. *Id.* And, because Figure 69E depicts only one flexible portion, there is no second flexible portion in which the additional layers could be found. JDX-0008C, ¶¶112, 115.

3. Ground 3: Williams in View of Aoyama Does Not Render Obvious Claims 7 or 9

Ground 3 uses Williams as the primary reference, and enlists Aoyama to supply “the materials and thicknesses of the PCB layers” that Williams lacks. JDX-0008C, ¶228; JDX-0001C, ¶¶4465-4469. But Dr. Mortazavi offers no non-hindsight reason why a POSITA would combine Williams’s large-area distributed board with Aoyama’s single-flexible-portion bare board. JDX-0008C, ¶¶222-23, 229. And the combination fails to disclose the limitations not disclosed or

rendered obvious by the references individually. *See* Sections VII.C.2 and VII.C.5; *see also* JDX-0008C, ¶¶220-21, 229.

a. No Motivation To Combine and No Reasonable Expectation of Success

Dr. Mortazavi’s motivation to combine rests on generic observations—that Williams and Aoyama are in the same field, that Aoyama teaches material selection for matched coefficients of thermal expansion, that Aoyama teaches thinner insulating layers in flexible portions, and that the combination would have yielded “predictable results.” JDX-0001C, ¶¶4467, 4469. None of it identifies why a POSITA working with Williams’s architecture would have abandoned Williams’s own design choices in favor of Aoyama’s, and Dr. Mortazavi identifies no deficiency in Williams’s design that would have prompted a POSITA to look to Aoyama. JDX-0008C, ¶¶222-23, 229.

Indeed, Williams is directed away from the claimed context. As explained for Ground 2, Williams addresses the fabrication of PCBs “occupying large areas” and distinguishes its methods from those “limited to small PCBs, typically the size of cell phones and smaller.” JDX-0008C, ¶¶82, 229; RX-0076, 76:17-22, 76:30-31. Because Claim 7 recites a “portable communication device,” a POSITA designing the compact rigid-flex transmission circuit the claims require would not have started from Williams. JDX-0008C, ¶¶82, 146.

Aoyama’s teachings are also not transportable into Williams’s architecture. Aoyama’s material and thickness disclosures describe a standalone flexible wiring board whose terminal ends are embedded within separately fabricated rigid boards, as described in further detail below, in the context of Ground 5. JDX-0008C, ¶¶170-71; RX-0074, 5:56-58, 6:17-23, 9:50-62, Figs. 1, 5. Williams, by contrast, describes a rigid-flex PCB as “one single heterogeneous printed circuit board” formed by lamination, in which each flex PCB is merged into the rigid PCB. JDX-0008C, ¶¶183-86; RX-0076, 1:47-51, 8:7-23, 71:26-40. A POSITA would not have transplanted layer

dimensions from an embedded-board construction into a merged, laminated one and expected the claimed layer relationships to follow. JDX-0008C, ¶¶187, 229.

There would have been no reasonable expectation of success. Adopting Aoyama’s flexible-portion materials and thicknesses would have carried with it Aoyama’s flexible stackup, in which conductive-paste shield layers (136, 137) sit on thin insulation layers (134, 135) directly over the signal-carrying conductive layers (132, 133)—a construction that cannot maintain the controlled 50 Ω impedance that antenna feed lines in a portable communication device require. JDX-0008C, ¶¶166-67; RX-0074, 4:13-36, 5:25-32. The ’789 Patent’s purpose was to route high-frequency antenna signals through the narrow interior of a portable device in place of coaxial cables. JX-0004, 1:28-2:34. A POSITA pursuing that objective would not have expected success by importing a flex stackup that degrades impedance control. JDX-0008C, ¶167.

b. Thinner Interior Flexible Layer—Limitations [7.c][v][c] and [7.c][vi][c] (rigid portions include “a second insulating layer located on or above the first conduction layer and having a first thickness”; flexible portions include “a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness”)

Limitations [7.c][v][c] and [7.c][vi][c] require a comparison between two specific interior layers: the “second insulating layer” of each flexible portion, positioned between two conduction layers, must be thinner than the corresponding interior “second insulating layer” of each rigid portion. JDX-0008C, ¶¶91-92. This is the principal feature Dr. Mortazavi asks Aoyama to supply. JDX-0001C, ¶¶4467, 4469. It does not.

Williams does not disclose this relationship, for the reasons given in Ground 2: its Background compares standalone flex boards to standalone rigid boards, not layers within one rigid-flex PCB; its flexible insulating layers (e.g., 801A) run continuously through both regions; and its QRF passage describes printed deposits for via formation, not the claimed layer

relationship. JDX-0008C, ¶¶95-98; RX-0076, 6:42-49, 12:54-67.

Aoyama supplies nothing that cures those defects. Dr. Mortazavi's Aoyama-based thickness evidence is that Aoyama sets its insulation layer thickness at 50-100 μm and that it is "inherent that each layer in Aoyama has a thickness." JDX-0001C, ¶¶4029-4031, 4066-4071; RX-0074, 6:58-63. But, in Williams's design, the flex PCB forms a continuous layer that runs through both the rigid and flexible portions. JDX-0008C, ¶97; RX-0076, 77:1-3. Using Aoyama's disclosed thickness range for an insulating layer spanning both rigid and flexible regions does not establish that a specific interior flexible layer is thinner than a specific interior rigid layer. JDX-0008C, ¶¶91-92, 229..

- c. **Different Exterior Materials—Limitations [7.c][v][a], [7.c][v][e], [7.c][vi][a], and [7.c][vi][e] (rigid portions include first and third insulating layers "including [a] first material" forming their exterior surfaces; flexible portions include first and third insulating layers "including a second material different from the first material" forming their exterior surfaces)**

Williams does not specify the composition of the layers Dr. Mortazavi maps. For the flexible portions, Williams describes laminating "protective caps comprising insulating material" without naming any polymer or compound, so Dr. Mortazavi draws his flexible-portion material from Williams's Background list of conventional standalone flex substrates and from the QRF fabrication discussion. JDX-0008C, ¶¶88, 104; JDX-0001C, ¶¶3650, 3654; RX-0076, 6:42-49, 81:55-60, 86:5-17. Neither passage describes the structural layers of the embodiment he maps, and a generic list of candidate materials does not disclose that a specific "second material" forms both exterior surfaces of each flexible portion while a different "first material" forms both exterior surfaces of each rigid portion. JDX-0008C, ¶¶88, 104.

Aoyama's polyimide coverlays and resin-based rigid materials do not fill that gap. Those

materials are attributes of Aoyama's separately fabricated flexible and rigid wiring boards, and Dr. Mortazavi never explains how substituting them into Williams's merged laminate would yield the same "first material" on *both* exterior surfaces of *each* rigid portion and a different "second material" on *both* exterior surfaces of *each* flexible portion. JDX-0008C, ¶¶104, 170-75, 229; RX-0074, 4:16-20, 4:31-36, 5:25-32, 6:58-63.

d. Remaining Limitations of Claim 7 and the Additional Flexible-Portion Layers of Claim 9

Two further gaps remain, and Aoyama is not offered to cure either. "*Electrically connected*" conduction layers—*Limitations [7.c][vi][b], [7.c][vi][d]*. Dr. Mortazavi's Williams theory rests on the propositions that certain flex metal layers are "continuous" through the rigid and flexible portions and that Williams discloses vias that "can" connect various metal layers; neither establishes that any specific via *does* connect the specific layer pairings he maps. JDX-0008C, ¶¶107, 112; JDX-0001C, ¶¶3661-62. As with the Ring 1, the presence of vias in a stack-up cannot establish that a given conduction layer in one portion is electrically connected to a given conduction layer in another without a layout file or a cross-section. JDX-0008C, ¶¶111-12. Williams supplies no layout diagrams, and Aoyama supplies none for Williams's architecture. JDX-0008C, ¶112.

Additional Flexible-Portion Layers—limitations [9.a] and [9.b]. Claim 9 requires a fourth insulating layer and a third conduction layer in the recited order in *each* flexible portion. JDX-0008C, ¶¶113-15. Williams offers only a menu of inconsistent alternative mappings that relabel and reorient the same layers, and Dr. Mortazavi identifies no single, consistent Williams embodiment in which each of two flexible portions includes the claimed additional layers in the claimed order. JDX-0008C, ¶¶114-15; JDX-0001C, ¶¶3517-3521, 3715-3722. Aoyama, which has only one flexible portion, cannot supply that structure. JDX-0008C, ¶¶229, 231-32. Ground

[REDACTED]

5 accordingly does not render Claim 7 or Claim 9 obvious. JDX-0008C, ¶229.

4. Ground 4: Williams in View of Naganuma Does Not Render Obvious Claims 7 or 9

In Ground 4, Williams is the primary reference, and Naganuma is enlisted to supply the layer materials and thicknesses that Williams lacks. JDX-0008C, ¶230; JDX-0001C, ¶¶4470-4474. But the combination fails to disclose the limitations not disclosed or rendered obvious by the references individually. *See* Sections VII.C.2 and VII.C.4.d below; *see also* JDX-0008C, ¶¶220-21, 231. Dr. Mortazavi’s motivation for this ground is, word for word, the rationale he gives for the Williams-Aoyama combination of Ground 3. JDX-0001C, ¶¶4469, 4474; JDX-0008C, ¶231. That interchangeability is itself the defect: a rationale that applies identically to any rigid-flex reference identifies no reason a POSITA would have made *this* combination. JDX-0008C, ¶222.

a. Naganuma Is an Assembly of Three Separately Fabricated Boards, Not a PCB With Integrated “Portions”

Limitations [7.c][i]-[iv] require “portions” of one integrated board rather than separately fabricated boards later joined together. JDX-0008C, ¶¶180-82. Naganuma discloses the opposite. As shown in Fig. 1, it is a composite wiring board assembled from three independently manufactured components: a first rigid wiring board 11, a second rigid wiring board 12, and a flexible wiring board 13. JDX-0008C, ¶177; RX-0075, ¶72, Fig. 1.

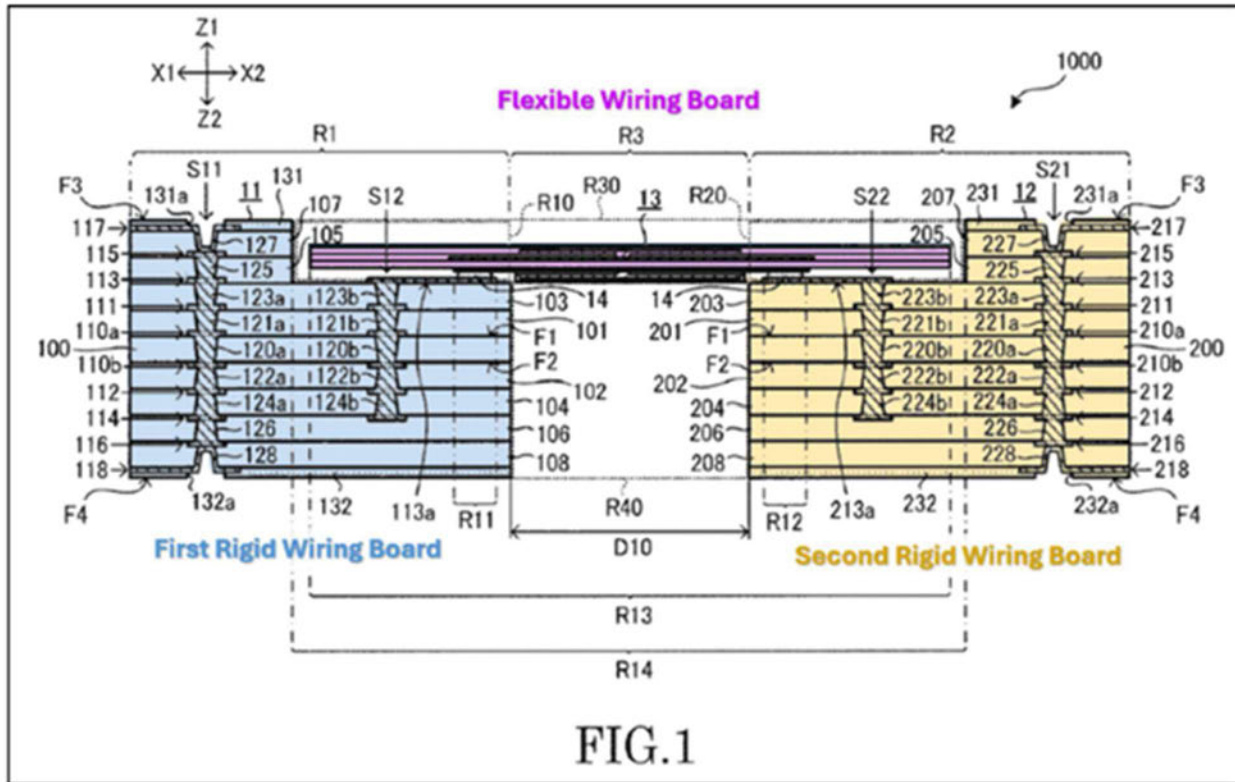


FIG. 1

JDX-0008C, ¶177 (reproducing RX-0075, Fig. 1 (annotated)). Naganuma states that each board is “collectively formed” in its own panel and then “separated from the assemblies” to produce individual units, and the boards are joined only afterward through bonding material at discrete pad interfaces using ACF, solder, or similar bonding agents. JDX-0008C, ¶¶177-79; RX-0075, Abstract, ¶¶6-7, 107-08, 130-31, 137, 142, 183, Figs. 3A-3B.

[illegible]

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manufacturing, not a single PCB with integrated “portions.” JDX-0008C, ¶179; JDX-0050C, 118:16-22 (Oura’s expert, Dr. Mercier agreeing that one of the reasons a rigid-flex PCB is considered “one singular structure” is “because it’s got continuous conductive layers running through the rigid and flex portions”).

By contrast, Williams explains that in a rigid-flex PCB “the entire matrix including both the rigid PCB portions and the flex PCB interconnections merged into the rigid PCB elements comprises one single heterogeneous printed circuit board,” with “portions” referring to regions of that unified board formed by lamination. JDX-0008C, ¶¶183-86; RX-0076, 1:47-51, 8:7-23, 71:26-40. Naganuma’s three separately fabricated, pad-bonded boards cannot be treated as “portions” of one PCB under the very reference Dr. Mortazavi pairs with it. JDX-0008C, ¶187.

b. No Motivation To Combine and No Reasonable Expectation of Success

Dr. Mortazavi asserts that Williams and Naganuma are in the same field of endeavor; that Naganuma provides “express teachings on the selection of materials for flexible PCB portions” and on matched coefficients of thermal expansion; that Naganuma teaches using thinner insulating layers in flexible portions than in rigid portions; that both references teach solder resist for rigid exterior surfaces and polyimide for flexible exterior surfaces; and that the combination would have yielded predictable results. JDX-0001C, ¶¶4472-4474. Three problems defeat that rationale.

First, the rationale is generic. It rests on the observation that both references relate to rigid-flex PCBs, which does not explain why a POSITA would have extracted particular layer material and thickness teachings from Naganuma and grafted them onto Williams’s architecture, and Dr. Mortazavi identifies no deficiency in Williams’s own design that would have prompted a POSITA to seek improvement from Naganuma. JDX-0008C, ¶¶222-23. *Second*, Williams is directed away from the claimed context, as explained above in connection with Grounds 2, 4, and 5. JDX-0008C,

¶¶82, 231; RX-0076, 76:17-22. *Third*, the two references describe structurally incompatible constructions: Naganuma's layer specifications are attributes of independently manufactured boards joined at pads after fabrication, while Williams's "portions" are regions of one laminated board. JDX-0008C, ¶¶177-79, 183-87. A POSITA would not have expected that transplanting dimensions from the former into the latter would produce the claimed layer relationships. JDX-0008C, ¶¶187, 231.

c. Thinner Interior Insulating Layer in the Flexible PCB Portions (Limitations [7.c][v][c] and [7.c][vi][c])

Williams does not disclose this comparative relationship for the reasons given above in connection with Ground 2: its Background statement compares standalone flex boards to standalone rigid boards; its flexible insulating layers (e.g., 801A) run continuously through both regions rather than existing as separately dimensioned structures; its QRF passage describes printed insulator deposits used to define via locations rather than the claimed layer relationship; and Williams nowhere states that Figure 69E is drawn to scale. JDX-0008C, ¶¶95-98, 101; RX-0076, 6:42-49, 12:54-67.

Naganuma does not cure that gap. Dr. Mortazavi's Naganuma-based thickness evidence is that Naganuma's figures "show" a second insulating layer with a thickness and that it is "inherent that each layer in Naganuma has a thickness." JDX-0001C, ¶¶4225-4227, 4278-4281. Naganuma never states the thickness of the mapped interior insulating layer in its flexible board as compared to the corresponding layer in its rigid boards, and the proposition that every layer has *some* thickness cannot establish the comparison the claim requires. JDX-0008C, ¶¶91-92, 231. Nor could the comparison transfer even if Naganuma had made it: its flexible and rigid wiring boards are separate boards with independent stackups, so their relative layer thicknesses say nothing about the interior layers of the rigid and flexible *portions* of a single laminated board. JDX-0008C,

¶¶177-79, 231.

- d. **Different Exterior Materials—Limitations [7.c][v][a], [7.c][v][e], [7.c][vi][a], and [7.c][vi][e] (rigid portions include first and third insulating layers “including [a] first material” forming their exterior surfaces; flexible portions include first and third insulating layers “including a second material different from the first material” forming their exterior surfaces)**

As explained in connection with Ground 2, Williams does not identify the composition of the layers Dr. Mortazavi maps as forming the exterior surfaces, and the material lists he borrows come from Williams’s Background discussion of conventional standalone flex PCBs and from the separate QRF discussion rather than from the embodiment he maps. JDX-0008C, ¶¶88, 104; RX-0076, 6:42-49, 81:55-60, 86:5-17. Naganuma does not supply the missing distinction either. Naganuma states only that its insulation layers can be made of various resins, including epoxy, polyester, polyimide, and phenol resin, or formed with multiple layers of different materials. JDX-0001C, ¶4223; RX-0075, ¶90. A list of candidate resins that may be used anywhere in the board does not disclose that a specific “first material” forms both exterior surfaces of each rigid portion while a different specific “second material” forms both exterior surfaces of each flexible portion. JDX-0008C, ¶¶104, 231.

- e. **Remaining Limitations of Claim 7 and the Additional Flexible-Portion Layers of Claim 9**

The remaining disputed features fail for reasons already given.

Alternating rigid and flexible PCB portions—Limitations [7.c][i]-[iv]. Naganuma’s three separately fabricated, pad-bonded boards are not “portions” of one PCB, and Naganuma as depicted contains only a single flexible board between two rigid boards, so it does not disclose the second flexible PCB portion in contact with the second rigid portion. JDX-0008C, ¶¶177-79, 231.

Electrically connected conduction layers—Limitations [7.c][vi][b], [7.c][vi][d]. Naganuma is

[REDACTED]

not offered to cure this feature, and Williams’s own disclosure establishes only that certain layers are “continuous” and that vias “can” connect layers—not that any specific via connects the specific layer pairings mapped—which is precisely the showing Oura’s corporate representative confirmed cannot be made from a stack-up without a layout file or a cross-section. JDX-0008C, ¶¶107, 111-12; JDX-0025C, 125:8-127:6.

Additional Flexible-Portion Layers—limitations [9.a] and [9.b]. Neither Williams nor Naganuma discloses a fourth insulating layer and a third conduction layer arranged as claimed in *each* of two flexible portions, and combining two references that each lack that structure does not supply it. JDX-0008C, ¶¶114-15, 231. Ground 4 accordingly does not render Claim 7 or Claim 9 obvious. JDX-0008C, ¶231.

5. Ground 5: Aoyama Does Not Render Obvious Claims 7 or 9

a. Aoyama Does Not Disclose or Suggest a Second Flexible PCB Portion (Limitation [7.c][iv])

Claim 7 requires a single PCB containing two rigid and two flexible portions arranged in a rigid-flexible-rigid-flexible sequence, each in contact with the next. Aoyama does not disclose that four-part sequence. JDX-0008C, ¶¶164-167. Aoyama’s completed board contains at most one flexible portion between two rigid portions—rigid-flexible-rigid, not rigid-flexible-rigid-flexible, as required. *Id.*; RX-0074, Figs. 1-2, 36 .

Consistent with that disclosure, Dr. Mortazavi maps the claimed first and second rigid PCB portions to Aoyama’s rigid wiring boards 11 and 12, and the claimed first flexible PCB portion to Aoyama’s flexible wiring board 13 (blue, below). JDX-0001C, ¶¶3966-3993; RX-0074, 4:1-12, 5:33-40, Figs. 1, 2.

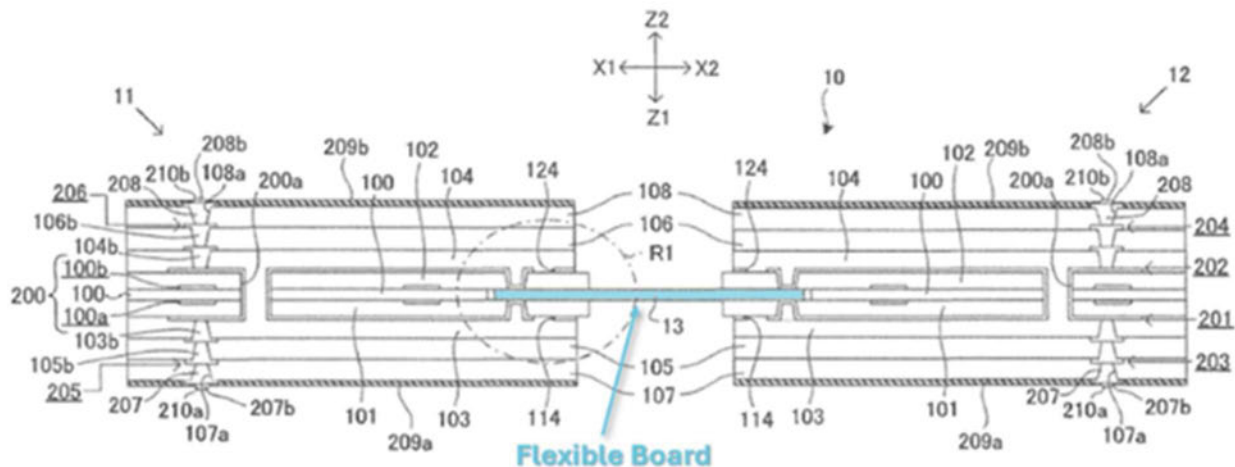


FIG.1

RX-0074, Fig. 1 (annotated). But Dr. Mortazavi identifies no structure in Aoyama corresponding to the claimed “second flexible PCB portion.” Instead, he argues that “it would have been obvious to modify Aoyama to include a second flexible PCB portion.” JDX-0001C, ¶¶3996-3997. But Aoyama provides neither a reason to make that modification nor a reasonable basis to expect that simply appending another flexible section would produce the claimed architecture. JDX-0008C ¶¶165-167.

As support for his obviousness theory, Dr. Mortazavi points to Aoyama’s statement that its board “may be used as a circuit board for cell phones or the like by being mounted onto other wiring boards.” JDX-0001C, ¶¶4090-4110; RX-0074, 7:62-66. Read in context, that passage teaches the opposite of his proposed modification: Aoyama contemplates that its rigid-flex-rigid board is a discrete, self-contained subsidiary module—a component that plugs into a larger system via external connection terminals (207b, 208b)—not a board that may itself be extended with additional segments to become the device’s main signal-routing backbone. JDX-0008C, ¶¶165-66; RX-0074, 7:62-67. That modular approach makes engineering sense for a cell phone. JDX-0008C ¶¶166-167. As Dr. Warren explains, antenna feed lines carry high-frequency RF signals,

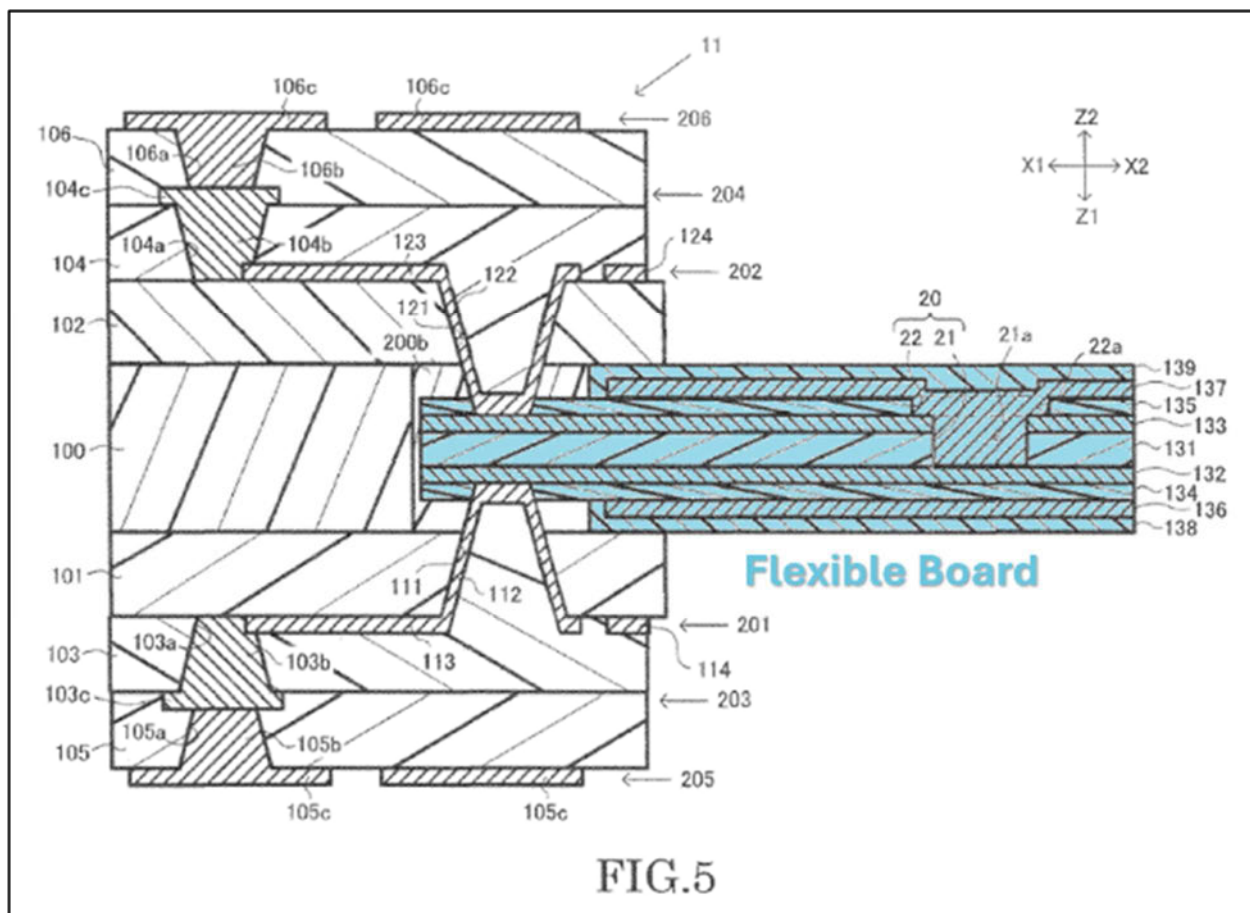
[REDACTED]

which must maintain precisely controlled characteristic impedance (typically 50 Ω) along their entire path. *Id.* ¶166. Aoyama's board, however, is not designed for impedance-controlled RF transmission. Its flexible section includes conductive-paste shield layers (136, 137) separated from the signal-carrying conductive layers (132, 133) by only thin insulating layers (134, 135). *Id.*; RX-0074, 4:13-36, 5:25-32. This configuration—while well-suited for shielding digital interconnects (RX-0074, 11:60-12:9)—would create excessive parasitic capacitance that would drive the characteristic impedance of any signal line far below the 50 Ω target required for RF transmission, causing severe impedance mismatch and signal degradation at the frequencies used in cellular communication. JDX-0008C, ¶¶166-67.

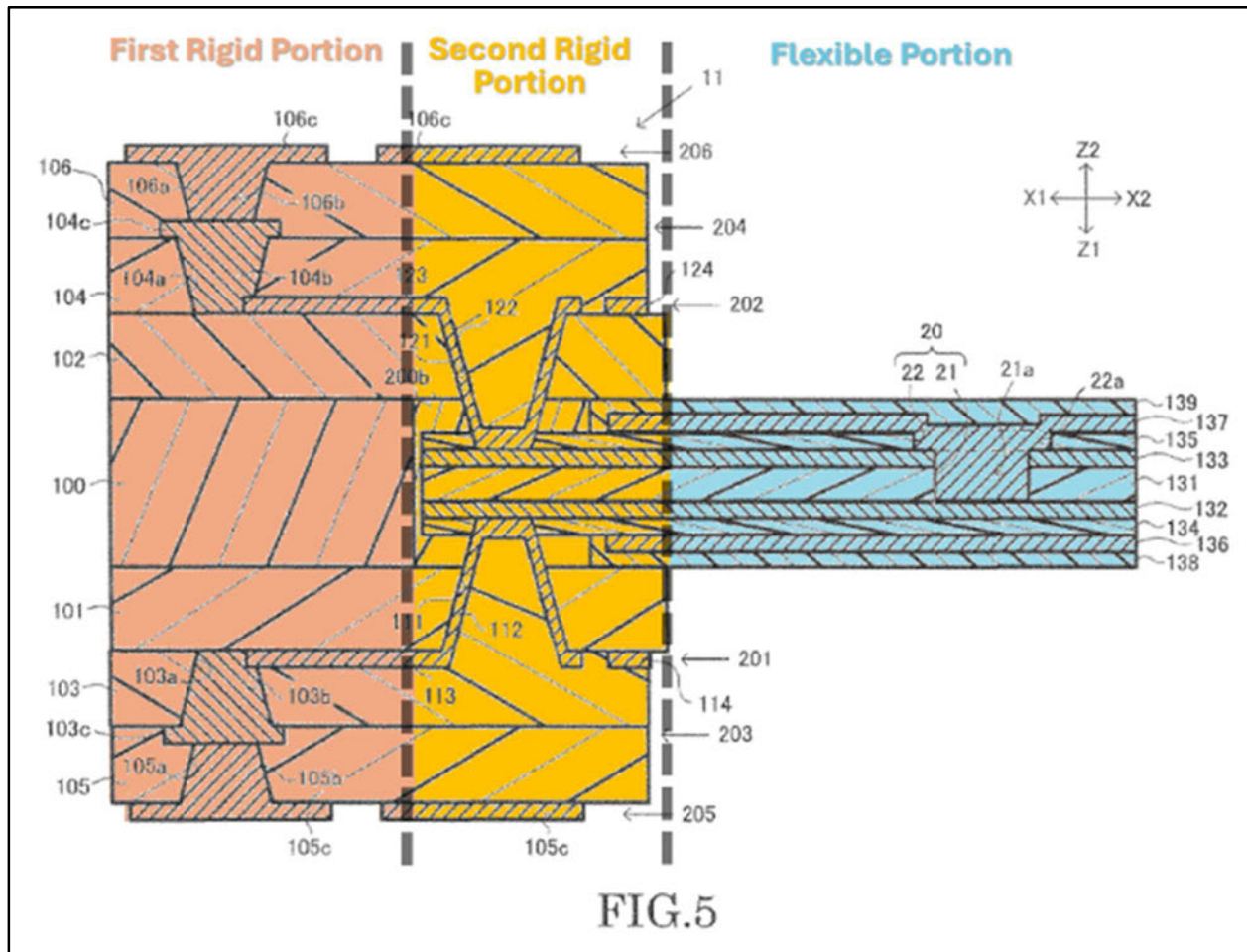
b. Aoyama Does Not Teach the Claimed Rigid and Flexible PCB Portions (Limitations [7.c][i]-[iv])

Claim 7 defines each rigid and flexible PCB portion by its own laminated structure, including insulating layers that form that portion's exterior surfaces. JDX-0008C, ¶¶168-69. Aoyama discloses a different structure: a separately fabricated flexible wiring board 13 whose terminal ends are inserted into, sandwiched within, and polymerized with rigid wiring boards 11 and 12 during lamination. JDX-0008C, ¶¶170-71; RX-0074, 5:56-58, 6:17-23, 9:50-62, Figs. 1, 5.

That embedded construction defeats Dr. Mortazavi's mapping. If the embedded ends are part of the alleged flexible portion, then the coverlays he identifies as exterior surfaces are buried inside the rigid-board laminate in those areas—i.e., they are internal layers, not exterior surfaces (as illustrated below). JDX-0008C, ¶¶172-73.



JDX-0008C ¶170 (reproducing RX-0074, Fig. 5 (annotated)). On the other hand, if the embedded ends are considered part of the rigid portions, then Aoyama no longer has a rigid-flexible-rigid sequence: the presence of the embedded flex creates a distinct rigid region having a different layer stack (including a different number of conductive layers), as compared to the rigid region without the embedded flex, as illustrated below. JDX-0008C, ¶¶174-75. That conclusion follows Dr. Mortazavi's own approach: in his [REDACTED] analysis, he treats regions with different metal-layer counts as distinct PCB regions. JDX-0007C ¶¶672-675; JDX-0052C, 554:2-555:2.



JDX-0008C ¶174 (reproducing RX-0074, Fig. 5 (annotated)). Either way, Aoyama does not disclose the claimed arrangement. JDX-0008C, ¶175.

6. Ground 6: Aoyama in View of Williams Does Not Render Obvious Claims 7 or 9

Ground 6 attempts to add Williams to supply the second flexible PCB portion missing from Aoyama. JDX-0008C, ¶234; JDX-0001C, ¶¶4480-4482. This is impermissible hindsight: Aoyama is a bare board with a single flexible portion, and Dr. Mortazavi proposes to add on a second flexible portion drawn from a different reference for no reason other than that Claim 7 recites one. JDX-0008C, ¶235. Williams also does not cure Aoyama's remaining deficiencies—it lacks the comparative interior-layer thicknesses and the specified exterior materials. JDX-

0008C, ¶¶229, 235. In addition, the combination of Aoyama and Williams fails to disclose the limitations of the claims that neither reference discloses or renders obvious individually (*see* Sections VII.C.2 and VII.C.5) and that Oura’s proposed combination does not address. JDX-0008C, ¶¶220-21, 234-34.

a. No Motivation To Combine and No Reasonable Expectation of Success

Dr. Mortazavi’s stated motivation to combine—that Williams is in Aoyama’s field of endeavor, it discloses multiple alternating rigid and flexible portions, and that a POSITA extending Aoyama’s board would have found it obvious that the second flexible portion would share the first portion’s layer structure—assumes the very structure it is supposed to supply. JDX-0001C, ¶¶4480-4482; JDX-0008C, ¶235.

Aoyama itself points the other way. As explained for Ground 3, Aoyama’s board is a discrete, self-contained module that plugs into “other wiring boards” through external connection terminals—not a multi-segment main board to be extended with additional flexible and rigid segments. JDX-0008C, ¶¶164-66; RX-0074, Abstract, 7:62-67, Figs. 1, 2, 36. A POSITA would have understood Aoyama’s board to be mounted onto a larger system, not enlarged into one. JDX-0008C, ¶¶165-66.

Williams, for its part, is directed away from the claimed context. As explained for Ground 2, Williams discloses a fabrication method for “PCBs occupying large areas,” and expressly distinguishes its methods from those “limited to small PCBs, typically the size of cell phones and smaller.” JDX-0008C, ¶¶82, 229; RX-0076, 76:17-22, 76:30-31; JDX-0052C, 486-88. A POSITA designing the compact rigid-flex transmission circuit of the ’789 Patent would not have looked to Williams to extend Aoyama. JDX-0008C, ¶¶82, 146.

Dr. Mortazavi’s fallback—that a thinner interior insulating layer in the flexible portions

[REDACTED]

aids flexibility while a thicker layer in the rigid portions aids rigidity—is a generic design observation that identifies no reason a POSITA would have selected these two references to arrive at the specific claimed architecture with a reasonable expectation of success. JDX-0008C, ¶229; JDX-0001C, ¶4469.

**b. Alternating Rigid and Flexible PCB Portions
(Limitations[7.c][i]-[iv])**

As explained for Ground 5, Aoyama discloses at most one flexible portion, and its flexible wiring board 13 terminates *inside* the two rigid boards rather than meeting them at an interface. JDX-0008C, ¶¶164, 170-75. Williams cannot cure either defect.

As explained above, Williams discloses distributed arrays of rigid PCB “islands” interconnected by redundant flex connectors—the architecture Williams designed for large-area, body-conforming products such as the “LightPad” phototherapy device, not the compact device Claim 7 requires. JDX-0008C, ¶¶82, 86; RX-0076, Abstract, 68:3-31, 88:26-38, Figs. 27F, 99. That architecture is unsuited to the claimed device. Williams gives each rigid PCB redundant parallel current paths so that “a breakage of a line will not necessarily affect the operation of the rigid PCB,” requiring duplicating power and signal lines—consuming substantial board area and volume, incompatible with a portable communication device’s constraints. JDX-0008C, ¶87; RX-0076, Abstract, 68:3-31.

Adding a second flexible segment also would not cure Aoyama’s separate structural defect. Aoyama’s flexible wiring board 13 is separately fabricated, and its terminal ends are embedded within the rigid wiring boards during lamination—so the coverlays Dr. Mortazavi maps as the flexible portion’s exterior surfaces are buried inside the rigid-board laminate and are not exterior surfaces. JDX-0008C, ¶¶170-73; RX-0074, 5:56-58, 6:17-23, 9:50-62, Figs. 1, 5. Adding a second embedded flexible board using that same connection methodology would replicate the defect

rather than cure it. JDX-0001C, ¶¶3997-3998; JDX-0008C, ¶¶173-75.

c. Different Exterior Materials, Thinner Interior Flexible Layer, and Electrically Connected Layers (Limitations [7.c][v][c], [7.c][vi][c], [7.c][vi][b], [7.c][vi][d])

Dr. Mortazavi attempts to rely on Williams to fill Aoyama's remaining gaps, but Williams lacks these features too. As explained for Ground 2, Williams does not specify the materials used for the layers Dr. Mortazavi maps as the flexible portions' exterior surfaces, beyond "protective caps comprising insulating material," and Dr. Mortazavi fills that gap with Williams's Background and QRF discussion, neither of which describes the embodiment he maps. JDX-0008C, ¶¶88, 104; RX-0076, 6:42-49, 81:55-60, 86:5-17.

Williams does not disclose the comparative thickness relationship. Its Background statement comparing standalone flex boards to standalone rigid boards, does not address layers within one rigid-flex PCB. JDX-0008C, ¶¶95-97; RX-0076, 6:42-49. Dr. Mortazavi's remaining thickness evidence is the visual appearance of layer 801A in Figure 69E, as shown in Section VII.C.2.b above, and he could not identify anything in Williams stating that its figures are drawn to scale. JDX-0008C, ¶101; JDX-0001C, ¶3674; JDX-0052C, 503-04. On electrical connectivity, Williams provides no layout diagrams, and Dr. Mortazavi points to none. JDX-0008C, ¶¶107, 112.

Aoyama cannot fill those gaps either. Whatever materials and thicknesses Aoyama specifies are attributes of a standalone flexible wiring board and separately fabricated rigid boards, not of flexible and rigid *portions* of one integrated PCB. JDX-0008C, ¶¶170-75; RX-0074, 4:16-20, 5:25-32, 6:58-63, Figs. 1, 5. Importing those values into Williams's architecture therefore does not establish that a specific interior flexible layer is thinner than its rigid-portion counterpart, or that a specific "first material" forms both exterior surfaces of each rigid portion while a different

“second material” forms both exterior surfaces of each flexible portion. JDX-0008C, ¶¶229, 235.

d. Additional Flexible-Portion Layers (Limitations [9.a] and [9.b])

Claim 9 requires that *each* flexible portion include a fourth insulating layer and a third conduction layer in the recited. JDX-0008C, ¶¶113-15. Ground 4 fails as to Claim 9 for the threshold reason that the combination does not disclose the Claim 7 architecture from which Claim 9 depends. JDX-0008C, ¶¶231-32, 234-35. Neither reference discloses the additional layers arranged as claimed in each of two flexible portions, and combining two references that each lack that structure does not supply it. JDX-0008C, ¶¶114-15, 227, 235; JDX-0001C, ¶¶3517-3521, 3715-3722.

7. Ground 7: The Combination of the Galaxy S5, Mow, and Kim Does Not Render Obvious Claim 6

Oura advances only a single invalidity ground against Claim 6: a three-reference combination of Galaxy S5, Mow, and Kim. Claim 6 depends from Claim 1, so the combination of the Samsung Galaxy S5, Mow, and Kim must render obvious every limitation of Claim 1 before reaching the additional limitations of Claim 6. JDX-0008C, ¶¶237, 267. It does not for the reasons detailed below.

a. No Motivation to Combine or Reasonable Expectation of Success

Oura’s combination proceeds in three steps: Dr. Mortazavi starts with the Galaxy S5—a commercial smartphone with a rigid-flex sub PCB—as the base platform, he then turns to Mow to supply the conductive housing portions required by limitation [1.a], and he relies on Kim to supply the specific arrangement of signal and ground lines recited in limitation [1.c][v][a]. JDX-0001C, ¶¶4486-4498, 4508, 4554-60. His stated reason for assembling them is that the three references address “different but related aspects of portable communication device design.” JDX-0001C,

[REDACTED]

¶4491.

That rationale defeats itself. The fact that each reference solves a distinct, unrelated technical problem shows a POSITA would not naturally have looked to all three together absent hindsight. JDX-0008C, ¶238. Mow's core contribution is real-time antenna adjustability, not PCB design or signal routing; a POSITA consulting Mow would be seeking antenna tuning solutions. JDX-0008C, ¶240; RX-0110, Abstract, ¶¶6-7. Kim discloses a purely flexible board with a coplanar signal-line arrangement using a central shielding part—fundamentally different from a rigid-flex PCB with alternating rigid and flexible portions—and does not address rigid-flex architectures, housing designs, or antenna integration at all. JDX-0008C, ¶¶241-43; RX-0112, ¶¶7, 8, 10, 43, 46, 52. Transplanting Kim's architecture into a multi-segment rigid-flex PCB would have required substantial redesign, not the straightforward application of known techniques. JDX-0008C, ¶243.

Dr. Mortazavi confirmed the disconnect at deposition, acknowledging that neither Mow nor Kim addresses PCB design or high-frequency transmission in a smartphone and explaining only that “that's why you have this motivation to combine”—offering the references' silence on the claimed application as the reason to combine them. JDX-0052C, 528:1-16; JDX-0008C, ¶¶238, 255-56.

Even assuming a POSITA could have attempted the combination, there would have been no reasonable expectation of success, because Kim's architecture is incompatible with the S5 flex it would have to be installed in. [REDACTED]

[REDACTED]

[REDACTED]

Kim's low-loss design depends on an air interface that the [REDACTED]. Kim reduces signal loss by arranging its signal and ground lines coplanar on a single flexible layer around a central shielding part and exposing the signal lines to an air layer, minimizing parasitic capacitance and high-frequency loss. JDX-0008C, ¶248; RX-0112, ¶¶7, 26-29, 43, 46. The S5's

[REDACTED]

flex offers no such interface: its outer surfaces are conductive shield sitting against the stack, so a Kim-style line set installed there would be capacitively loaded on both sides rather than exposed to air. JDX-0008C, ¶248. The shields would function as unaccounted-for ground planes altering the characteristic impedance of the transmission lines and would reintroduce the parasitic capacitance Kim’s air-layer design exists to eliminate, so a POSITA would obtain none of the benefit Dr. Mortazavi invokes as the reason to adopt Kim. *Id.*

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

A POSITA would also face a compounding problem. [REDACTED]

[REDACTED]—that Kim’s architecture provides no way to accommodate, and adding them would require a substantially wider flex portion that may not fit within the S5’s constrained internal geometry. JDX-0008C, ¶¶249, 252-53; RX-0085C, 30134, 30136; RX-0112, Fig. 1.

The S5’s own design choices confirm the point. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] What remains is hindsight: a POSITA working on the S5 had no reason to abandon its internal antennas for Mow’s

housing antennas, and a POSITA reading Kim had no reason to apply its teachings to a rigid-flex PCB. JDX-0008C, ¶¶255-56.

- b. **Limitation [1.a] (“a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;”)**

Dr. Mortazavi concedes the S5 does not disclose this limitation, acknowledging that “to the extent the S5 does not explicitly disclose” conductive housing portions operating as antennas, “a POSITA would find it obvious to modify the S5 to include this feature.” JDX-0001C, ¶4508. The S5 is a commercial smartphone with a plastic housing and internal antennas. JDX-0008C, ¶239; RX-0078, SAMS382-0034414.

The screenshot displays the Samsung Galaxy S5 product page. At the top, the title "Samsung Galaxy S5" is shown. Below it, a smartphone image is displayed alongside key specifications: Released 2014, April; 145g, 8.1mm thickness; Android 4.4.2, up to 6.0, TouchWiz UI; 16GB/32GB storage, microSDXC; 5.1" 1080x1920 pixels; 16MP 2160p; 2GB RAM Snapdragon 801; 2800mAh. To the right, there is a fan badge for 2623 fans and a hit count of 69,652,622. Below the specifications, there are tabs for REVIEW, OPINIONS, COMPARE, PICTURES, and 360° VIEW. A section titled "ALL VERSIONS" lists G900F, G900I, G900M, G900A, and G900T, with a note about their regional availability. A table follows, detailing the phone's network capabilities, launch date, and body specifications. The "BODY" section highlights the "plastic back, plastic frame" in a yellow box.

ALL VERSIONS	G900F	G900I	G900M	G900A	G900T
Versions: G900F (Europe), G900I (Asia, Australia), G900M (Vodafone), G900A (AT&T), G900T (T-Mobile), G900W8 (Canada), G900K/G900L/G900S (Korea)					
NETWORK	Technology	GSM / HSPA / LTE			
LAUNCH	Announced	2014, February. Released 2014, April			
	Status	Discontinued			
BODY	Dimensions	142 x 72.5 x 8.1 mm (5.59 x 2.85 x 0.32 in)			
	Weight	145 g (5.11 oz)			
	Build	Glass front (Gorilla Glass 3), plastic back, plastic frame			
	SIM	Micro-SIM			
	IP67 dust/water resistant (up to 1m for 30 min)				

Galaxy S5 housing and internal antenna configuration, RX-0078, SAMS382-0034414

[REDACTED]

(highlighted). Incorporating Mow’s housing-integrated antenna design would not have been a minor adjustment—it would have required replacing the plastic frame and back cover with a conductive metal structure segmented by dielectric gaps, and redesigning at least two antennas and the antenna-interfacing circuitry, fundamentally altering the S5’s industrial design, manufacturing process, and internal component layout. JDX-0008C, ¶239. A POSITA would have had no reason to undertake that redesign of a commercially successful product. JDX-0008C, ¶239.

- c. **Limitation [1.c][v][a] (“a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion ... the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.”)**

This limitation requires a pair of signal lines, each electrically connected to an antenna, that traverse the full rigid-flex-rigid-flex assembly from the first PCB portion to the fourth. JDX-0008C, ¶¶258, 263. Dr. Mortazavi presents two alternative mappings of the S5 sub PCB into five distinct portions and relies on Kim to supply the antenna signal lines traversing them. JDX-0001C, ¶¶4536, 4554-60.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

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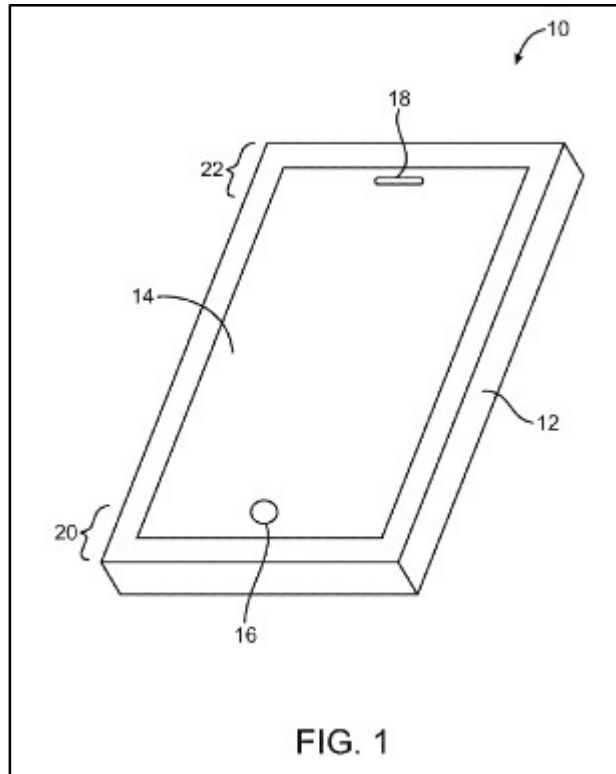
[REDACTED]

[REDACTED]

[REDACTED]

But limitation [1.c][v][a] is not satisfied by a route a POSITA *could* have taken; it requires that the signal lines actually traverse all four claimed PCB portions. JDX-0008C, ¶¶258, 263. Dr. Mortazavi’s mapping therefore depends on a POSITA rejecting the shorter, lower-loss path and routing the second antenna’s line the long way around, adding attenuation for no engineering benefit. JDX-0008C, ¶¶263-64. That is contrary to Kim, the reference he invokes to supply these lines, which is designed so that “the signal loss may be minimized.” JDX-0008C, ¶263; RX-0112, ¶¶7, 26-29.

A POSITA would instead route the second antenna’s line by the shortest available path, which would not traverse all four claimed portions. JDX-0008C, ¶264. Substituting Mow’s housing antennas does not change the analysis: Mow contemplates antennas at separated locations on the housing periphery, and a POSITA would keep each antenna’s feed contact on the rigid portion nearest that antenna rather than relocating one to the opposite end of the sub PCB and increasing signal loss for no benefit. JDX-0008C, ¶¶265-66; RX-0110, ¶52, Fig. 1.



Mow Fig. 1 showing upper and lower antenna regions 20 and 22. The combination therefore does not disclose limitation [1.c][v][a].

- d. **Limitations [6.a] (“a second PCB on which the wireless communication circuitry is located,”) and [6.b] (“wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.”)**

Claim 6 depends from Claim 1 and is valid for at least the reasons Claim 1 is valid. JDX-0008C, ¶267. Claim 6’s ground-member limitation also presupposes the very structure the combination of S5, Kim, and Mow lacks: it requires that “the first ground line and the second ground line”—the ground lines recited in limitation [1.c][v][a] as flanking the two antenna signal lines across all four PCB portions—be electrically connected with a ground member. Because the combination does not disclose the underlying ground-line and signal-line architecture traversing

[REDACTED]

the four claimed portions, it cannot disclose the additional connection Claim 6 requires. JDX-0008C, ¶¶263-64, 267.

VIII. SECONDARY CONSIDERATIONS OF NON-OBVIOUSNESS

One of the requirements for an obviousness analysis is the evaluation of any secondary considerations, which can establish that a patent claim is not obvious. *See Graham v. John Deere Co.*, 383 U.S. 1, 17-18 (1966); *Crocs, Inc. v. Int’l Trade Comm’n*, 598 F.3d 1294, 1310 (Fed. Cir. 2010) (reaffirming that secondary considerations “can be the most probative evidence of non-obviousness”).

Samsung and its experts Dr. Welch and Warren, have shown extensive evidence of long-felt need, commercial, success, and praise by others as secondary considerations supporting non-obviousness. In response, Oura’s invalidity expert Dr. Mortazavi did not identify or analyze any secondary considerations, stating only that he had “not identified any” objective indicia of non-obviousness for the ’789 Patent alone. JDX-0008C, ¶292; JDX-0001C, ¶¶3166-68. Oura’s TDI expert, Dr. Mercier presented opinions only on the “nexus” aspect of secondary considerations, but only to opine that the PDI products are not “coextensive” with the Asserted Patents, meaning a “nexus” cannot be *presumed* for the secondary considerations of commercial success and praise, though a nexus may still be established through other evidence. JDX-0006C, ¶¶209-225. Dr. Mercier’s opinions leave long-felt need un rebutted and there is a nexus for commercial success and praise.

A. Long-Felt But Unmet Need

The long-felt but unmet need for the claimed inventions of the Asserted Patents further supports non-obviousness. *See Apple Inc. v. Samsung Elecs. Co.*, 839 F.3d 1034, 1056-57 (Fed. Cir. 2016). As explained below, the ’359 Patent addresses the long-felt need to improve gesture

[REDACTED]

recognition in wearable devices, the '675 and '677 Patents address the long-felt need to improve the battery life of such devices, and the '789 Patent addresses the long-felt need for transmitting high-frequency signals through the narrow internal spaces of portable devices while reducing the material cost, labor cost, and elevated defect rate associated with traditional coaxial cables. JDX-0009C, ¶505; JDX-0008C, ¶295.

1. The Sensor Patents

Before the '359 Patent's prior art date of August 20, 2013, health monitoring and sensing systems that relied on gestures to determine when to initiate and end measurements of biological signals risked "misrecognizing the intention of the user," because an incidental arm movement, for example, could be mistaken for a deliberate hand gesture. JDX-0009C, ¶506; JX-0001, 1:34-42. Improved mechanisms for gesture recognition were important long before the '359 Patent was filed, and scientific literature pre-dating the '359 Patent acknowledges that "despite all the previous work, a reasonable solution to the gesture recognition problem has not been found yet," CX-1401, 11842, and there is a need for "improving the recognition accuracy" of gesture recognition. CX-1402, 5788; JDX-0009C, ¶¶508-09.

The asserted claims of the '359 Patent directly address this long-felt need to improve gesture recognition for wearable devices through a multi-sensor system that cross-validates sensor data across multiple categories of information to recognize a user's intent "with a greater precision" and "improve user experience." JDX-0009C, ¶507, 510; JX-0001, 1:43-45. As explained above in Section VI.C, Oura's asserted prior art, including Kuroda II and Suzuki, do not disclose this cross-validation solution. JDX-0009C, ¶507. Further, one of Oura's own patents recognizes the long-felt need for improved gesture recognition:

[T]here may be hundreds of different gestures that wearable devices may be looking for at any given time. As such, wearable devices may be expected to process large

quantities of motion data and compare the motion data to the hundreds of potential gestures, which may result in excessive processing power and increased battery consumption.

JDX-0009C, ¶511; CX-1365, 2:27-32. Not only does Oura recognize the long-felt need for improved gesture recognition, but Oura also acknowledges “the whole thrust and focus of the ’359 patent is how are we detecting and determining intentional motions and movements of a user to control an external device.” JDX-0009C, ¶511; JX-0039, 112:2-14. As such, Oura confirms both that improved gesture recognition is a long-felt need in wearable devices and that the ’359 Patent explicitly focuses on addressing this long-felt need. JDX-0009C, ¶511.

The ’675 and ’677 Patents address a second long-felt but unmet need: existing wearable devices had difficulty measuring biometric information over extended periods of time given the substantial battery power consumed by biometric sensors. JDX-0009C, ¶512; JX-0003, 1:44-63; JX-0002, 1:47-65. Articles have highlighted this need for a smart wearable with a long battery life, including one from Forbes explaining that “[t]he combination of LCD screens, constant wireless pinging, and limited real estate for batteries makes battery life a unique challenge for smartwatches. Considering these constraints, the fact that some of these products can last an entire day is a minor miracle. But it’s also not good enough.” JDX-0009C, ¶513; CX-1370, OURA1468_00096164.

The ’675 and ’677 Patents solve this problem by selectively operating a biometric sensor based on a specified condition—for example, suspending measurement when excessive movement would render the data unreliable, thereby conserving battery life until conditions are suitable for accurate sensing. JDX-0009C, ¶¶512, 514; JX-0003, 1:44-63, 31:24-58, 32:28-33; JX-0002, 1:47-65, 31:24-58, 32:28-33. As explained above in Sections IV.C and V.C, none of Oura’s asserted prior art discloses this selective-activation solution; at best, Presura and Ahmed⁴⁸⁶ teach only

[REDACTED]

changing a sensor's sampling schedule, not selectively activating or deactivating sensors based on motion. JDX-0009C, ¶512.

As explained below in Section VIII.B.1, numerous articles note the surprisingly impressive accuracy and usability of the HR and SpO2 measurement features of the patent-practicing Galaxy Ring and Watch Products where those features are directly correlated with, and attributable to, the inventions of the '359, '675, and '677 Patents. JDX-0009C, ¶515.

2. The '789 Patent

The '789 Patent is directed to resolving existing problems with small, portable electronic devices that employ PCBs and have traditionally utilized coaxial cables for high-frequency signal transmission, including the high potential defect rate associated with manufacturing such systems given that they require many assembly steps. JDX-0008C, ¶293; JX-0004, 1:28-48. The patent improves over these prior methods by providing a transmission line capable of effectively transmitting noiseless high-frequency signals in narrow internal spaces of portable terminals such as smartphones, without affecting other modules or requiring expensive coaxial cables with their associated material and labor costs. JDX-0008C, ¶294. The '789 Patent achieves this by disclosing a rigid-flex PCB structure with carefully arranged ground lines, signal lines, and conductive vias that reduce inductance-induced line loss. *Id.*; JX-0004, 1:52-2:34.

The need that the '789 Patent addressed long predated its June 13, 2016 priority date. JDX-0008C, ¶295. For years before the priority date, the smartphone industry was driven toward ever-thinner devices packed with ever more functionality, each of which consumed scarce internal volume. *Id.* As more modules were mounted in the narrow internal spaces of these devices, the long-standing reliance on coaxial cables to carry high-frequency signals—with their material cost, added assembly steps, and elevated defect rate in confined spaces—became an increasingly acute

[REDACTED]

problem: precisely the problem the '789 Patent identifies. *Id.*; JX-0004, 1:28-48. That this need persisted and intensified over the years preceding the priority date, without the industry arriving at the claimed rigid-flex transmission-circuit solution, is objective evidence of a long-felt but unmet need that the '789 Patent addressed. JDX-0008C, ¶295.

The prior art confirms both the long-felt need and the failure of others to achieve the claimed solution. *Id.*, ¶296. None of the references and systems Dr. Mortazavi relies upon—including Williams, Aoyama, Naganuma, Galaxy S5, and Ring 1—discloses the claimed integrated architecture. *Id.* Those references reflect a common set of shortcomings: they continued to rely on coaxial cables to carry high-frequency signals, or they disclosed only generic or bare rigid-flex boards lacking the claimed rigid-flexible-rigid-flexible structure with alternating conductive and insulating layers, and using signal lines in the conductive layers to transmit high frequency signals. *Id.* That the contemporaneous state of the art did not arrive at the claimed transmission circuit is objective evidence that the '789 Patent addressed a need that others had not met. *Id.*

The '789 Patent's claimed architecture also achieved results that the prior approaches did not. *Id.*, ¶297. By replacing coaxial cables with an integrated rigid-flex PCB having arranged ground lines, signal lines, and conductive vias, the claimed invention transmits noiseless high-frequency signals through the narrow internal spaces of portable communication devices while reducing the material cost, labor cost, and elevated defect rate associated with coaxial cables and their small connectors. *Id.*; JX-0004, 1:28-2:34. Those benefits—reliable high-frequency transmission together with reduced size, cost, and defect risk—reflect more than the predictable use of known PCB components, and they are realized in the commercially successful Galaxy Phone Products. JDX-0008C, ¶297.

B. Commercial Success; Praise by Others

Commercial success of Samsung's DI Products supports non-obviousness. *Demaco Corp. v. F. Von Langsdorff Licensing Ltd.*, 851 F.2d 1387, 1391 (Fed. Cir. 1988) (“[t]he commercial response to an invention is significant to determinations of obviousness”); JDX-0009C, ¶¶516-31; JDX-0008C, ¶¶284-92. Commercial success is typically shown through “significant sales in a relevant market.” *J.T. Eaton & Co. v. Atl. Paste & Glue Co.*, 106 F.3d 1563, 1571 (Fed. Cir. 1997) (citation omitted). Industry praise of the claimed invention also weighs in favor of non-obviousness. *Apple Inc. v. Samsung Elecs. Co.*, 839 F.3d at 1053. As explained below, the Samsung DI Products have been highly commercially successful and have received considerable industry praise as a result of their incorporation of the claimed inventions.

1. The Sensor Patents

As explained further below in Section IX, Samsung is a “leading provider of wearable devices in the United States,” with 22% of U.S. respondents of a September 2025 Statista report indicating that they used a Samsung brand eHealth tracker or smartwatch products. JDX-0002C, ¶74; JDX-0009C, ¶516; CX-1369. From January 2022 through September 2025, SEA sold more than [REDACTED] Galaxy Smartwatches in the U.S., generating more than [REDACTED] in net sales revenue. JDX-0002C, ¶61; JDX-0009C, ¶516. From the launch of the Galaxy Ring in July 2024 through December 31, 2025, SEA sold more than [REDACTED] Galaxy Rings in the U.S., generating more than [REDACTED] in net sales revenue. JDX-0002C, ¶63; JDX-0009C, ¶516.

The commercial success of these WDI Products is attributable to the claimed features of the '677, '675, and '359 Patents, including sleep detection, HR measurement, SpO2 measurement, and improved battery life. JDX-0009C, ¶517. Consumer criteria and preferences for purchasing wearable devices place high emphasis health and fitness monitoring as well as sleep tracking. JDX-

[REDACTED]

0002C, ¶76; CX-1514, -514-515. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Sales of WDI Products have continued to grow in recent years, and this growth is at least in part, attributable to the claimed features of the '359, '677, and '675 Patents. JDX-0009C, ¶¶518-20. [REDACTED]

[REDACTED]

[REDACTED] An August 2025 Smart Wearable Study conducted by NielsenIQ found that the top four features of smart rings were HR monitoring, sleep tracking, daily activity monitoring, and exercise and workout tracking, CX-1372C, -868, which are all enhanced by the improved gesture recognition and battery-saving measures provided by the Asserted Claims. JDX-0009C, ¶¶523, 517.

Many third-party publications and Samsung press reports have also recognized the success of the Galaxy Ring and Galaxy Watch products, with one outlet characterizing the Galaxy Ring as a potential “market disruptor” that “has resonated with consumers,” CX-1390, -306, and a press release highlighting that the Galaxy Watch “received numerous compliments from consumers and tech experts about the improved product quality.” CX-1391, -312.

These articles also praise these products and specifically highlight the features relevant to the Asserted Patents, such as monitoring sleep, SpO2, and HR. JDX-0009C, ¶526. Samsung studies and press reports as well as consumers have similarly praised features of the Samsung WDI

[REDACTED]

Products directly attributable to the inventions of the Asserted Patents such as improved HR and SpO2 measurement for their accuracy and user-friendliness. JDX-0009C, ¶¶526-30; CX-0037, -084-087; CX-1392-CX-1400.

The above facts, taken together, highlight a strong nexus between the commercial success and praise for the WDI Products and the Asserted Patents' claimed features. JDX-0009C, ¶525.

2. The '789 Patent

From January 1, 2022 through December 31, 2025, SEA sold more than [REDACTED] domestic-industry Galaxy phones in the United States, representing almost [REDACTED] [REDACTED] SEA sold in the United States during this period. JDX-0008C, ¶285; JDX-0002C, ¶68. During the same period, the Galaxy phones generated more than [REDACTED] in net sales revenue—approximately [REDACTED] SEA generated. *Id.* Ms. Rowe further explains that the domestic-industry Galaxy S-series phones are Samsung's "flagship smartphones," and that Samsung has comprised approximately 20% to 30% of all U.S. smartphone sales since 2020 (and approximately 25% by 2020). JDX-0008C, ¶285; JDX-0002C, ¶¶70, 234. These sales volumes, revenues, and market shares reflect substantial and sustained commercial success of the patent-practicing Galaxy Phone Products.

Samsung's commercial success in mobile devices is long-standing and independently documented. JDX-0008C, ¶286. Samsung became the global leader in mobile-phone shipments in 2011 and has maintained that position for fourteen consecutive years through 2024. *Id.*; CX-1508, -044, -081. Reflecting that sustained success, the brand value of Samsung Electronics exceeded \$100 billion USD, ranking Samsung as the world's fifth most valuable brand for the fifth consecutive year. CX-1508, -051. The Galaxy S-series smartphones that practice the Asserted DI Claims are Samsung's flagship devices within this commercially successful mobile portfolio.

[REDACTED]

JDX-0008C, ¶286. Independent third-party publications and industry reports likewise recognize the commercial success and market leadership of the Galaxy Phone Products. *Id.*, ¶¶287-88; CX-1420-CX-1426.

The commercial success of the PDI Products has a nexus to the claimed features of the '789 Patent. JDX-0008C, ¶288. The products are slim, high-performance smartphones that integrate cellular and other wireless functionality into a compact, durable housing. As explained in Section VIII.A.2, the '789 Patent's claimed rigid-flex transmission circuit—with its alternating rigid and flexible PCB portions and arranged ground and signal lines—enables high-frequency signals to be routed reliably through the narrow internal spaces of such devices without the added bulk, cost, and defect risk of coaxial cables. *Id.* The compactness, reliability, and structural integrity that reviewers and the marketplace associate with the Galaxy phones are attributable, at least in part, to the claimed architecture, which supports a nexus between commercial success and the claimed features of the '789 Patent. *Id.*

Independent industry publications and technical reviewers have likewise praised the design and engineering of Galaxy phones, including the compact form factor, structural durability, and reliable performance that the '789 Patent's claimed architecture helps enable. JDX-0008C, ¶289; CX-1427-CX-1432. Samsung's Galaxy product portfolio has also drawn sustained recognition for a decade of hardware innovation that depends on dense, space-efficient internal architectures. JDX-0008C, ¶290; CX-1440, -001-013. The space-efficiency achieved notwithstanding additional features and components is attributable to the compact, reliable high-frequency signal routing that the '789 Patent's claimed rigid-flex transmission circuit provides. JDX-0008C, ¶290.

As with commercial success, the industry praise that the PDI Products have received has a nexus to the claimed features of the '789 Patent. *Id.*, ¶291. Reviewers repeatedly single out the

[REDACTED]

products’ slim, minimalist form factor and their reliable wireless performance—precisely the attributes that the claimed rigid-flex transmission circuit enables by routing high-frequency signals reliably through a compact, space-constrained housing without utilizing coaxial cables. *Id.*; CX-1427-CX-1432. Praise directed to these attributes is therefore probative of the non-obviousness of the claimed features, and not of unrelated aspects of the products. JDX-0008C, ¶291.

IX. DOMESTIC INDUSTRY ECONOMIC PRONG

The economic prong of the domestic industry requirement is satisfied when, with respect to the articles protected by the asserted patents, there exists in the United States one of three independently sufficient showings: “(A) significant investment in plant and equipment; (B) significant employment of labor or capital; or (C) substantial investment in its exploitation, including engineering, research and development, or licensing.” 19 U.S.C. § 1337(a)(3). As shown below, Samsung has satisfied the economic prong of the domestic industry requirement under sub-prongs (A) and (B); neither Oura nor its expert, John Jarosz, disputes that Samsung has an economic domestic industry in the United States. *See, e.g.*, JX-0049C at 28 (“[Q.] Is it your opinion that Samsung has no domestic industry in this case? A. No, I don't think I've drawn that specific opinion....”).

The Commission undertakes a “holistic review of all relevant considerations” examining “the articles of commerce, and the realities of the marketplace.” *Lashify Inc. v. Int’l Trade Comm’n*, 130 F.4th 948, 963 (Fed. Cir. 2025); *Wuhan Healthgen Biotechnology Corp. v. Int’l Trade Comm’n*, 127 F.4th 1334, 1339 (Fed. Cir. 2025). A complainant demonstrates significance by showing how its activities are important to the articles protected by the patent in the context of its own operations, the marketplace, and the industry in question. *Certain Printing & Imaging Devices & Components Thereof*, Inv. No. 337-TA-690, Comm’n Op. at 27–28 (Feb. 17, 2011).

[REDACTED]

The scope of cognizable activities under sub-prongs (A) and (B) is broad: *Lashify* confirmed that sub-prongs (A) and (B) do not cabin the range of enterprise activities whose costs may be counted—sales, marketing, customer support, and operations all qualify. *Lashify*, 130 F.4th at 963.

The relevant date for assessing domestic industry is the filing date of the complaint. *Philip Morris Prods. S.A. v. Int’l Trade Comm’n*, 63 F.4th 1328, 1341 n.4 (Fed. Cir. 2023). Because companies do not ordinarily track investments on a product-by-product basis, the Commission accepts reasonable allocation methodologies appropriate to a complainant’s circumstances and supported by the evidence. *Certain Mobile Device Holders and Components Thereof*, Inv. No. 337-TA-1028, Comm’n Op. at 18 (Mar. 22, 2018).

SEA, a wholly-owned subsidiary of SEC, is headquartered in Englewood Cliffs, New Jersey. JDX-32C, Ex. 11C (“Martinez Decl.”), ¶1; JDX-0015C, 157; CX-1441, 21, 67, 354; JDX-0045C, 20; RX-0411C; CX-1534. SEA is the Samsung entity that sells Samsung products in the United States.¹⁴ JDX-0002C, ¶28; CX-1441, JDX-0045C, 7; RX-0411C; CX-1534; CX-1553, 19; JDX-0015C, 48-49; JDX-32C, Ex. 11C, ¶11. Christopher Martinez, Director of Marketing for SEA, will testify that SEA is involved in numerous development and commercialization activities, including engineering, research, development, testing, customer service, importation, sales, marketing, and distribution of products in the United States, including wearables and smartphones. JDX-2C, ¶28C; CX-1441; JDX-0045C, 7, RX-0411C; CX-1534; CX-1553, 19; JDX-0015C, 48-49; JDX-32C, Ex. 11C, ¶1. It has over 40 facilities across the U.S. and employs thousands of personnel tasked with commercializing U.S. products, including its smartphones, wearables, and

¹⁴ Contrary to Oura’s assertion, SEA is authorized to practice the patents in the U.S. *See, e.g., Certain Products Containing Interactive Program Guide and Parental Control Technology*, Inv. No. 337-TA-845, Final Initial Determination at 276 (June 7, 2013).

[REDACTED]

smart rings in the U.S. JDX-32C, Ex. 11C, ¶3. [REDACTED]

[REDACTED]

[REDACTED] JDX-0015C, 91, 140, 157; JDX-2C, ¶91; JDX-0032, ¶1.

Samsung has made, and continues to make, significant and/or substantial long-term and ongoing economic investments in the United States for the domestic industry products. Samsung's investments in plant and equipment, and labor or capital are economically significant within the meaning of sub-prongs (A) and (B).

SEA's extensive domestic business operations for the domestic industry products include

[REDACTED]

[REDACTED]. JDX-0002C, ¶16; JDX-0015C, 130; JX-0047, 10-11; CX-1469C. [REDACTED]

[REDACTED] JDX-0002C, ¶16; JDX-0015C, 130; JX-0047, 10-11; CX-1469C. The DI Products in this matter include Samsung Galaxy smartwatches, Galaxy rings (collectively the WDI Products), and Galaxy smartphones (the PDI Products). JDX-0002C, ¶16; JDX-0015C, 130; JX-0047, 10-11; CX-1469C.

Samsung's DI Products belong to the Samsung mobile division [REDACTED], which also includes non-practicing tablets, earbuds, and other smartphones. JDX-0002C, ¶17; RX-0369C; JX-0047, 9-10; JDX-0015C, 61-63. [REDACTED]

[REDACTED] JDX-0002C, ¶17; RX-0369C; JX-0047, 9-10; JDX-0015C, 61-63. Given that Samsung's mobile products include all smartphones and certain tablets, both of which make up a significant portion of U.S. unit sales in their respective categories, the DI Products represent significant sales, and are important products for Samsung. CX-1556; CX-1524; JDX-0002C, ¶17; RX-0369C; JX-0047, 9-

[REDACTED]

10; JDX-0015C, 61-63.

A. Allocation Methodology

In the ordinary course of business, Samsung generally records costs by [REDACTED]. JDX-0002C, ¶94; JDX-0015C, 26-27, 35-36, 57-58. Samsung employs a [REDACTED]. JDX-0002C, ¶95; JDX-0015C, 58, 82-83, 101-103, 111. [REDACTED] JDX-0002C, ¶95; JDX-0015C, 95-96. This includes the DI Products. JDX-0002C, ¶95. Then, expenses associated with that work can be allocated based on the [REDACTED] associated with the DI Products as a portion of all mobile products. At a high level, Samsung's investments in labor and capital for the DI Products may be calculated by allocating labor or capital costs recorded under SEA's [REDACTED] divisions. JDX-0002C, ¶95; JDX-0015C, 94. Facility expenses are allocated by the same cost centers. JDX-0002C, ¶163; JDX-0032, ¶2.

Contrary to the arguments presented by Mr. Jarosz, the employees represented in Samsung's spreadsheets as [REDACTED] or based off the allocation model DI specific products are [REDACTED]. JDX-0002C, ¶95; JDX-0015C, 131-132. Samsung's U.S. mobile products include the following categories: [REDACTED]. JDX-0002C, ¶95; JDX-0015C, 61-63. Overall, these products are associated with a [REDACTED]. JDX-0002C, ¶95; JDX-0015C, 58, 61-63, 82-83, 94-98, 101-103, 111, 131-132; RX-0369C; JX-0047, 9-10. Employees and expenses associated

[REDACTED]
[REDACTED]
with work done on, e.g., [REDACTED]

[REDACTED]. As Mr. Jarosz acknowledged, Samsung, like many companies (including Oura), does not categorize or track its employees or expenses on a product-by-product basis. JDX-49C. Instead, Samsung tracks those expenses and employees [REDACTED]. This [REDACTED] addresses the realities of Samsung's employees' work—that such work often relates to multiple products within a product ecosystem inside the business unit. For instance, Mr. Martinez will testify that [REDACTED]. JDX-15C.

The starting point for the calculations of SEA's investments under sub-prongs (A) and (B) is a document that shows total expenses for SEA, including [REDACTED]. JDX-0002C, ¶95; CX-1469C at tab [REDACTED]; JDX-0015C, 93-94. The data in this spreadsheet comes from an underlying financial database at Samsung and is the same data SEA records and uses in the ordinary course of business. *Id.* As Mr. Martinez will testify, the high-level cost centers associated with this work are [REDACTED]. JDX-0002C, ¶96; JDX-0015C, 93-94, 97-98; CX-1469C at tab [REDACTED] (*see, e.g.,* [REDACTED]). High-level cost centers associated with [REDACTED]. JDX-0002C, ¶95; JDX-0015C, 58, 61-63, 82-83, 94-98, 101-103, 111, 131-132; RX-0369C; JX-0047, 9-10.

[REDACTED]
[REDACTED]
[REDACTED]

[REDACTED]

[REDACTED]

these departments were identified as having SEA employees whose functions related to mobile products, including the DI Products. JDX-0002C, ¶97; JDX-0015C, 57-58 95-97; JDX-53C. JX-0047, 10-11; CX-1469C at tabs [REDACTED] [REDACTED]

[REDACTED]

By cost center, Samsung's document CX-1469C provided monthly costs for the period January 1, 2021 through November 30, 2025. JDX-0002C, ¶98; CX-1469C at tab [REDACTED] (*see, e.g.*, [REDACTED]). Samsung's labor or capital costs are generally categorized as: [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

JDX-0002C, ¶98, Ex. 4; CX-1469C at tab [REDACTED] [REDACTED]. Samsung's data in CX-1469C also includes an [REDACTED] [REDACTED], where the headcount for each month from January 31, 2021 through November 30, 2025 can be derived. JDX-0002C, ¶99; CX-1469C at tabs [REDACTED] [REDACTED] JDX-0015C, 108-109.

After identifying labor or capital costs that are specific to Samsung's U.S. mobile products, Samsung's expert, Ms. Julia Rowe, performed a sales-based allocation to isolate amounts that are attributable to the DI Products. Using a sales allocation to allocate investments to the DI Products is [REDACTED] and a reasonable approach here. JDX-0002C, ¶100; JDX-0015C, 58, 82-83, 101-103, 111.A [REDACTED] was most appropriate for each of

[REDACTED]

[REDACTED]

JDX-0002C, ¶101; JDX-0015C, 122-123. While this includes both [REDACTED]
[REDACTED], Samsung does not organize its business strictly by product or function,
and the included employees work on [REDACTED]. *Id.* On
average, Samsung's U.S. [REDACTED] from the domestic industry products comprised
approximately [REDACTED] of Samsung's total U.S. mobile product [REDACTED]. JDX-0002C, ¶101, Ex.
5.0. This is summarized below:

[REDACTED]

Allocating in proportion to Samsung's [REDACTED] is reflective of SEA's business
practices. JDX-0002C, ¶101; JDX-0015C, 58, 82-83, 101-103, 111 [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. Sales-based allocations described herein apply
the ratios in the above table.

Facility expenses are allocated by the same cost centers described above with regards to

[REDACTED]

labor and capital. JDX-0002C, Exs. 1.1, 3.0; JDX-32C, Ex. 11C, ¶2. Therefore, the approach for plant and equipment is very similar to the approach described above for labor and capital. The expenses allocated to plant and equipment include [REDACTED]

[REDACTED] JDX-0002C, ¶164, Ex. 4.0. These expenses are allocated to the different groups or cost centers of Samsung in the ordinary course of business. JDX-0002C, ¶164. The amounts described herein are based on this internal allocation methodology. *Id.* These expenses are further allocated to the domestic industry products using the same sales-based methodology described above with regards to labor or capital.

Mr. Jarosz's challenges to the above allocation strategy are misplaced. First, Mr. Jarosz himself admitted repeatedly that he does not "understand the Samsung records at all." JDX-49C, 39-42, 44-46. When asked at his deposition regarding these concerns, Mr. Jarosz demonstrated his lack of familiarity with the underlying data, how to filter the data to obtain an appropriate base of expenses, and how to apply the [REDACTED] methodology to obtain the results that Ms. Rowe obtained. *Id.*

Second, Ms. Rowe's analysis and findings are supported by her report and SEA data. The calculation of investments and application of sales-based allocation methodology herein is accurate and reflective of SEA's business practices. The expenses and cost centers relied upon are sufficiently tied to the DI Products and constitute qualifying DI investments. As described below with regards to both labor or capital and plant and equipment, Ms. Rowe's DI analysis relied on an investment base of a proper scope, and the underlying activities of the selected groups and cost centers relate to the DI Products.

[REDACTED]

B. Samsung's Domestic Investments in Labor or Capital Satisfy Section 1337(a)(3)(B)

Ms. Rowe will testify that Samsung has made, and will continue to make, significant domestic investments in labor or capital with respect to the DI Products. This includes the cost of employees dedicated to the development of Samsung products and services, U.S.-specific market strategy and distribution, U.S.-specific product packaging, and customer service. SEA employs [REDACTED] direct employees and contractors, and invests [REDACTED] dollars in product strategy, management, planning, research, development, engineering, testing, compliance, quality control, technical support, sales, marketing, distribution, warehousing, service, warranty, repair, and replacement for Samsung mobile products in the U.S., including the WDI and PDI Products. JDX-0002C.

Over the period leading up to the complaint in this Investigation, from January 1, 2022 through November 30, 2025, Samsung invested over [REDACTED] in labor and capital associated with the domestic industry products practicing the '789 patent, and [REDACTED] in labor and capital associated with the domestic industry products practicing the '677 patent, the '675 patent, and the '359 patent. JDX-0002C, ¶18, Exs. 1.0, 1.2, 9.2, 10.1. These amounts are significant in the context of Samsung's mobile product investments. The labor and capital investments in the '789 patent make up [REDACTED] of SEA's labor and capital investments in all mobile products, and the labor and capital investments in the '359, '675, and '677 patent make up [REDACTED] of SEA's labor or capital investments in all mobile products. *Id.* A detailed analysis and allocation of SEA's labor or capital investments, along with an explanation of each department's work and employees, is provided below.

1. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-32C, Ex. 11C, ¶10; JDX-0015C, 128-129, 161-162. As of the end of November 2025, approximately [REDACTED] full-time employees worked in [REDACTED] in the US. JDX-0002C, ¶107, Ex. 9.3; JDX-32C, Ex. 11C, ¶10. From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED] [REDACTED] in [REDACTED] in the United States. JDX-0002C, ¶108, Ex. 3.1. Samsung's labor and capital investments in the [REDACTED] were primarily comprised of [REDACTED]. *Id.*

In total, Samsung invested approximately [REDACTED] in domestic investments in labor or capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶112, Ex. 3.0. PDI Product amounts are allocable to the '789 patent, and the WDI Product amounts are allocable to the other asserted patents. *Id.* Applying a sales-based allocation to Samsung's domestic investments in labor or capital for the [REDACTED] yields [REDACTED] allocable to the '789 patent;

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

2.

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶113; JDX-32C, Ex. 11C,

¶7. [REDACTED]

[REDACTED] JDX-0002C, ¶114; CX-1695C, 15-16. [REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶115; JDX-0015C, 9. [REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶115; JDX-0015C, 9-10. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶116; JDX-32C, Ex. 11C,

¶¶18-24; JDX-0045C, 13. [REDACTED] JDX-0002C, ¶115;

[REDACTED]

JDX-0015C, 9-10.

The [REDACTED] team is located in the United States. JDX-0002C, ¶117. As of the end of November 2025, [REDACTED] U.S. full time employees worked in this group. JDX-0002C, ¶117; JDX-32C, Ex. 11C, ¶8. From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED]. JDX-0002C, ¶117, Exs. 9.3, 3.2. Samsung's labor and capital investments in the [REDACTED] group were primarily comprised of [REDACTED]. JDX-0002C, ¶118, Ex. 3.2. [REDACTED]. *Id.*

In total, Samsung invested more than [REDACTED] in domestic investments in labor and capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶121, Ex. 3.0. Applying a sales-based allocation to Samsung's domestic investments in labor and capital for the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED], during the relevant timeframe. JDX-0002C, ¶¶112-121.

[REDACTED]

[REDACTED]

[REDACTED]

3. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶122; JDX-0015C, 114-

115, 121-122.

[REDACTED]

[REDACTED]

[REDACTED] JDX-0045C, 12.

All employees classified as belonging to the [REDACTED] are based in the United

States. JDX-0002C, ¶123; JDX-32C, Ex. 11C, ¶6; JDX-0045C, 11-12. As of the end of November

2025, [REDACTED] U.S. full time employees worked in this group. JDX-0002C, ¶123; JDX-32C, Ex. 11C, ¶7. From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED]. JDX-0002C, ¶123. Samsung's labor and capital investments in the [REDACTED] group are primarily comprised of [REDACTED].

In total, Samsung invested approximately [REDACTED] in domestic investments in labor and capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶127, Ex. 3.0. Applying a sales-based allocation to Samsung's domestic investments in labor and capital for the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED].

[REDACTED]. *Id.*

4. [REDACTED]

Samsung also has [REDACTED] employees focused on mobile products (including the DI Products) in the United States. *Certain Printing & Imaging Devices & Components Thereof*, Inv. No. 337-TA-690, Comm'n Op. at 27 (Feb. 17, 2011). [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶129; CX-1469C.

As of the end of November 2025, [REDACTED] U.S. full time employees worked in the U.S. [REDACTED] functions described above. JDX-0002C, Exs. 9.3, 3.4; JDX-32C, Ex. 11C, ¶9. From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED] [REDACTED]. JDX-0002C, Exs. 9.3, 3.4. Samsung’s labor and capital investments in the [REDACTED] group were primarily comprised of [REDACTED] [REDACTED]. JDX-0002C, Ex. 3.4.

In total, Samsung invested approximately [REDACTED] in domestic investments in labor and capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶134, Ex. 3.0. Applying a sales-based allocation to Samsung’s domestic labor and capital investments in [REDACTED] yields [REDACTED] allocable to the ‘789 patent, and amounts allocable to the ‘359 patent, the ‘677 patent, and/or the ‘675 patent totaling [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

5. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*

As of the end of November 2025, [REDACTED] U.S. full time employees worked in [REDACTED] in the United States. JDX-0002C, Exs. 9.3, 3.5; JDX-32C, Ex. 11C, ¶10-11. From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED]

[REDACTED]. *Id.* Samsung's labor and capital investments in the [REDACTED] group were primarily comprised of [REDACTED]

[REDACTED]. JDX-0002C, Ex. 3.5.

In total, Samsung invested approximately [REDACTED] in domestic investments in labor

[REDACTED]

and capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶140, Ex. 3.0. Applying a sales-based allocation to Samsung's domestic labor and capital investments in the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]

[REDACTED]

[REDACTED]. JDX-0002C, ¶¶133-140; JDX-0015C, 133-136.

[REDACTED]

6. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

As of the end of November 2025, [REDACTED] U.S. full time employees worked in this group. JDX-0002C, ¶144, Ex. 9.3; JDX-32C, Ex. 11C, ¶10. From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED]. *Id.* Samsung’s labor and capital investments in the [REDACTED] group are primarily comprised of [REDACTED]. JDX-0002C, ¶145, Ex. 3.6; JDX-0015C, 153-155.

In total, Samsung invested approximately [REDACTED] in domestic investments in labor and capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶148, Ex. 3.0. Applying a sales-based allocation to Samsung’s domestic labor and capital investments in the [REDACTED] group yields [REDACTED] allocable to the ‘789 patent, and amounts allocable to the ‘359 patent, the ‘677 patent, and/or the ‘675 patent totaling [REDACTED]. JDX-0002C, ¶¶141-149.

[REDACTED]

[REDACTED]

[REDACTED]

7. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*

As of the end of November 30, 2025, [REDACTED] U.S. full time employees worked in this group. *Id.* From January 1, 2022 through November 30, 2025, SEA invested approximately [REDACTED] in [REDACTED]. *Id.* Samsung's labor and capital investments in the [REDACTED]

[REDACTED]

[REDACTED] *Id.*

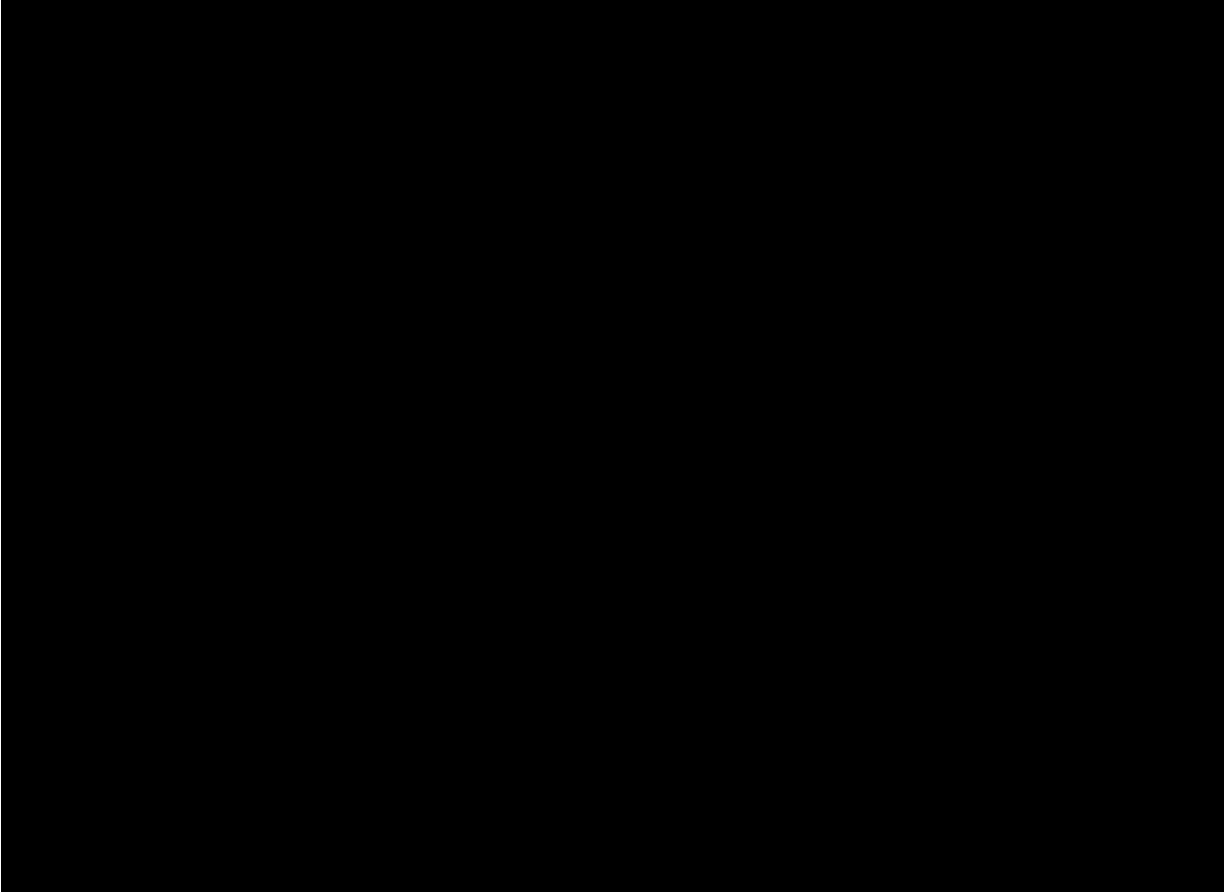
In total, Samsung invested approximately [REDACTED] in domestic investments in labor and capital in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶160, Ex. 3.0. Applying a sales-based allocation to Samsung's domestic investments in labor and capital for the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]

[REDACTED]. JDX-0002C ¶¶147-161.

[REDACTED]

8. SEA's Total Domestic Labor and Capital Investments Related to the DI Products

Samsung's domestic investments in labor and capital for the domestic industry products for the '789 patent are summarized below:



JDX-0002C, Exs. 1.1, 3.0.

Samsung's domestic investments in labor and capital for the domestic industry products for the other asserted patents are summarized below:

[REDACTED]

C. Samsung's Domestic Investments in Plant and Equipment Satisfy Section 1337(a)(3)(A)

Samsung has made, and will continue to make, significant domestic investments in plant and equipment with respect to the domestic industry products. Over the period January 1, 2022 through November 30, 2025, Samsung invested approximately [REDACTED] in plant and equipment associated with the domestic industry products practicing the '789 patent, and [REDACTED] in plant and equipment associated with the domestic industry products practicing the '677 patent, the '675 patent, and the '359 patent. JDX-0002C, ¶19, Ex. 1.0. These amounts are significant in the context of Samsung's mobile product investments. *Id.* The domestic plant and equipment investments in the '789 patent make up [REDACTED] of SEA's plant and equipment investments

[REDACTED]

in all mobile products, and the domestic plant and equipment investments in the '359, '675, and '677 patents make up [REDACTED] of SEA's plant and equipment investments in all mobile products. *Id.*

The expenses allocated to plant and equipment include [REDACTED]

[REDACTED]

[REDACTED]. JDX-0002C, ¶164, Ex. 4.0. These expenses are allocated to the different groups or cost centers of Samsung in the ordinary course of Business, and are allocated by the same cost centers as described with regards to labor and capital. JDX-0002C, ¶¶163-164, Ex. 4.0. The amounts described below are based on this internal allocation methodology. JDX-0002C, ¶164, Ex. 4.0. These expenses are further allocated to the domestic industry products using the same sales-based methodology described with regards to labor and capital.

The following groups include plant and equipment investments related to the domestic industry products: [REDACTED]

[REDACTED]. JDX-0002C, ¶97.

1. [REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶165, Ex. 2.1.

The C&S team is primarily located in [REDACTED]. JDX-0002C, ¶166; Martinez Decl. ¶10; JDX-0045C, 9, 16; JDX-0015C, 161-162, 165-166; JDX-0045C, 21; RX-0411C; CX-1592C, 856, 859, 872-873; CX-1594C; CX-1597C, 958. [REDACTED]

[REDACTED]

office totals [REDACTED] and housed over [REDACTED] employees as of July 2025. *Id.* The [REDACTED]. *Id.* There are also [REDACTED] employees located in [REDACTED]. JDX-0002C, ¶167; JDX-0015C, 130-131; JDX-0045C, 22; RX-0411C. [REDACTED]

[REDACTED]

[REDACTED]. *Id.*

In total, Samsung invested approximately [REDACTED] in domestic plant and equipment in C&S between January 1, 2022 and November 30, 2025. JDX-0002C, ¶169, Ex. 2.0. Applying a sales-based allocation yields [REDACTED] allocable to the ‘789 patent, and amounts allocable to the ‘359 patent, the ‘677 patent, and/or the ‘675 patent totaling [REDACTED]. JDX-0002C, ¶¶168-169, Ex. 2.0.

[REDACTED]

2. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

JDX-0002C, ¶170, Ex. 2.2.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶171; JDX-0045C, 9, 12, 14-16, 20; RX-0411C; JDX-0015C, 160-161; CX-1510. [REDACTED]

[REDACTED] *Id.*

In total, Samsung invested approximately [REDACTED] in plant and equipment in [REDACTED] [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶173, Ex. 2.0. Applying a sales-based allocation yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]

[REDACTED]. JDX-0002C, ¶¶172-173, Ex. 2.0.

3. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶174, Ex. 2.3. [REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶175; JDX-0045C, 9, 14-16, 20-21;

RX-0411C; JDX-0015C, 12-14, 128-129, 160-161, 166-169; CX-1593C; CX-1596C; CX-1550.

Other locations include:

I [REDACTED]

[REDACTED]

[REDACTED]

I [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

I [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

I [REDACTED]

[REDACTED]

[REDACTED]

In total, Samsung invested approximately [REDACTED] in plant and equipment in [REDACTED]

[REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶177, Ex. 2.0.

Applying a sales-based allocation to Samsung’s domestic investments in plant and equipment for

[REDACTED]

the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]. JDX-0002C, ¶¶173-178; JDX-0045C, 9, 14-16, 20-21; RX-0411C; JDX-0015C, 12-14, 128-129, 160-161, 166-169; CX-1593C; CX-1596C; CX-1550.

[REDACTED]

4. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

JDX-0002C, ¶178, Ex. 2.4. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]. JDX-0002C, ¶179; JDX-0045C, 9, 14, 16, 20-21; RX-0411C; JDX-0015C, 157, 168-169; CX-1593C; CX-1596C; CX-1550; JDX-0015C, 166, 169. [REDACTED] has [REDACTED] and housed [REDACTED] employees as of July 2025. *Id.*

[REDACTED]

In total, Samsung invested approximately [REDACTED] in plant and equipment in [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶181, Ex. 2.0. Applying a sales-based allocation to Samsung's domestic investments in plant and equipment for the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]. JDX-0002C, ¶¶177-182; JDX-0045C, 9, 14, 16, 20-21; RX-0411C; JDX-0015C, 157, 168-169; CX-1593C; CX-1596C; CX-1550; JDX-0015C, 166, 169.

5. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶182, Ex. 2.5. [REDACTED]

[REDACTED] JDX-0002C, ¶183, Ex. 2.0; JDX-0015C, 160-161.

In total, Samsung invested approximately [REDACTED] in plant and equipment in [REDACTED] [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶184, Ex. 2.0. Applying a sales-based allocation to Samsung's domestic investments in the [REDACTED]

[REDACTED]

group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]. JDX-0002C, ¶¶180-186; JDX-0015C, 160-161, 173; JDX-0045C, 18.

[REDACTED]

6. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶185, Ex. 2.6. [REDACTED]

[REDACTED] JDX-0002C, ¶186; JDX-0045C, 18, 21; JDX-0015C, 173; RX-0411C. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] . *Id.*

In total, Samsung invested approximately [REDACTED] in plant and equipment in [REDACTED] [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶188, Ex. 2.0.

[REDACTED]

Applying a sales-based allocation to Samsung's domestic investments in plant and equipment for the [REDACTED] group yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]

[REDACTED]

JDX-0002C, ¶¶184-188; JDX-0045C, 18, 21; JDX-0015C, 160-161, 173; RX-0411C.

7. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶189, Ex. 2.7. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶190; JDX-0015C, 12-14, 173; JDX-0045C, 21; RX-0411C; CX-1598C; CX-1599C; CX-1602C; CX-1601C; CX-1604C. [REDACTED]

[REDACTED] *Id.*

[REDACTED]

In total, Samsung invested approximately [REDACTED] in plant and equipment in [REDACTED] [REDACTED] between January 1, 2022 and November 30, 2025. JDX-0002C, ¶192. Applying a sales-based allocation to Samsung's domestic investments in plant and equipment for the [REDACTED] groups yields [REDACTED] allocable to the '789 patent, and amounts allocable to the '359 patent, the '677 patent, and/or the '675 patent totaling [REDACTED]. JDX-0002C, ¶¶189-192; JDX-0015C, 12-14, 173; JDX-0045C, 21; RX-0411C; CX-1598C; CX-1599C; CX-1602C; CX-1601C; CX-1604C.

[REDACTED]

8. SEA's Total Domestic Investments in Plant and Equipment Related to the DI Products

Samsung's domestic investments in plant and equipment for the domestic industry products for the '789 patent are summarized below:



JDX-0002C, Exs. 1.1, 2.0.

Samsung's domestic investments in plant and equipment for the domestic industry products for the '359, '677, and '675 patents are summarized below:

JDX-0002C, Exs. 1.1, 2.0.

D. Samsung's Domestic Investments in Labor and Capital (Comprehensive)

In a more comprehensive view of labor and capital (where capital includes Samsung's domestic investments in plant and equipment), Samsung's domestic investments in labor and capital (comprehensive) in the '789 patent between January 1, 2022 through November 30, 2025 were approximately [REDACTED]:

[REDACTED]

JDX-0002C, Exs. 1.1, 2.0.

In a more comprehensive view of labor and capital (where capital includes Samsung's domestic investments in plant and equipment, Samsung's domestic investments in labor and capital (comprehensive) in the '359, '677, and '675 patents between January 1, 2022 through November 30, 2025 were more than [REDACTED]:

JDX-0002C, Ex. 1.1.

E. Samsung's Domestic Investments Are Significant And Substantial

Samsung's domestic investments in the DI Products are significant and substantial in the context of SEA's mobile division, in the context of Samsung's foreign expenditures, and in the context of the industry. Samsung's domestic investments are expected to continue.

The domestic investments in each of the asserted patents are significant. JDX-0002C, ¶22, Ex. 9.0. Samsung's domestic investments in labor or capital (comprehensive) represent approximately [REDACTED] for the DI Products, both for the '789 patent (Galaxy S22-S25 smartphones), and the '359, '675, and '677 patents (Galaxy Watches 4-8 and Ultra and Galaxy Ring). *Id.* When compared to SEA's overall investments in its mobile business, the domestic industry investments in the '789 patent make up over [REDACTED] of Samsung's investments in domestic plant and equipment and domestic labor or capital. JDX-0002C, ¶23, Ex. 9.0. Domestic

investments in the '359, '657, and '677 patents represent approximately [REDACTED] of Samsung's domestic investments in plant and equipment and labor or capital. *Id.*

1. Samsung's Domestic Investments Are Significant and Substantial In the Context of the Domestic Industry and SEA's Mobile Division

Altogether, Samsung's U.S. [REDACTED] from the DI Products comprised approximately [REDACTED] of Samsung's total U.S. mobile product [REDACTED]. JDX-0002C, Ex. 5.0.

For the '789 patent, Samsung's U.S. [REDACTED] from the smartphone DI Products comprised approximately [REDACTED] of Samsung's U.S. mobile product [REDACTED]. JDX-0002C, Ex. 5.0. For the other asserted patents, Samsung's U.S. [REDACTED] from the WDI Products comprised approximately [REDACTED] of Samsung's U.S. mobile product [REDACTED]. *Id.* The DI Products include a new product line for Samsung, the Galaxy Ring. The Galaxy Ring is Samsung's first smart ring product. *Id.* While the Samsung Galaxy Ring represents approximately [REDACTED] of Samsung's mobile product [REDACTED] for the period July 1, 2024 through December 31, 2025, it also represents a new form factor to the Samsung mobile product family. JDX-0002C, ¶203, Ex. 5.0; CX-1542.

Each of the net revenue percentages for the Samsung DI Products is significant when considered in the context of Samsung's overall U.S. mobile sales. Samsung mobile products make up a significant percentage of sales within the United States in general. Since 2020, Samsung has comprised approximately 20% to 30% of U.S. smartphone sales. JDX-0002C, ¶204; CX-1556; CX-1524; JDX-0015C, 33-34, 61-63; RX-0369C; JX-0047, 9-10. In addition to the Samsung Galaxy S series smartphones which are PDI Products, Samsung also sells the Z series line of foldables and the A series line, which is a lower-priced device. *Id.* Samsung tablets, some of which belong to Samsung's mobile product and group, made up 24% of tablet ownership in the United States as of September 2025. *Id.* In the context of these sales, the WDI Products making

[REDACTED]

up approximately [REDACTED] of overall [REDACTED] is significant. *Id.*

The work done by SEA with respect to the DI Products is significant and critical for the sale of these products in the United States. JDX-0002C, ¶205, Ex. 5.0; JDX-0015C, 49-51. [REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶206; JDX-0015C, 9-10, 36-37. [REDACTED]

[REDACTED] *Id.* The work done by SEA on the DI Products continues through the [REDACTED]. JDX-0002C, ¶207; JDX-0015C, 9-11.

[REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶208; JDX-0015C, 43, 45-46. [REDACTED]

[REDACTED]

The investments made by Samsung in the DI Products in labor and capital comprise a significant portion of Samsung's U.S. [REDACTED] of DI Products. Samsung's domestic investments in labor and capital (comprehensive) for the DI Products totaled more than [REDACTED] for the '789 patent and more than [REDACTED] for the other asserted patents from the period January 1, 2022 through November 30, 2025. JDX-0002C, ¶210, Ex. 1.1. Samsung's domestic investments made up approximately [REDACTED] of Samsung's [REDACTED] of PDI Products practicing the '789 patent, and [REDACTED] of Samsung's [REDACTED] of WDI Products practicing the other asserted patents. JDX-0002C, ¶¶210-211, Exs. 1.1, 9.0, 9.2, 10.0, 10.1.

Samsung investments in plant and equipment for the DI Products also comprise a

[REDACTED]

significant portion of Samsung's operating expenses associated with the DI Products sold in the United States. Samsung financial documents recording the sales of the DI Products also include

[REDACTED]. JDX-0002C, ¶212, Exs. 7.2, 7.3; JDX-0015C, 82-83, 86. [REDACTED]

[REDACTED] JDX-0002C, ¶¶212-213, Exs. 7.2, 7.3; JDX-0015C, 81-83, 86, 107; RX-0409C.

[REDACTED] JDX-0002C, ¶213, Ex. 9.1; JDX-0015C, 81-82, 107; RX-0409C. Samsung's domestic investments in plant and equipment made up approximately [REDACTED] of Samsung's [REDACTED] of PDI Products practicing the '789 patent, and [REDACTED] of Samsung's [REDACTED] of WDI Products practicing the other asserted patents. JDX-0002C, ¶¶213-214, Ex. 9.1; JDX-0015C, 51-52, 81-82, 107-108; RX-0409C.

Samsung's domestic investments in labor or capital (comprehensive) made up approximately [REDACTED] of Samsung's [REDACTED] of PDI Products practicing the '789 patent, and [REDACTED] of Samsung's [REDACTED] of WDI Products practicing the other asserted patents. JDX-0002C, ¶216; JDX-0015C, 81-83, 107-108; RX-0409C. This is indicative of a relatively significant amount of dedicated activity that is applicable to the DI Products, [REDACTED]

[REDACTED] *Id.*

The investments made by Samsung in the DI Products in plant and equipment comprise a significant portion of Samsung's total investments directed to the [REDACTED]

[REDACTED]

[REDACTED] divisions of SEA. JDX-0002C, ¶218, Ex. 9.1. SEA's overall plant and equipment investments in mobile products also include [REDACTED]. *Id.* Samsung's domestic investments in plant and equipment for the DI Products associated with the '789 patent comprised approximately [REDACTED] of total domestic investments in plant and equipment in 2022, and this grew to approximately [REDACTED] in 2024. JDX-0002C, ¶¶218-219, Ex. 9.1. [REDACTED]

[REDACTED]

[REDACTED]

The investments made by Samsung in the DI Products in labor or capital (comprehensive) comprise a significant portion of Samsung's total labor and capital investments directed to the [REDACTED] of SEA. JDX-0002C, ¶220, Ex. 9.0. SEA's domestic labor and capital investments, too, [REDACTED]. *Id.* Samsung's domestic investments in labor and capital (comprehensive) for the PDI Products associated with the '789 patent comprised approximately [REDACTED] of total domestic investments in labor and capital and plant and equipment in 2022, and this grew to approximately [REDACTED] in 2024. JDX-0002C, ¶¶220-221, Ex. 9.0. [REDACTED]

[REDACTED]

[REDACTED]

Further, the significance of Samsung's investments in the context of mobile services may be seen by comparing the relative headcounts of the employees dedicated to the DI Products to the overall headcount [REDACTED] divisions of SEA. The headcount attributable to the DI Products comprised more than [REDACTED] of the overall domestic headcount for SEA employees working on mobile products as of the end of November 2025. JDX-0002C, ¶¶222-

[REDACTED]

224, Ex. 9.3; CX-1469C; JDX-0015C, 108-109.

2. Samsung's Domestic Investments Are Significant and Substantial In the Context of Samsung's Foreign Expenditures

[REDACTED]

[REDACTED] JDX-0002C, ¶225; JDX-0015C, 91. [REDACTED]

[REDACTED] *Id.* [REDACTED]

[REDACTED] *Id.* [REDACTED]

[REDACTED]

Samsung's domestic investments in the DI Products are significant in the context of foreign costs associated with the DI Products. The investments made by Samsung in the DI Products in labor and capital (comprehensive) comprise a significant portion of Samsung's U.S. cost of goods sold for the DI Products. Samsung's domestic investments in labor and capital (comprehensive) for the DI Products totaled more than [REDACTED] for the '789 patent and more than [REDACTED] for the other asserted patents from January 1, 2022 through November 30, 2025. JDX-0002C, ¶227, Exs. 1.1, 9.0. Samsung's domestic investments made up almost [REDACTED] of Samsung's [REDACTED] for DI Products sold in the U.S. practicing the '789 patent, and [REDACTED] of Samsung's [REDACTED] for DI Products sold in the U.S. practicing the other asserted patents. JDX-0002C, ¶¶227-228, Exs. 1.1, 9.0.

The DI Products are qualitatively and quantitatively significant to Samsung in a global context. SEC is one of the largest providers by volume of wearables and smartphones in the world, and SEA is one of the top providers by volume of wearables and smartphones in the United States. JDX-0002C, ¶229; JDX-32C, Ex. 11C, ¶1-2; CX-1695C, 15; JDX-0015C, 37-38, 49. The U.S. is a major market for Samsung with respect to the DI Products. *Id.* Samsung's net revenues

[REDACTED]

associated with the '789 patent made up approximately [REDACTED] of Samsung's [REDACTED] associated with the '789 patent from the DI Products. Samsung's net revenues associated with the other asserted patents made up approximately [REDACTED] of Samsung's [REDACTED] associated with the other asserted patents from the DI Products. JDX-0002C, ¶¶230-231; CX-1696C, 24; JDX-0021C, 44-46, 74-75. Given that Samsung broadly operates in 76 countries, having over [REDACTED] for a group of products associated with one specific country is significant. JDX-0002C, ¶232; CX-1512.

3. Samsung's Domestic Investments Are Significant and Substantial in the Context of the Industry

Samsung's DI Products are an important part of the U.S. mobile product competitive environment. By 2020, Samsung smartphones comprised approximately 25% of U.S. smartphone sales, competing primarily with Apple, which comprised approximately 50% of smartphone sales. JDX-0002C, ¶234, Ex. 9.6; CX-1556; CX-1524; CX-1552. Since 2020, Samsung has remained relatively stable, comprising approximately 20% to 30% of U.S. smartphone sales. *Id.* The domestic industry smartphone products, the Galaxy S series, are Samsung's flagship smartphones. *Id.* Samsung is also leading provider of wearable devices in the United States. JDX-0002C, ¶235; CX-1558C; CX-1554; CX-1369.

The investments made by Samsung into the DI Products are qualitatively significant and substantial. SEA's U.S. activities and investments are critical to the local development, commercialization, and marketing of the DI Products in the United States. JDX-0002C, ¶236; CX-1696C, 24; JDX-0015C, 50-51.

Service to mobile product customers in the United States is also significant, as it enables Samsung to provide timely and effective service to its customers, and is an important component of Samsung's operations. JDX-0002C, ¶237; CX-1696C; JDX-32C, Ex. 11C, ¶11; JDX-0015C,

[REDACTED]

120, 140, 142-146; JDX-0045C, 49. The [REDACTED] team at SEA provides service to mobile product customers in the United States. *Id.* [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *Id.*

Samsung's U.S. presence extends further through its partnership with U.S.-based institutions in connection with the DI Products. Samsung's [REDACTED] activities related to [REDACTED] also include [REDACTED]

[REDACTED]

[REDACTED]. JDX-0002C, ¶238; CX-1696C, 25; CX-0959C.

4. Samsung's Domestic Investments Are Expected To Continue

Samsung's commitment to its products and customers in the United States are ongoing.

[REDACTED]

[REDACTED] JDX-0002C, ¶240; JDX-0015C, 22, 26-29; CX-1695C. [REDACTED]

[REDACTED]

[REDACTED] JDX-0002C, ¶241; JDX-0015C, 22, 27-28.

[REDACTED]

[REDACTED] *Id.*

X. REMEDY & BONDING

The appropriate remedies in this Investigation are a permanent limited exclusion order ("LEO") and a permanent cease-and-desist order ("CDO") against Respondents, their subsidiaries, related companies, and agents. These remedies are warranted and necessary to protect Samsung's DI from Respondents' unfair acts. Oura has presented no evidence to support any exemptions or

delays in these remedial orders.

A. Limited Exclusion Order (LEO)

The appropriate remedy in the event of a violation finding under 19 U.S.C. § 1337(d)(1) is a permanent LEO directed to each Respondent (including its subsidiaries, predecessors, affiliates, agents, successors, and assigns) barring and excluding from entry into the U.S. wearable devices that infringe one or more of the Asserted Claims. *See Certain Fitness Devices, Streaming Components Thereof, & Sys. Containing Same*, 337-TA-1265, Comm’n Op. at 75 (Mar. 23, 2023) (holding that the LEO applies to covered articles unless products are “explicitly adjudicated as noninfringing”). If the Commission finds a violation of Section 337, an exclusion order “shall” issue against all of Respondents’ infringing products. *See* 19 U.S.C. § 1337(d)(1).

Respondents have not carried their burden to show that the LEO should have delays, carveouts, or exemptions. *See, e.g., Certain Automated Media Library Devices*, Inv. No. 337-TA-746, ID at 122, 2012 ITC LEXIS 1823, at *207 (June 20, 2012). For instance, Respondents have asserted that any LEO should contain a twelve-month transition period but have not provided any factual basis to justify implementation of such a period here. *See, e.g., Certain Mobile Devices, Associated Software, & Components Thereof*, Inv. No. 337-TA-744, Comm’n Op., 2012 WL 3715788, at *15 (June 5, 2012). Indeed, Respondents have no non-accused ring products to which they could shift production and no noninfringing alternative has been established. *See id.* Respondents also have not identified any third-parties who would be adversely affected or damaged by an LEO. *See Certain Mobile Devices*, 2012 WL 3715788, at *15. Further, there is no evidence that consumer demand cannot be satisfied by competing ring products sold by Samsung or other licensed entities. *Id.*

Respondents have also failed to establish that service and repair exceptions are warranted.

[REDACTED]

[REDACTED]

“The Commission [] has declined to issue service and repair exemptions to limited exclusion orders where the record contains insufficient evidence regarding the burdens and expenses that would be imposed on third parties in the absence of an exemption.” *Certain Wet Dry Surface Cleaning Devices*, Inv. No. 337-TA-1304, 2023 WL 3093539, *2-3 (ITC Apr. 7, 2023) (collecting cases). Such is the scenario here. Although Respondents have provided testimony that Accused Products may be subject to warranty obligations, Respondents have failed to establish how an LEO would render Respondents unable to satisfy any such obligations and have not shown any detrimental effect on customers without a service and repair exception. *Id.* [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] *See, e.g., JX-27C (Murrin), 77:11-78:9, 90:23-91:9.*

In view of the dearth of information about what the actual impact may be on consumers, and no evidence that consumer issues cannot be adequately redressed through refunds, a service and warranty repair exemption is not warranted. *See, e.g., Certain Magnetic Data Storage Tapes and Cartridges Containing the Same (II)*, Inv. No. 337-TA-1076, ID at 174 (Oct. 25, 2018), *affirmed by Comm’n Op. at 61-62* (June 20, 2019) (denying exemption because respondent failed to provide sufficient evidence). However, if one is issued, the cutoff date should be “the date the Commission’s orders issue” when “the public is put on notice of a violation.” *Certain Light-Based Physiological Measurement Devices & Components Thereof*, 337-TA-1276, Comm’n Op. at 107 n.72 (Nov. 14, 2023) (EDIS ID 808521) (dissenting views of Commissioner Kearns).

B. Cease and Desist Order (CDO)

The appropriate remedy in the event of a violation finding under 19 U.S.C. § 1337(d)(1) is

[REDACTED]

additionally a CDO directed to each Respondent barring the importing, selling for importation, marketing, demonstrating, distributing, repairing, refurbishing, offering for sale, selling after importation, or transferring inventory to the United States of all wearable devices that infringe one or more of the Asserted Claims. Respondents have commercially significant inventories, warranting CDOs to prevent Respondents from undercutting the remedy provided by an LEO. *See, e.g., Certain Digital Models, Digital Data, & Treatment Plans for Use in Making Incremental Dental Positioning Adjustment Appliances, the Appliances Made Therefrom, & Methods of Making the Same*, Inv. No. 337-TA-833, Comm’n Op. at 147 (Apr. 9, 2014). “[T]here is no lower limit on the number of articles a domestic respondent must have in inventory before that inventory can be found to be ‘commercially significant’” and the presence of units “held in ‘distribution centers’ demonstrates the units are meant for and capable of being distributed to consumers throughout the United States thereby harming” a complainant. *Certain Agricultural Vehicles and Components Thereof*, Inv. No. 337-TA-487, Comm’n Op. at 14 (Sept. 24, 2004); *Certain Audio Players and Components Thereof (II)*, Inv. No. 337-TA-1330, ID on Violation and RD at 194 (Sept. 15, 2023).

Respondents hold commercially significant domestic inventories of its Accused Products warranting a CDO. Specifically, Oura does not dispute that it maintains commercially significant inventory of the accused Oura Ring products in the US. JX-74C at 2-4. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

C. Bonding

If the Commission enters an LEO or CDO, a respondent may continue to import and sell its products during the PRP under a bond in an amount “sufficient to protect the complainant from any injury.” 19 U.S.C. § 1337(j)(3). A bond is warranted here to protect Samsung from injury. There is direct competition between Samsung’s DI Products and the Accused Products. For instance, Samsung considers the Accused Products to be direct competitors. RX-156C, RX-242C, JDX-2C, 33. As a result, there is a likelihood of injury to Samsung in the form of displaced sales, reduced market share, and/or price undercutting.

In the co-pending 1468 Investigation, Oura and its economic expert, John Jarosz, contended that [REDACTED] [REDACTED] set an established royalty rate for bond of [REDACTED] [REDACTED]. *See* CX-1697C, ¶¶201-203. Although Samsung disagrees that the alleged “established royalty rate” is so established, and disagrees with the underlying rationale for applying such a royalty rate to the accused products in the 1468 Investigation, to the extent the ALJ in that Investigation recommends and the Commission adopts Oura’s proposed rate as the established rate associated with the Oura Ring products, that rate should apply in this Investigation to the Galaxy Ring product as well.

XI. CONCLUSION

As explained herein, the evidence will show that a violation of Section 337 has occurred, and the ALJ should recommend and appropriate LEO, CDO, and bond to protect Samsung’s domestic industry and curtail infringement of the Asserted Claims.

Dated: August 10, 2026

Respectfully submitted,

/s/ D. Sean Trainor

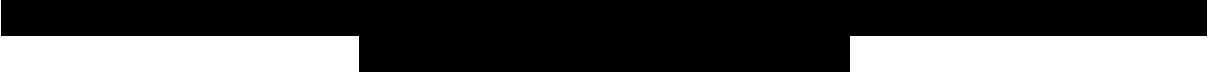
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CERTIFICATE OF WORD COUNT

The undersigned certifies that this brief complies with the word limitation set forth in Order No. 11 regarding pre-hearing briefs. The certified word count includes footnotes and excludes tables of contents, authorities, abbreviations, and asserted claims, words in figures, the word count certificate, signature block, and the certificate of service.

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The undersigned certifies that on August 10, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

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